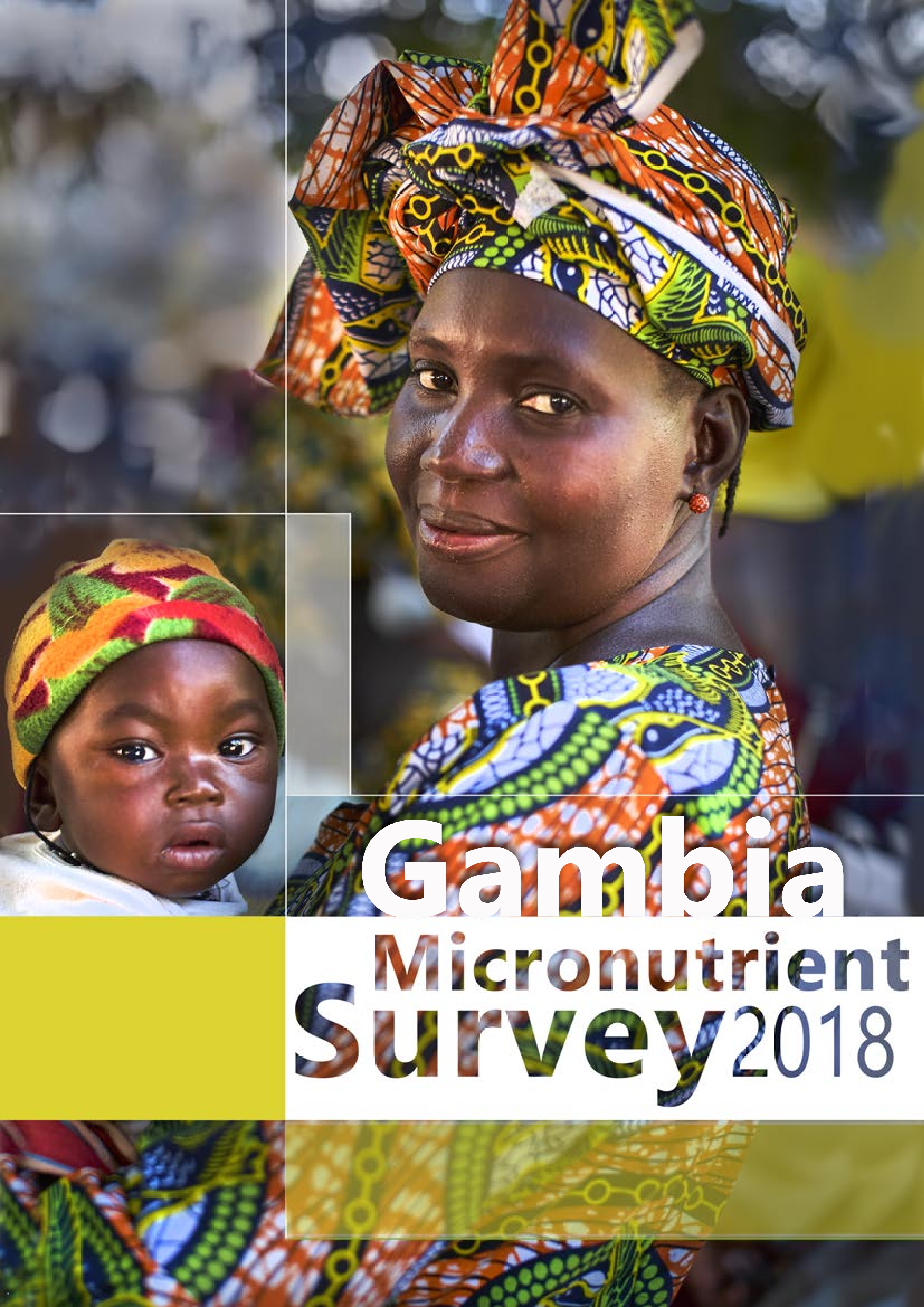
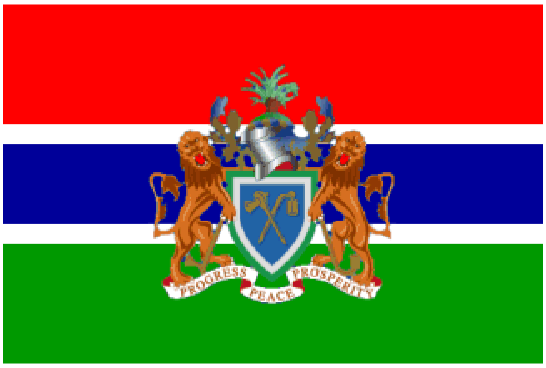


Gambia Micronutrient Survey 2018

THE GAMBIA MICRONUTRIENT SURVEY 2018 (GMNS)



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NaNA
National Nutrition Agency

unicef

Food and Agriculture Organization of the United Nations

THE WORLD BANK

WFP
United Nations
World Food Programme

UNFPA

MRC Unit The Gambia

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Gambia Bureau of Statistics
GBOS

GroundWork
BUILDING GROUNDS FOR NUTRITION & HEALTH

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ABBREVIATIONS

AGP	α-1-acid glycoprotein
BMI	Body mass index
CRP	C-reactive protein
DHS	Demographic and Health Survey
EA	Enumeration area
ELISA	Enzyme linked immunosorbent assay
ERC	Ethical Review Committee, Gambia Health Service
EU	European Union
GMNS	The Gambia Micronutrient Survey
HAZ	Height-for-age z-score
Hb	Hemoglobin
HbA1c	Glycated hemoglobin
HCAZ	Head circumference-for-age z-score
LRI	Lower respiratory infection
ID	Iron deficiency
IDA	Iron deficiency anemia
MICS	Multiple Indicator Cluster Survey
MDG	Millennium Development Goal
MoH	Ministry of Health
MUAC	Mid-upper arm circumference
NCD	Non-communicable disease
NPW	Non-pregnant women (15-49 years)
ppm	Parts per million
PSC	Preschool-age children (0-59 months)
PW	Pregnant women
RBP	Retinol-binding protein
RDT	Rapid diagnostic test
TfR	Transferrin receptor
UNICEF	United Nation's Children Fund
VAD	Vitamin A deficiency
WAZ	Weight-for-age z-score
WHO	World Health Organization
WHZ	Weight-for-height z-score

EXECUTIVE SUMMARY

Introduction

In The Gambia, nationally-representative data on micronutrient status and diet-related non-communicable diseases (NCDs) are limited and/or outdated. The last national micronutrient survey on iron and vitamin A status was conducted in 1999 reporting iron deficiency prevalence between 20-26% and vitamin A deficiency prevalence of 16-64% in preschool-age children and pregnant and lactating women. The MICS 2000 found that only 4% of children 6-59 months were given vitamin A supplements in the previous six months, which may partly explain the high prevalence of vitamin A deficiency (64%) in this age group. An iodine status assessment was also done prior to 2000 and revealed a median urinary iodine concentration of 42 µg/L, which classified Gambian school-age children as moderately iodine deficient. From 2000 to 2010, household coverage with iodized salt increased from 9% to 63.6%; however, the MICS 2010 found that only 23% of salt was adequately iodized. In addition, the emergence of diet-related NCDs is becoming more and more relevant in The Gambia. Increased prevalences of diabetes and cardiovascular diseases have been reported between 2008 and 2013, and there are indications that hypertension and other related forms of malnutrition are increasingly common health problems.

Objectives

The objective of The Gambian National Micronutrient Survey (GMNS) was to obtain updated and reliable information on the prevalence of micronutrient deficiencies and nutrition-related non-communicable diseases in children 0-59 months of age[1], non-pregnant women 15-49 years of age, and pregnant women in The Gambia. The aim is that these results will enable stakeholders to formulate evidence-based interventions to improve the nutritional status of these vulnerable groups.

Key nutrition indicators assessed included the prevalence of anemia and malaria parasitemia in children, non-pregnant women, and pregnant women; deficiencies of iron and vitamin A in children and non-pregnant women; prevalence of diabetes in non-pregnant women; prevalence of hypertension and iodine status in non-pregnant and pregnant women; childhood wasting, stunting and undernutrition; undernutrition in non- pregnant and pregnant women; child and adult overweight and obesity.

Other variables that may potentially influence or cause various types of micronutrient deficiencies, such as individual and household food consumption patterns and knowledge about fortified products. For households, the coverage of adequately iodized (≥15 ppm iodine) salt by quantitative measurement of iodine content was the primary indicator

[1] Blood samples were only collected from children 6-59 months of age, and thus micronutrient status biomarkers and anemia and malaria status are not provided for children 0-5 months of age.

assessed. Further, the GMNS sample of households is a subsample of the MICS. Some indicators examined from data collected by the MICS such as socio-economic status, household food consumption patterns, intake of micronutrient supplements or fortified/fortifiable foods, and household sanitation and hygiene were used to conduct sub-group analyses.

Design

The GMNS 2018 is a cross-sectional stratified survey based on a probability sample to produce estimates that have acceptable precision for priority indicators of nutritional status in children 0-59 months of age and non-pregnant women. In addition, nutrition related non-communicable diseases in women of reproductive age were assessed using two different strata (urban/ rural) across Gambia. The sample selection was done separately in each of the 14 strata defined by LGA and, within each LGA, urban and rural areas.

A two-stage sampling procedure was conducted to randomly select households. The MICS 2018 served as sampling frame. As a first stage, enumeration areas (EAs) or clusters within each sub stratum were randomly selected with probability proportional to population size from the 390 EAs enrolled in the MICS. As household sizes vary by region, different numbers of households were selected by LGA. For most LGAs, only a subsample of the households enrolled by the MICS in each cluster was included in the GMNS.

The households were randomly selected from the MICS household list by using simple random sampling. The GMNS 2018 survey was nationwide in scope, and collected data at the cluster level and from four target groups: 1) households, 2) children aged 0-59 months (6-59 months for blood biomarkers), 3) non-pregnant women of child-bearing age (15-49 years of age), and 4) pregnant women.

Results

In this executive summary, only national estimates are presented in Table 1. However, results in this table refer readers to the corresponding tables in the report that contain more detailed results.

Table 1. Summary results of the Gambia Micronutrient and Nutrition Survey 2018

Target group	Indicator[a]	Result	Table[b]
Households			
	Iodine in salt	10.8 %	Table 6
Children 6-59 months			
	Anemia	50.4 %	Table 18
	Mild anemia	28.1 %	Table 17
	Moderate anemia	21.4 %	Table 17
	Severe anemia	0.9 %	Table 17
	Iron deficiency	59.0 %	Table 18
	Iron deficiency anemia	38.2 %	Table 18
	Vitamin A deficiency (RBP)	18.3 %	Table 21
	Malaria (RDT)	<1 %	---
	Malaria (PCR)	8 %	Table 16
	Stunting (i.e. HAZ < -2)	15.7 %	Table 11
	Wasting (i.e. WHZ < -2)	5.8 %	Table 12
	Underweight (i.e. WAZ < -2)	10.6 %	Table 13
	Small head circumference (i.e. HCAZ < -2)	7.4 %	Table 14
Non-pregnant women (15-49 years)			
	Anemia	50.9 %	Table 32
	Mild anemia	26.7 %	Table 31
	Moderate anemia	22.9 %	Table 31
	Severe anemia	1.3 %	Table 31
	Iron deficiency	41.4 %	Table 32
	Iron deficiency anemia	28.0 %	Table 32
	Vitamin A deficiency (RBP)	1.8 %	Table 34
	Median urinary iodine concentration	143.1 µg/L	Table 35
	Malaria (RDT)	<1 %	---
	Malaria (PCR)	6.0 %	Table 30
	Hyperglycemia	8.1 %	Table 36
	Hypertension	10.6 %	Table 37
	Underweight (BMI<18)	15.4 %	Table 29
	Overweight (BMI>25)	18.3 %	Table 29
	Obesity (BMI>30)	11.1 %	Table 29
Pregnant women			
	Anemia	56.8 %	Table 41
	Median urinary iodine concentration	113.5 µg/L	Table 42
	Malaria (RDT)	2.3 %	---
	Hypertension	9.6 %	Table 43
	Underweight (MUAC)	7.5 %	Table 40

[a] See text of method section for case definitions;
[b] Refer to the table indicated for more detailed analysis of the outcome, including group-specific results by age, region, wealth quintiles and other analyses.

Discussion

More than 90% of the surveyed households consume vegetable oil. While few women are vitamin A deficient and may, at present, not require vitamin A-fortified oil, oil fortification can potentially address vitamin A deficiency in children in these households. Iron deficiency is a severe problem in women and children in The Gambia. Survey findings show that rice and wheat flour constitute suitable iron fortification vehicles to reduce iron deficiency in women and children as both are consumed by the vast majority of surveyed households. Moreover, the home production of wheat flour and rice is low, which should allow a wide coverage of fortified products.

This survey's findings demonstrate a decline of stunting, wasting and underweight compared to previous assessments. Despite this relatively low national prevalence, undernutrition is more prevalent in certain sub-groups. High prevalences can be found in the rural population and LGAs with a high proportion of rural population. Highest prevalence of malnutrition is found in Kuntaur, denoting a severe public health problem according to WHO in that LGA. Further, stunting and underweight, but not wasting is more prevalent in children living in poorer households. Stunting is highly associated with inadequate household sanitation in The Gambia.

Microcephaly, which affects brain size and thus cognitive development, is present in 7.4% of the surveyed children. The survey did not allow distinguishing between primary (congenital) microcephaly (the brain fails to grow during pregnancy) and secondary (perinatal) microcephaly (head circumference normal at birth, but fails to grow appropriately thereafter). Our data suggests that undernutrition, which has previously been identified as the major cause of microcephaly, is highly associated with microcephaly in Gambia as a significantly higher proportion of stunted, wasted and underweight surveyed children also suffered from microcephaly.

Undernutrition in women, both pregnant and non-pregnant, does not appear to be of major concern in The Gambia. By far, the larger problem in this group is overweight and obesity, affecting about one-third of surveyed non-pregnant women. In non-pregnant women, obesity is associated with hyperglycemia, overweight, and intermediate hyperglycemia. Moreover, significantly more overweight and obese women suffer from hypertension, compared to normal weight and underweight women.

About 50% of surveyed children suffer from anemia, the majority of them having concurrent iron deficiency. The current analysis clearly suggests that although there are multiple factors contributing to anemia, anemia in Gambian children is mainly nutritionally-induced. Overall anemia prevalence is lower than reported in 2013, which could be due to the low malaria prevalence and low presence of illness found during GMNS/MICS and/or technical differences of the hemoglobin analyzers. Contrarily, the prevalence of anemia found in non-pregnant women and pregnant women are similar to the prevalence found in 2013.

The national prevalence of iron deficiency (ID) in children and women is high. Compared to the survey in 1999 the prevalence in children almost tripled and doubled in women. Of note, a direct comparison of the results is not possible since the survey in 1999 a) did not adjust ID for increased concentrations of AGP and CRP; b) used a different cut-off for ID in children and women; c) only included lactating and pregnant women. The main reasons for the high prevalence of ID are low intake of iron and/or low bioavailability of iron in the diet. The majority of the population, especially the poor subsist on a plant-based diet, which is low in bioavailable iron and high in iron absorption inhibitors. Very few women had ever heard about fortified wheat flour, fortified vegetable oil, or had seen the Enrichi logo. The fortification of wheat flour is nascent in Gambia, and as such, the low awareness of fortified wheat flour products is understandable. However, the low awareness of fortified vegetable oil and the Enrichi logo, which are both present in the Gambia marketplace, suggests that the marketing to date of fortified products has not translated to consumer awareness.

VAD affects more than 18% of children in Gambia and is almost non-existent in non-pregnant women. Although the prevalence in children is lower than the prevalence reported in 1999, there are still subgroups where VAD poses a severe public health problem such as children 24-59 months of age, male children and children living in rural areas. The GMNS did not find an association between VAD and routine vitamin A supplementation in the past 6 months, but did find significantly lower levels of VAD in children that received routine vitamin A supplements in the past 3 months. This finding is intuitive as studies have found that vitamin A supplementation results in elevate blood retinol levels for approximately 3 months. Nonetheless, it is notable that relatively few children received vitamin A supplements; only 31% and 16% of children <5 years of age received a vitamin A supplement six or three months prior to the GMNS, respectively. Vitamin A supplements were more frequently given to children 6-23 months of age, which likely explains the lower VAD prevalence found in these younger children.

Although only a small proportion of the salt in Gambia is adequately iodized the high median urinary iodine concentration in non-pregnant women demonstrates an absence of iodine deficiency in this group. Nonetheless, median urinary iodine concentrations are below levels of adequacy in certain LGAs and in women living in poor households. The situation is more severe in pregnant women, as in this population group the median urinary iodine concentration indicates iodine deficiency; especially for pregnant women living in rural areas and in households with inadequately iodized salt.

HbA1c was used to determine hyperglycemia in women, reflecting plasma glucose levels of the last 8-12 weeks. Nationally the prevalence of hyperglycemia in non-pregnant women is 8.1%, which is somewhat lower than in European women (9.6%), but higher than the estimated prevalence in African countries in 2013 (about 5%). The survey found a high prevalence of intermediate hyperglycemia, though these findings have to be used with caution as there is no consensus on the cut-off for intermediate hyperglycemia. Both, hyperglycemia and intermediate hyperglycemia are associated with nutritional status, and significantly more obese women suffer from hyperglycemia and significantly more women with intermediate hyperglycemia are overweight. Similar to hyperglycemia, hypertension in Gambian women is related to nutritional status and age, which is in accordance with literature.

Recommendations

1. Reduce undernutrition in children: The GMNS did not collect data on IYCF indicators, thus their impact on nutritional status in Gambian children requires further research. However, the survey findings show that inadequate sanitary conditions lead to increased rates of stunting and underweight and thus, WASH programs targeting at increasing access to safe drinking water and adequate sanitation facilities should be strengthened.

2. Reduce anemia and iron deficiency in children and women: Results suggest that the main type of anemia in women and children is nutritional anemia. Iron deficiency is the major contributor to anemia in children and woman. Vitamin A deficiency is also highly associated with anemia, but contributes far less than iron deficiency due to its low prevalence, especially in women. Although anemia, ID and IDA affect the whole population, programs to reduce their prevalence in Gambia should prioritize activities in rural areas of certain LGAs, targeting poor households. Interventions should include the promotion of age-appropriate infant and young child feeding practices and the promotion of the consumption of iron-rich foods and iron supplements for women. Moreover, to decrease the prevalence of ID, mandatory iron fortification programs should be implemented along with programs that target vulnerable groups, such as iron/folic acid supplementation targeted at adolescent girls.

3. Reduce vitamin A deficiency in children: While vitamin A deficiency affects almost 20% of children in The Gambia, the severity of this deficiency is significantly higher in some LGAs. To address this severe public health problem, multiple approaches should be used. First, The Gambia's vitamin A supplementation program should be improved to strengthen children's immune systems. Second, to increase vitamin A stores, mandatory vitamin A fortification programs should be implemented, such as vitamin A fortification of vegetable oil. Third, programs to improve consumption of vitamin A-rich foods, other than fortified products, should be pursued.

4. Reduce the prevalence of overweight and obesity in women: Nearly one-third of non-pregnant women were classified as either overweight or obese, with the highest proportions in urban area and among women residing in wealthier households. As overweight and obesity are risk factors of type-2 diabetes mellitus and are associated with hypertension and some forms of cancer, there is an imperative need to educate urban women about approaches to maintaining healthy weight to prevent the prevalence of overweight and obesity from rising further.

5. Increase median urinary iodine concentration: Only a small proportion of the salt consumed in The Gambia is adequately fortified, leading to low median urinary iodine concentrations in some population sub groups. One way to tackle this is increasing the coverage with adequately iodized salt by optimizing the regulatory monitoring system. As salt iodization in Gambia was made mandatory in 2006, there already exists a programmatic foundation to increase coverage of iodized salt. An improved surveillance system will have to be accompanied by Social and Behaviour Change Communication (SBCC) campaigns to increase the awareness and with it demand for iodized salt, targeting mainly the poor population in rural areas. Thus, salt intake, iodine intake and median urinary iodine concentrations should be regularly monitored to ensure adequate iodine status in all population groups.

1. INTRODUCTION

1.1. Country overview and nutritional situation in Gambia

Countries in sub-Saharan Africa are highly affected by malnutrition, including micronutrient deficiencies [1], underweight, and stunting. These adverse health outcomes inhibit progress in human and economic development. Women of reproductive age, particularly during pregnancy and lactation, exhibit increased nutrient needs and are therefore vulnerable population groups. Additionally, because of the critical link between maternal and child nutrition, public health efforts have turned to ensuring adequate mother and child nutrition during the first 1,000 days of life: from conception to a child's second birthday. In additional to under-nutrition, over-nutrition and nutrition-related non-communicable diseases are on the rise affecting more and more people on the African continent and contributing to the burden of disease [2]. WHO estimated that almost 50% of adult disease burden can be attributed to non-communicable diseases in low income countries [3].

The first and only Demographic Health Survey (DHS) in the Republic of The Gambia (The Gambia hereafter) was conducted in 2013, and found that under-5-mortality had decreased in the 15 previous years from 89 to 54 deaths/1000 live births [4]. Reductions in other child mortality indicators have also been observed, and were reported to have met or even exceeded Millennium Development Goal (MDG) Target #4 [5]. The proportion of underweight among children under 5 years of age has decreased slightly, from 20.3% in 2003 to 16.2% in 2013. However, the MDG target for underweight was not met. The prevalence of stunting and wasting remained mostly unchanged in the same time period: in 2015, a SMART survey estimated that 23% of children 0-59 months of age were stunted and 10% were wasted; the 2013 DHS found prevalences of 25% and 12%, respectively [4], while the 2012 SMART survey reported respective prevalences of 21% and 10%; in 2000, the Multiple Indicator Cluster Survey (MICS) found prevalences of 19% and 8% [5,6].

Among children, the problem of overweight currently appears limited, as the 2013 DHS estimated a prevalence of 3%. On the other hand, about a quarter (23%) of adult women are overweight [4] according to the 2013 DHS. The 2015 SMART survey found similar results, with 24.1% of women classified as overweight or obese [5].

In The Gambia, nationally-representative data on micronutrient status and diet-related non-communicable diseases (NCDs) are limited and/or outdated. The last national micronutrient survey on iron and vitamin A status was conducted in 1999 [6]; it estimated the deficiency prevalence in preschool-age children and pregnant and lactating women. The prevalence of iron deficiency for all groups ranged from 20% to 26% (see [6]; note that these prevalence estimates were adjusted for some measure of acute phase response). In children and pregnant women, the majority of anemia was associated with iron deficiency, while in lactating women, about half of anemia cases were also iron deficient [6]. In 2013, 17% of children <5 years of age were reported to have received iron supplementation, while 96% of pregnant women received iron supplements [4].

In 1999, less than one-fifth (16%) of lactating women were affected by vitamin A (VA) deficiency, while one-third (34%) of pregnant women and almost two-thirds (64%) of children 12-59 months were deficient in vitamin A. For a similar time period, the MICS 2000 found that only 4% of children 6-59 months were given vitamin A supplements in the previous six months [7], which may partly explain the high prevalence of vitamin A deficiency. The situation has apparently improved; the 2013 DHS reported a vitamin A supplementation coverage in the previous six months of 69%.

An iodine status assessment was also assessed prior to 2000 and revealed a median urinary iodine concentration of 42 µg/L, which classified Gambian school-age children as moderately iodine deficient [8]. Goiter assessment using palpation was also conducted and nationally, 16% of children had goiter of any degree. From 2000 to 2010, household coverage with iodized salt increased from 9% to 63.6%; however, the MICS 2010 found that only 23% of salt was adequately iodized (i.e. >15ppm iodine) based on rapid test kit results. The 2013 DHS found that 76% of household salt was iodized, but no results of adequately iodized salt coverage were reported.

The various forms of malnutrition vary according to population groups and geography. To illustrate: The Local Government Areas (LGA) of Kuntaur, Janjanbureh and Basse had the highest anemia prevalence in both women and children and stunting prevalence in children in 2013 [4]. Kerewan and Mansakonko LGAs also displayed considerably lower household coverage of iodized salt. These five LGAs tended also to be worse off in terms of children's intake of vitamin A and iron rich foods. In contrast, in Banjul and Kanifing, which are better off for most of aforementioned indicators, the prevalence of overweight and obese women are double those of other LGAs.

Though not following the same patterns across the different LGAs, the emergence of diet-related NCDs is becoming more and more relevant in The Gambia. Omoleke reported increased caseloads of diabetes and cardiovascular diseases between 2008-13 [2], and noted that hypertension and other sequelae of malnutrition may also be common health problems.

1.2. Rationale for the survey

While some data on the nutritional and micronutrient status of women and children are available, there is no comprehensive assessment of nutritional status measuring both micro- and macro-nutrient indicators in Gambia. The prevalence of anemia and under- and over-nutrition has been collected as part of DHS and Multiple Indicator Cluster Surveys (MICS), but these surveys do not assess key risk factors of anemia, such as micronutrient deficiencies, malaria, and inflammation. Although DHS and MICS often measure hemoglobin concentration and estimate anemia prevalence, it is not possible to accurately extrapolate the underlying risk factors of anemia. While anemia is often presumed to be largely-caused by iron deficiency

(ID), this assumption has been challenged by recently-conducted micronutrient surveys in the region [9,10].

As malnutrition and related NCDs are posited to be among the leading causes of morbidity and mortality in The Gambia, a current and thorough investigation of the nutritional status of vulnerable groups is warranted. The Gambia Micronutrient Survey 2018 (GMNS) is expected to increase the understanding of the extent of micronutrient deficiencies and will enable the government and international agencies to monitor and evaluate the current status of national nutrition programs (e.g. salt iodization) and to plan future nutrition programs (e.g. additional food fortification).

1.3. Primary objectives and indicators

The overall goal of this survey was to obtain updated and reliable information on the current micronutrient status and diet-related NCDs of women 15-49 years of age and of micronutrient status of children 6-59 months of age in The Gambia. To develop evidence-based policies to improve the nutritional status of these target groups and to implement programs to reduce the prevalence of malnutrition in the country.

The GMNS 2018 was nationwide in scope, and collected data from four target groups: 1) households, 2) children aged 0-59 months, 3) non-pregnant women of child-bearing age (15-49 years of age), and 4) pregnant women. Indicators collected varied by population groups and are detailed below.

1. To measure the prevalence and severity of anemia in children 6-59 months of age, non-pregnant women of child-bearing age, and pregnant women based on blood hemoglobin concentrations.
2. To assess the prevalence and severity of iron deficiency in children 6-59 months of age and non-pregnant women of child-bearing age by measuring serum ferritin concentration (adjusted for the presence of inflammation) and by measuring serum transferrin receptor (TfR).
3. To assess the vitamin A status of children 6-59 months of age and non-pregnant women of child-bearing age by measuring retinol-binding protein (RBP) in serum (adjusted for the presence of inflammation).
4. To measure the population iodine status by measuring the median urinary iodine concentration among non-pregnant women 15-49 years of age and pregnant women.
5. Estimate the current prevalence of hyperglycemia among women of reproductive age measured using glycated hemoglobin (HbA1c).
6. To assess the prevalence of malaria in children 6-59 months of age and non-pregnant women of child-bearing age by using a rapid antigen test kit and molecular diagnostic for the presence of *Plasmodium falciparum* parasites and other *Plasmodium* species.

7. Estimate the current prevalence hypertension of women of reproductive age measured using blood pressure.
8. To assess the household coverage of adequately iodized salt using a quantitative measure of salt iodine content.

1.4. Secondary objectives and indicators

The GMNS also investigated other variables that may influence various types of malnutrition or play a causative role. This questionnaire-based data included household and individual food consumption patterns, physical activity, and knowledge and practice regarding fortified foods.

In addition, the GMNS collected anthropometric data to:

1. Estimate the current prevalence of acute malnutrition (wasting), chronic malnutrition (stunting), and overweight and obesity in children 0-59 by calculation of the weight-for-height and height-for-age z-scores.
2. Estimate the long-term effect of undernutrition in children 0-59 months by measuring head circumference.
3. Estimate the current prevalence of chronic energy deficiency and overweight and obesity in non-pregnant women by calculating body mass index (BMI).
4. Estimate the current prevalence of undernutrition in pregnant women by measuring mid-upper arm circumference (MUAC).

Further, the GMNS sample of households is a subsample of the MICS and included only respondents who have also been part of the MICS. Thus, for data analysis, the MICS data was merged with data from the GMNS and additional variables were used during the analysis and interpretation of key biomarkers. Indicators examined from data collected by the MICS include assessment of socio-economic status, household food consumption patterns, infant feeding and breastfeeding practices, intake of micronutrient supplements or fortified/fortifiable foods, and household sanitation and hygiene. The GMNS did not report on these variables individually, but used data collected by the MICS to conduct sub-group analyses.

Table 2. Overview of key biomarker indicators, by population group*

Biomarker Indicators	Sample	Children	NPW	PW	HH
Hemoglobin (anemia status)	Whole Blood	+	+	+	
Malaria infection and speciation	Whole Blood	+	+	+	
HbA1c (hyperglycemia)	Whole Blood		+	+	
Ferritin (iron status)	Serum	+	+		
Soluble transferrin receptor (iron status)	Serum	+	+		
Retinol binding protein (Vitamin A status)	Serum	+	+		
AGP and CRP (Inflammation)	Serum	+	+		
Urinary iodine excretion (population iodine status)	Urine		+	+	
Blood pressure	Examination		+	+	
Height	Examination	+	+		
Weight	Examination	+	+		
MUAC	Examination	+		+	
Head circumference	Examination	+			
HH salt iodine concentration (salt iodization)	Salt sample				+

NPW = non-pregnant women; PW = pregnant women; HH = households

2. METHODOLOGY

2.1. Survey design

The GMNS was a cross-sectional stratified survey. Results were generated for two explicit strata: urban and rural areas in The Gambia. In addition, implicit sampling was done by LGA. As a result, the GMNS sample was drawn separately from the urban and rural areas in each of the 8 LGAs. This resulted in 14 strata as two LGAs, Banjul and Kanifing, were entirely urban. Each of the 14 strata was treated as an independent sampling universe, and enumeration areas (EAs) for each stratum were selected from the 390 MICS sample EAs in the corresponding stratum. The number of urban and rural EAs selected from each LGA was determined proportionately based on the distribution of the households within each LGA, except in Banjul and Kanifing where there were only urban EAs (see list of selected EAs in appendix 7.1).

2.2. Sample size determination

An *a priori* sample size for the entire survey and each stratum was based on the estimated prevalence, the desired precision around the resulting estimate of prevalence, and the expected design effect for priority indicators of nutritional status in children 0-59 months of age and non-pregnant women 15-49 years of age. Calculations assumed an expected household response rate of 95%. Individual response rates for interview questions and anthropometric measurements were assumed to be 95%, while response rates for phlebotomy were assumed to be lower (85% for women and children). All calculations used Fisher's formula for estimating the minimum sample size for estimating prevalence:

$$n = \frac{Z^2{}_{\alpha/2}P(1-P)}{d^2} \times DEFF \times \frac{100}{CombinedRR}$$

Where:

$Z_{\alpha/2}$ = Z-value corresponding to the 95% confidence level

P = Assumed prevalence

d = Desired precision expressed as the half confidence interval in decimal form

DEFF = Design effect

RR = Overall response rate (household response x individual response) expressed as a decimal

The GMNS aimed to select 1,156 households from the MICS sample to ensure sufficient sample size of households, children, and women. In each selected household all children 0-59 months of age, non-pregnant women of reproductive age and pregnant women were recruited, and questionnaire data was collected. Blood samples were taken from all children found in the households. For women, the GMNS commenced by selecting all women in the households. However, this sampling approach was modified midway through data collection as the number of women and children residing in the households was larger than estimated, resulting in more blood samples than expected. Thus, during the second half of data collection, blood and anthropometry data were taken only from women and children in two-thirds of the households. This different likelihood of selection was accounted for during data analysis.

2.3. Sampling procedure

GMNS clusters were exclusively drawn from the 390 EAs selected as part of the 2018 MICS. The MICS was a nationally-representative survey, and within each LGA, the sample EAs were allocated to the urban and rural strata approximately in proportion to the number of households in the sampling frame based on The Gambia Census 2013. The 2018 MICS contained 14 strata comprised of urban areas in Banjul and Kanifing, respectively, and urban and rural areas in the remaining six LGAs. Within each EA, the MICS selected 20 households.

The GMNS selected a sub-set of EAs from the MICS using the same stratification approach. As such, the GMNS is nationally representative, and representative of each LGA. and individuals who participated in the MICS were enrolled in the GMNS, with the exception of children born between MICS and GMNS data collection who were included in the GMNS and children ageing-out of the target age range, who were excluded from the GMNS.

The average household size, and thus the average number of eligible women and children residing in the household differs substantially by LGA. As the MICS only selected 20 households per EA, the GMNS enrolled a different number of households per cluster depending on the household size in the LGA in order to recruit the minimum number of women and children. In order to obtain roughly equal numbers of children and women in each LGA, different numbers of EAs had to be included per LGA. Regarding households, the GMNS selected 20 households in Banjul, Kanifing and Brikama; 15 households in Mansakonko, Kerewan and Janjanbureh; 12 households in Kuntaur and 10 households in Basse (see Table 3). Regarding EA selection, 13 EAs were selected for Banjul, 10 for Kanifing and 8 EAs for the remaining LGAs. Thus, the GMNS was conducted in 71 EAs selected randomly from the 390 EAs already selected as part of the 2018 MICS (see Table 3). The selected clusters are shown in the map below and in Section 7 (Appendices).

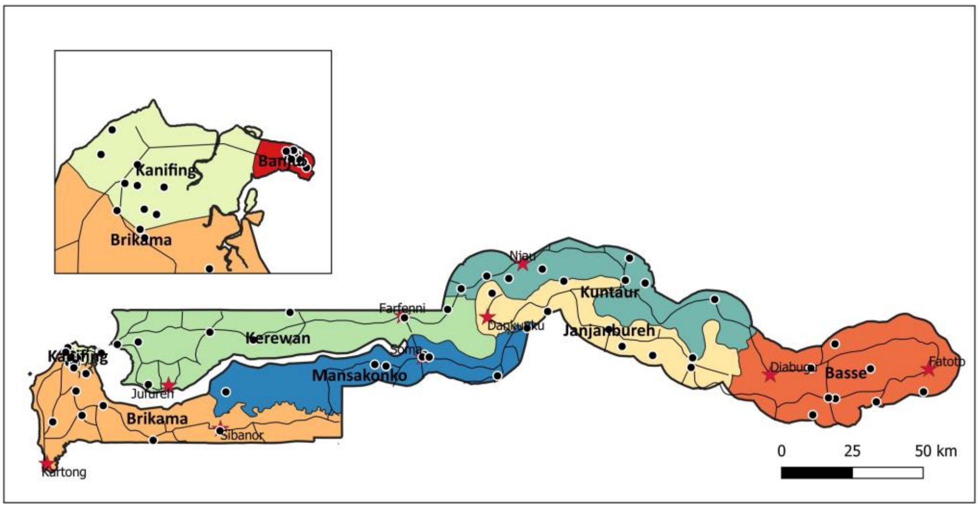
Table 3. Number of EAs per LGA, households per EA and expected number of Children and NPW

LGA	Average number of Children/ HH[a]	Average number of NPW/ HH[a]	Number of EAs	Number of HH/ EA	Total number of HH	Total number of Children	Total number of NPW
Banjul	0.67	1.32	13	20	260	175	293
Kanifing	0.9	1.67	10	20	200	184	284
Brikama	1.22	1.66	8	20	160	196	226
Mansakonko	1.45	1.77	8	15	120	176	180
Kerewan	1.56	1.76	8	15	120	188	180
Kuntaur	2.09	2.22	8	12	96	193	181
Janjanbureh	1.65	2.13	8	15	120	203	217
Basse	2.58	3.01	8	10	80	207	205
All	1.36	1.84	71		1156	1400	1765

[a] Numbers obtained from MICS 2010

For children, the selection of 1,156 households was expected to result in the recruitment of about 1,260 survey subjects for the entire survey sample, with about 1,020 of them expected to have their biological specimens collected. The selection of 1,156 households was expected to result in the recruitment of more than 1,550 non-pregnant women, with about 1,400 of them expected to have their biological specimens collected.

With this sample size and sampling scheme, the precision obtained for most indicators in most target groups were equal to or only slightly less than the desired precision. The exception to this was pregnant women, as it is difficult to recruit a large number of pregnant women in household-based surveys. Even in the highest fertility populations, pregnant women are still relatively rare in the general population. As a result, usable precision among pregnant women can be obtained only on a national level. The estimates for pregnant women in each stratum will have very low precision and will therefore not be calculated.



Map. Location of 71 clusters included in The Gambia Micronutrient Survey

Table 4 below lists the inclusion criteria for enrollment into the survey for each of the target groups. Note that only households and targeted individuals who participated in the MICS were enrolled in the GMNS, with the exception of children born between the MICS and GMNS data collection, who were also included in the GMNS 2018. Thus, all children 0-59 months of age at the time of MICS data collection and children born between MICS and GMNS data collection were enrolled in the GMNS 2018. Note that for pregnant women, only one nationally representative estimate is generated for the prevalence of anemia, malaria, hypertension and underweight. For all groups, an individual was excluded if he/she was excluded by the MICS and if, for children 0-59, the parent or guardian refused; for girls 15-17 years, the girl herself or a parent refused; and for an adult woman 18 years of age and older, the individual herself did not give consent.

2.4. Ethical considerations

Ethical approval for the survey (R18014) was obtained from The Gambia Government/ MRC Joint Ethics Committee and the School of Medicine and Allied Health Sciences Research & Publication Committee (RePubliC); the approval letters are presented in Appendix 7.3.

For household interviews, oral consent was sought from the household head or in his/her absence, from another adult household member. The selected women and child caregivers were asked to provide written informed consent (see appendix 7.4 for information sheet and consent form) for themselves and their participating children. If any consenting survey participants were unable to read and write, the consent form was read out loud to them and a thumbprint or fingerprint was taken as evidence of consent in lieu of a signature. Alternatively, the respondent could assign a witness to sign on their behalf. The respondents were also told that they are free to withdraw from participation in the survey at any time, even after written consent had been given.

Table 4. Inclusion criteria by targeted population group

Target population	Inclusion criteria
Households	• Household head or spouse or other adult household member gives oral consent for survey data collection • Members currently residing in selected EA • Household was part of the MICS survey
Children 0-59 months	• Age 0-59 months at the time of MICS data collection and children born between MICS and GMNS data collection; 6-59 months for blood sampling • Child has been part of the MICS survey • Considered a household member by adults living in the household • Mother or caretaker provided written consent for questionnaire, anthropometric measurements, and collection of biologic specimens
Non-pregnant women	• Age 15-49 years at the time of MICS survey data collection • Participated in the MICS • Currently non-pregnant by self-report • Gives written consent for survey questionnaire, anthropometric measurements, and collection of biologic specimens • Considered a household member by other adults living in the household
Pregnant women	• Currently pregnant by self-report • Participated in the MICS • Gives written consent for survey questionnaire, anthropometric measurements, and collection of biologic specimens • Considered a household member by other adults living in the household

Urine and blood were collected in the survey. Collection of urine samples is non-invasive, and collection of blood samples was done via finger prick in women and children 12-59 months or heel prick children 6-11 months, which is considered minimally invasive, and only causes momentary discomfort. To protect small children and avoid undue stress, no blood samples were collected from children less than 6 months of age.

Individuals identified with malaria, severe anemia or severe acute malnutrition (children only) according to WHO criteria [11,12] were referred for treatment at the nearest health facility. The GMNS also referred all women with a MUAC<18.5 cm as this cutoff has been used in other countries in West Africa [13] to define acute malnutrition. The referral form is presented in Appendix 7.5.

2.5. Field work and data collection

2.5.1. Instrument pre-testing, training of survey teams and field testing

Prior to data collection, team members were thoroughly trained, and all survey instruments were pre-tested during the training. Survey staff also received extensive classroom training of each questionnaire, whereby interviewers and team leaders discussed each question, practiced reading the questions, and role-played interviews in local languages. In addition, instruction was provided on how to record, save, and upload data using tablet computers (Lenovo) programmed with CSPro (Version 7.1) used in the GMNS 2018 survey. The training consisted of classroom instruction and practice, laboratory practice and field testing of all survey procedures.

As part of classroom training, anthropometrists and phlebotomists were trained on anthropometric and blood and urine collection techniques. A standardization exercise was conducted for the survey anthropometrists, whereby two anthropometrists worked as a team and recorded the length/height and weight of 10 children, and their results were checked for precision as well as for accuracy when compared with those of the mean of all anthropometrists. Regarding phlebotomy, blood collection procedures were practiced, including training on labeling of samples, and maintenance of the cold chain when transporting blood specimens.

Laboratory technicians received class room and laboratory practice training in processing of samples, labeling of aliquots, pipetting procedures, and maintenance of the cold chain when transporting blood specimens.

Following classroom and laboratory training, one day of field pre-testing was undertaken in two EAs in Bwiam, which were not included in the survey sample. Example MICS household lists (see appendix 7.6) were developed prior to the field work to enable teams to identify household in the same manner that would be used during the actual fieldwork. The teams then conducted the community sensitization, household interviews including the collection of salt samples, individual interviews, and all anthropometry and phlebotomy procedures. Teams also practiced transportation of specimens to a laboratory, where blood and urine specimens were processed by laboratory technicians.

To ensure that all survey staff ultimately hired could implement the survey procedures and to provide cover for unexpected absences, more survey workers than required were enlisted for the training. The results of the anthropometry standardization exercise, and observations from the survey trainers were used to: a) identify the best-performing team member to be the team leader for each of the teams and b) identify survey workers that could not adequately understand and implement the survey procedures. These individuals were released and were not included in the fieldwork.

2.5.2. Information sharing between MICS and GMNS

During the implementation of field work, key information was shared by the MICS management team and the GMNS management team. In particular, the MICS Data Unit based in Dakar and GBOS shared basic household level information for each EA included in the GMNS after the data collection in the EA has been completed by the MICS. This information consisted of a list of all households enrolled in the MICS, the location of the households, the interview status of the MICS (e.g. accepted, refused, absent, household not found), the names of the household heads and the names of all women 15-49 years of age, pregnant women and children <5 years of age residing in the household at the time of data collection by the MICS. An example of this form is provided in appendix 7.6)

2.5.3. Community mobilization and sensitization

A series of community mobilization and sensitization activities were conducted by the various stakeholders for the GMNS. Community mobilization were conducted in a systematic and coherent manner via the national, regional, district and community structures.

At the central level, a task force was created with various stakeholders, who were tasked with the responsibility of developing audio/visual and key messages for the GMNS and signed contracts with all the community radios and held series of panel discussions on national television. The central level communicated to the regional levels through the Regional Health Directorates. In addition to these approaches, the Regional Health Promotion and Education Officer went to all the selected EAs ahead of the GMNS survey teams to sensitize the Alkalo and the members of the selected households. The District Chief were also sensitized about the survey and they were encouraged to inform the Village Alkalo.

Upon arrival of the teams in each EA, the team leader met with the relevant authorities to inform them again about the work and also seek their support.

2.5.4. Household identification and random selection of households

For each EA, a cluster map as well a household list containing the up to 20 households included in the MICS survey were provided by GBOS. The household lists and map contained the location of the household, name of the household head as well as the number and names of eligible women and children living in the households that were enrolled in the MICS. For the LGAs where the GMNS required less than the number of households enrolled by the MICS in each cluster, the team leader selected the required number of households at random using random number tables. After selection, the different households were assigned to the interviewers, and interviews with the head of households were scheduled.

2.5.5. Field work (interviews)

Each of the 8 teams was comprised of one team leader, one interviewer, one phlebotomist, two anthropometrists and one driver. The composition of each team is provided in appendix7.7. Data collection was conducted between 13th March and 4th May 2018, which is the dry season in The Gambia.

All reasonable attempts were made to recruit selected households. At least two repeat visits were made before dismissing a household as non-responding. No substitution of non-responding households was done as this had been taken into account during the sample size calculation.

For data collection at the household level, tablet computers were used for direct data entry. In addition to the questionnaires, a series of supporting instruments (loaded on the tablets, such as picture of the 'Enrichi' logo) were used to facilitate field work and ensure high quality of the field work. Skip patterns were built into the questionnaires as uploaded in the tablets, which increased the speed of the interviewing process as well as minimized inappropriate entries. Interviewers administered the household questionnaire first, followed by women questionnaires if the household had eligible women. No questionnaire data was collected for children, where only anthropometry measurements and blood samples were taken. Household and individual questionnaires were administered mainly in the local languages (e.g. Mandinka, Fula and Wollof) depending on the interviewee's preference. Links to all questionnaires (in English) are provided in appendix 7.8.

After the interviews, a labeled urine beaker was handed out to each woman and a sample of salt was collected from each household if they agreed to. For selected women and children, interviewers prepared and labeled a woman and/or child paper biological form and directed those participants (or their mothers) to a central location in the EA where the anthropometrists and phlebotomist were stationed.

2.5.6. Field work (anthropometry and phlebotomy)

Before conducting anthropometric measurements, anthropometrists received urine beakers from women and stored them in a cold box at +4 - 8°C and then took the anthropometric measurements from the selected children and women using standard methods [14]. For weight measurements, for children who could not stand by themselves, the mother or caregiver was first measured alone, then together with the child, so that the child's weight was obtained by subtracting the mother's weight from the combined mother-and-child weight (SECA). Children's height or length was measured by using a standard wooden height board (UNICEF, #S0114540 usually referred to as the Shorr Board). Head circumference was measured by using a circumference measuring tape (SECA 201CM). In addition, mid-upper arm circumference (MUAC) was measured in children 6-59 months of age by using MUAC tapes (UniMUAC tapes, Medicine sans Frontières UK). The feet of children were examined for edema, and edema was only considered present if it was bilateral. For non-pregnant women, weight was measured using the same scale as used for children. Height was measured using the same standard wooden height board as used for the children. For pregnant women, only their MUAC was measured by using the same MUAC tape as used for children.

Following the collection of anthropometric measurements, blood pressure was measured in all women by using an automatic blood pressure monitor (OMRON M300, Kyoto, Japan).

Capillary blood was collected from all children 6-59 months of age as well as in non- pregnant and pregnant women. For the finger prick or heel-prick (children 6-11 months of age), the site was cleaned with an alcohol pad and wiped dry with a sterile gauze pad. Following the lancet puncture (Becton Dickinson, Franklin Lakes, NJ, USA), the first drop was wiped away with sterile gauze. The second and third drops of blood were collected to measure hemoglobin concentration (HemoCue 301) and malaria status (SD Bioline, Malaria Ag P.f/Pan). In non-pregnant women the fourth drop was collected to measure hyperglycemia status (HbA1c; HemoCue). Thereafter, approximately 300-400 µl of blood was collected from children and non-pregnant women into a silica-coated Microtainer™. Microtainers were placed in a cool box containing cold packs to ensure they are stored cold but not frozen at ~4°C and in the dark until further processing later the same day.

Participants found to have severe acute malnutrition, severe anemia or malaria parasitemia were referred for treatment at the nearest health facility. Blood was not collected in a fasting state as this was unnecessary, since none of the biomarkers collected were sensitive to fasting.

At the end of each day the team leader collated all forms and reviewed all data collected. Interviewers were notified of any errors and/ or omissions, for which they were instructed to make the necessary corrections.

All sharp materials (e.g. lancets, hemocuvettes, etc) were placed in sharps boxes directly following blood collection to prevent any injury. Teams collected all biohazard waste in biohazard bags, which were brought to MRCG laboratories for appropriate biomedical disposal.

2.5.7. Cold chain and processing of blood samples

Following collection, all blood samples were placed on cold packs (approximately 4°C) until processing in the laboratories. The phlebotomists recorded the temperature inside of the cold box containing the cold packs every two hours.

Every evening blood and urine samples were transported to one of the MRC laboratories located in Fajara, Keneba, or Basse, and Farafenni General Hospital[2] for processing. Blood was centrifuged at 3,000 rpm for 10 minutes. Serum was subsequently aliquoted into appropriately labeled micro tubes and red blood cells (RBCs) remained in the microtainer. Blood and urine samples were stored at -20°C in Farafenni and at -80°C in Basse, Keneba and Fajara for the duration of the field data collection. Thereafter, all blood and urine samples were transported to the MRC Laboratory in Fajara and frozen at -80°C.

[2] MRC laboratory technicians were posted to the hospital in Farafenni for the duration of the GMNS field work.

Serum samples were then air freighted on dry ice (-70°C) to VitMin Laboratory in Germany, and urine samples were transported using ice packs frozen at -80°C to the Noguchi Memorial Institute for Medical Research in Ghana. All samples arrived frozen at their respective destinations.

2.5.8. Supervision of fieldwork

During the field work, intense supervision was conducted to prevent potential flaws in operation. In addition to team leaders, roaming supervisors were assigned to the different LGAs, who ensured that the correct survey procedures were followed. Furthermore, spot-check visits to teams were conducted by the Principal Investigator and Survey Coordinator. The Laboratories were frequently visited by the Laboratory Coordinator to ensure that samples received from the field are properly processed and stored.

2.6. Definitions of indicators and specimen analysis

2.6.1. Anthropometric indicators

Children 0-59

In children, undernutrition (including wasting, stunting, underweight and microcephaly) and overnutrition was defined using WHO Child Growth Standards [15]. Children with z-scores of ≤ -2.0 for weight-for-height, height-for-age, weight-for-age and head circumference-for-age were defined as wasted, stunted, underweight, or having microcephaly, respectively [15,16]. Moderate wasting, stunting and microcephaly were defined as a z-score less than -2.0 but greater than or equal to -3.0, and severe wasting, stunting, underweight and microcephaly were denoted by z-scores less than -3.0. Children with bilateral pitting edema in the feet and/or lower legs were automatically considered as having severe acute malnutrition, regardless of their weight-for-height z-score. Overnutrition was defined as a weight-for-height z-score greater than +2.0. Overweight was defined as a weight-for-height z-score of greater than +2.0 but less than or equal to +3.0 and obesity as a weight-for-height z-score greater than +3.0.

Mid-upper arm circumference (MUAC) was also taken for all children 6-59 months, and those with a MUAC<115 millimeters were classified as having severe acute malnutrition.

Non-pregnant women

Chronic energy deficiency and overnutrition in non-pregnant women were assessed by using BMI. We used the following most commonly used cut-off points for Body Mass Index (BMI) to define levels of malnutrition in non-pregnant women (19): <16.0 severe undernutrition, 16.0-16.9 moderate undernutrition, 17.0-18.4 at risk of undernutrition, 18.5-24.9 normal, 25.0-29.9 overweight and >30 obese.

Pregnant women

Because body weight in pregnancy is increased by the products of conception and extra body fluid, BMI is not a valid indicator of nutritional status. MUAC was used instead to measure the nutritional status of pregnant women. A MUAC of less than 23 cm was used to define a pregnant woman as undernourished [13].

2.6.2. Urinary iodine concentration

Urinary iodine concentration was determined using the ammonium persulfate/Sandell-Kolthoff reaction method [17] conducted at the newly-established Iodine Global Network Laboratory at the Noguchi Memorial Institute for Medical Research in Accra, Ghana. Technicians assessed the concentration of each specimen twice, and the mean of both runs was used as the specimen concentration. Internal quality control materials labelled as low, medium and high were run with specimens. Results from an analytical run were rejected if the value from the internal quality control material was not within the acceptable range.

2.6.3. Blood specimens

Malaria measurement

The qualitative assessment of malaria infection was done on whole blood by using the SD BIOLINE Malaria Ag P.f/Pan rapid diagnostic test (RDT) produced by Standard Diagnostics Inc (Gyeonggi-do, Republic of Korea), which detects *P. falciparum* and other *Plasmodium* species (e.g. *P. vivax, P. malariae or P. ovale*).

Red cell pellets were also used to further test for malaria infection using a molecular diagnostic technique: DNA was first extracted according to the instructions in the QIAMP DNA extraction mini kit manual [18]. Parasitemia was then determined using an ultra-sensitive PCR tool, the VarATS as described by Hoffman et al [19].

Anemia

Blood hemoglobin concentration was measured by using a HemoCue™ portable hemoglobinometer (Hb301, HemoCue, Ängelholm, Sweden). Quality control of the Hemocue devices was done daily using low and medium concentration liquid control blood commercially available from the device supplier. Control blood was kept in cold boxes (2-8°C) for the duration of the field work to prevent degradation.

Hyperglycemia

Glycated hemoglobin concentration was measured by using the Hemocue® HbA1c 501, a portable and fully automated point of care system (HemoCue, Ängelholm, Sweden). Quality control of the device was done using the “daily” and “monthly” check cartridges delivered with the device. In addition, daily quality control was done using control blood with normal (range 7.1%- 8.3%) and high (8.8%-10.2%) HbA1c concentration (HemoCue, Ängelholm, Sweden). Control blood was kept in cold boxes (2-8°C) for the duration of the field work to prevent degradation.

Iron (serum ferritin and TfR), acute phase proteins (CRP, AGP), and vitamin A (RBP)

Serum ferritin and TfR were used to assess iron status of all young children and non-pregnant women. Ferritin concentration has been recommended by the World Health Organization (WHO) as the marker of first choice for iron deficiency in population based surveys [20]. Because serum ferritin levels can be elevated in the presence of infection or other causes of inflammation, the acute phase proteins AGP and CRP were used to detect the presence of inflammation in survey subjects. For children and women, ferritin values were adjusted for inflammation using the correction algorithm developed by the *Biomarkers Reflecting Inflammation and Nutritional Determinants of Anemia* (BRINDA) Project (brinda-nutrition.org) [21,22], and these adjusted ferritin values were used to calculate the prevalence of iron deficiency.

Retinol Binding Protein (RBP) was used to assess vitamin A status of young children and non-pregnant women. RBP instead of serum retinol was used because it requires smaller quantities of serum, and is much cheaper to measure than retinol using HPLC. RBP concentration is highly correlated with serum retinol [23], which is the biomarker for vitamin A status recommended by WHO. As inflammation can depress RBP levels, the correction algorithm developed by the BRINDA project was used to adjust the RBP concentrations in children [21,22], and these adjusted RBP values were used to calculate the prevalence of vitamin A deficiency in children and women.

The BRINDA project does not recommend any adjustments to RBP concentrations in women, and as such, no adjustments were made to RBP in non-pregnant women. A comparison of distributions and deficiency prevalences resulting from adjusting ferritin and RBP using both the Thurnham [24,25] and BRINDA methods are presented for children and for women in the "additional tables and figures" appendices.

In addition to inflammation adjustments made to ferritin and RBP, acute phase proteins were used to categorize the various inflammation stages of individuals. Concentrations of CRP >5 mg/L and AGP >1 g/L denote acute and chronic inflammation, respectively. Using the approach developed by Thurnham [24,25], each individual was assigned to one of the following four inflammation categories: incubation (elevated CRP only), early convalescence (elevated CRP and AGP), and late convalescence (elevated AGP only), and no inflammation.

Serum ferritin, TfR, CRP, AGP, and RBP were analyzed using an enzyme linked immunosorbent assay (ELISA) technique [26]. The VitMin Laboratory in Germany, where these analyses were conducted, participates regularly in inter-laboratory quality assurance programs, such as the VITAL-EQA from the CDC.

2.6.4. Hypertension

Hypertension was defined as systolic blood pressure of ≥ 140 and a diastolic blood pressure of ≥ 90 in pregnant and non- pregnant women [27]. Blood pressure was measured on the left arm of each woman, while the woman was sitting in a chair.

2.6.5. Analysis of iodine in salt

Iodine in salt was analyzed quantitatively using the iCheck Iodine (BioAnalyt, Potsdam, Germany; [28,29]). Analyses were done at NaNA by trained technicians. Every tenth sample was measured in duplicate and a quality control sample was measured every day before the analyses to serve as a quality control mechanism.

Following The Gambia's Food Fortification and Salt Iodization Regulation (2006), salt iodine concentrations of 15 ppm was used to classify salt as "adequately fortified".

2.7. Data management and analysis

2.7.1. Data entry

Direct electronic data entry was done by using CSPro during the household and women interviews. For the parts of the individual questionnaires (biological form) that were completed by the anthropometrists and phlebotomist using a paper form, the interviewers entered the data into CSPro on the same or the following day on the additional data entry forms (biological form for children and women) pre-programmed in CSPro.

2.7.2. Data monitoring

Interview data was uploaded from the tablets to a Dropbox on a daily basis. Data were monitored continuously and in case of systematic errors made by several teams, all team leaders were immediately informed about the problem, so the problem was not repeated; sporadic errors were directly reported to the respective team leaders. For errors that the teams could address, they were requested to do so immediately, while still in a given EA. For errors that could not be addressed by the teams immediately, the national coordinator was informed about the problem and visited the respective teams to correct errors.

2.7.3. Data analysis

Data analysis was done using Stata/IC version 14.2. All analyses of questionnaire data, biomarkers and salt samples were conducted using a weighted analysis to account for the unequal probability of selection in the 14 sub-strata. However, tables presenting the summary of the survey sample accounted for unequal probability of selection primary sampling units, but each household, child, or woman were equally weighted to produce summary figures.

For continuous variables, means with standard deviations and medians with interquartile ranges were calculated. Regarding urinary iodine concentrations (UIC), median UICs were calculated for each target group overall and for subgroups in order to judge population iodine status against WHO criteria [30]. However, in order to judge the statistical precision of apparent differences among subgroups, a square root transformation of the UIC values created a variable which was normally distributed [31]. Linear regression was then used to calculate p values for apparent association between the geometric UIC means and each characteristic.

For categorical variables, proportions were calculated to derive the prevalence of various outcomes. The statistical precisions of all prevalence estimates were assessed by using 95% confidence limits which were calculated accounting for the complex sampling used in this survey, including the cluster and stratified sampling. All measures of precision, including confidence limits and chi square p values for differences, were calculated accounting for the complex cluster and stratified sampling used by the GMNS 2018.

Descriptive statistics were calculated for all children together and all women together (i.e., across all strata), for each stratum separately, and by sex (for children). The results are also presented by specific age sub-groups for non-pregnant women (15-19 years, 20-29 years, 30-39 years, and 40-49 years) and children (6-11 months, 12-23 months, 24-35 months, 36-47 months, and 48-59 months of age). For pregnant women, only national estimates for all ages were generated.

To geographically represent the locations of all selected EAs, GPS points (collected by the 2018 MICS) of the 71 GMNS EAs were merged with geographic information related to Gambia's administrative divisions using geographic analysis software (Quantum GIS 2.6; http://qgis.osgeo.org).

2.7.4. Case definitions of deficiency

The cut-off values for each biomarker indicator used to determine nutritional status for each participant are presented in Table 5. For hemoglobin concentration multiple cut-offs were used to classify the severity of anemia. For other indicators, a single cut-off is used to identify deficiency or abnormality. A wealth index was calculated by using the World Bank method [32]. This method utilizes data on each household's dwelling, water and conditions and facilities, and ownership of durable goods, and generates a composite index using principal component analysis. The wealth index was subsequently separated into quintiles to permit the sub-group analysis of various key indicators by a household's wealth quintile group.

Each household's latrine type was classified as "improved" or "unimproved" based on the methodology developed by the WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation and Hygiene (JMP) [33]. Sanitation adequacy is a composite measure, based on the household's toilet/latrine type, and the household's exclusive use or shared use of the toilet/latrine.

Table 5. Clinical cut-off points and classifications for biomarker indicators

Indicator	Cut-offs defining deficiency or abnormality		
Hemoglobin [11]	Severe	Moderate	Mild
Children 6-59 months of age and pregnant women	<70 g/L	70-99 g/L	100-109 g/L
Non-pregnant women	<80 g/L	80-109 g/L	110-119 g/L
Ferritin † [20]			
Children 6-59 months of age (PSC)		< 12 µg/L	
Non-pregnant women (NPW)		< 15 µg/L	
Transferrin receptor (TfR)			
PSC and NPW		>8.3 mg/L‡	
α1-acid-glycoprotein (AGP) [24,25]			
PSC and NPW		>1 g/L	
C-reactive protein (CRP) [24,25]			
PSC and NPW		>5 mg/L	
Retinol-binding protein (RBP) †, §			
PSC and NPW		<0.7 µmol/L§	
Glycated hemoglobin (HbA1c) [34]			
NPW and PW		≥6.5%	
Urinary iodine concentration [35]			
NPW		< 100 µg/L	
PW		<150 µg/L	

* The cut-off defining normal hemoglobin concentrations is typically adjusted for the number of cigarettes smoked per day following WHO guidelines.

† Ferritin and RBP values were adjusted for inflammation using appropriate algorithms [21,22].

‡ There is no generally agreed upon threshold for serum TfR, but the most commonly used commercial assay (Ramco) suggests the above threshold.

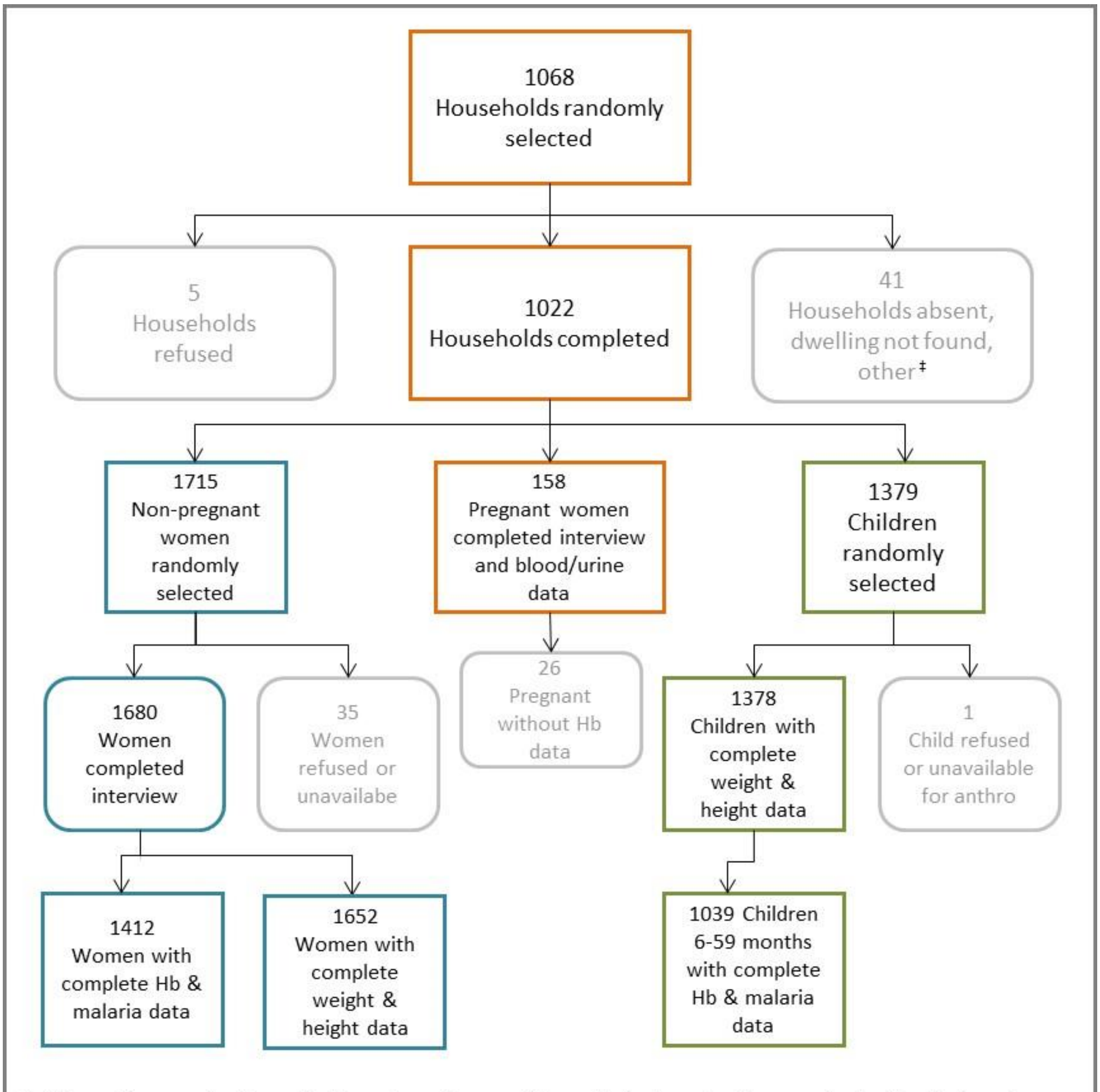
§ There is no general consensus on the cut-off point for RBP to be used to define vitamin A deficiency. Because of the 1:1 molar ratio of retinol and RBP in blood, many investigators use 0.70 µmol/L (the same cut-off used for serum retinol) although other cut-offs have been proposed [36].

3. RESULTS

3.1. Response rates for households, children, and women

Figure 1 below provides an overview of the number of respondents at the different survey stages, and illustrates that few households, women and children refused to participate in the survey. The GMNS anticipated enrolling 1156 households. As the study sample was drawn exclusively from households enrolled by MICS, for some clusters where households refused participating in the MICS or households selected by MICS were vacant, fewer households were available to be enrolled in the GMNS. Out of the 1156 households, 1068 (92.3%) were available to be enrolled in the GMNS. Of those, only 46 households (4.3%) refused to participate in the survey or could not be located. In aggregate, the GMNS enrolled 88.4% of anticipated households.

Among women residing in the participating households, 1715 non-pregnant women were identified, and only 2% of these women refused or were unavailable to participate. In these same households, 158 pregnant women were present, none of whom refused to participate. Among children, no caretakers refused to participate in the interview, and only 1 child refused to participate in the anthropometry component of the survey.



‡ Other = No competent household member at home at time of interview, dwelling vacant, dwelling destroyed, or dwelling not found.

Figure 1. Participation diagram of households, women and children, The Gambia 2018

3.2. Household level data (salt, rice, wheat flour, bread)

3.2.1. Salt iodine concentration

10.8% of households in Gambia consume adequately iodized salt (i.e. iodine concentrations ≥15 ppm), and mean salt iodine concentration is 8.1 ppm. The GMNS shows no significant differences in the proportion of adequately iodized salt samples by residence, LGA, or wealth quintile, however, the mean iodine concentration salt samples is slightly higher in Basse than other LGAs.

The range of iodine concentration in salt ranged from 2.7 to 65.0 ppm, suggesting that all salt samples contained some amount of iodine. However, as small levels of iodine are naturally occurring in sea salt, examining salt samples with less than 5 ppm iodine can approximate the proportion of non-iodized salt. The GMNS found that 51.3 % (95% CI:43.0, 59.6) of salt samples contained less than 5 ppm of iodine.

Table 6. Proportion of salt specimens with an iodine concentration ≥15 ppm in participating households, Gambia 2018

Characteristic	n	Proportion adequately fortified (%) [a]	(95% CI) [b]	P value [c]	Mean iodine concentration (ppm)	(95% CI) [b]	P value [c]
Residence							
Urban	47	10.6	(7.2, 15.4)	0.79	7.7	(6.9, 8.4)	0.18
Rural	44	11.4	(7.4, 17.2)		9.3	(7.1, 11.4)	
LGA							
Banjul	8	4.9	(2.8, 8.6)	0.24	5.7	(5.0, 6.5)	<0.001
Kanifing	16	14.4	(8.9, 22.3)		8.9	(7.7, 10.0)	
Brikama	22	10.4	(5.9, 17.6)		7.8	(6.5, 9.1)	
Mansakonko	2	1.6	(0.4, 5.8)		5.3	(4.2, 6.5)	
Kuntaur	8	8.1	(5.1, 12.6)		7.3	(6.0, 8.5)	
Kerewan	14	17.1	(6.4, 38.2)		8.7	(6.1, 11.2)	
Janjanbureh	8	7.6	(3.1, 17.2)		8.1	(6.2, 10.0)	
Basse	13	16.5	(6.3, 36.4)		11.5	(5.3, 17.7)	
Wealth Quintile							
Lowest	23	11.6	(6.4, 20.0)	0.69	8.8	(6.7, 10.8)	0.66
Second	18	6.9	(3.3, 14.1)		7.6	(6.0, 9.3)	
Middle	18	13.2	(6.2, 25.9)		9.0	(6.5, 11.5)	
Fourth	15	12.6	(6.4, 23.4)		7.6	(6.0, 9.1)	
Highest	17	10.3	(5.0, 19.9)		7.8	(6.6, 8.9)	
ALL HOUSEHOLDS	91	10.8	(8.0, 14.5)	--	8.1	(7.3, 8.9)	--

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] CI=confidence interval, calculated taking into account the complex sampling design.

[c] Chi-square p-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

As shown in Figure 2, 89% of salt samples contain less than 15ppm iodine, and few salt samples contain excessive (i.e. ≥ 45ppm) levels of iodine. Among salt samples with iodine concentrations less than 15 ppm, the median iodine concentration is 4.7 ppm (see Figure 3). Mean iodine concentration in salt is approximately 8 ppm, with slightly higher levels found in Basse. No statistical differences are observed in the prevalence of adequately iodize salt (i.e. ≥ 15 ppm) by residence, LGA, or wealth quintile.

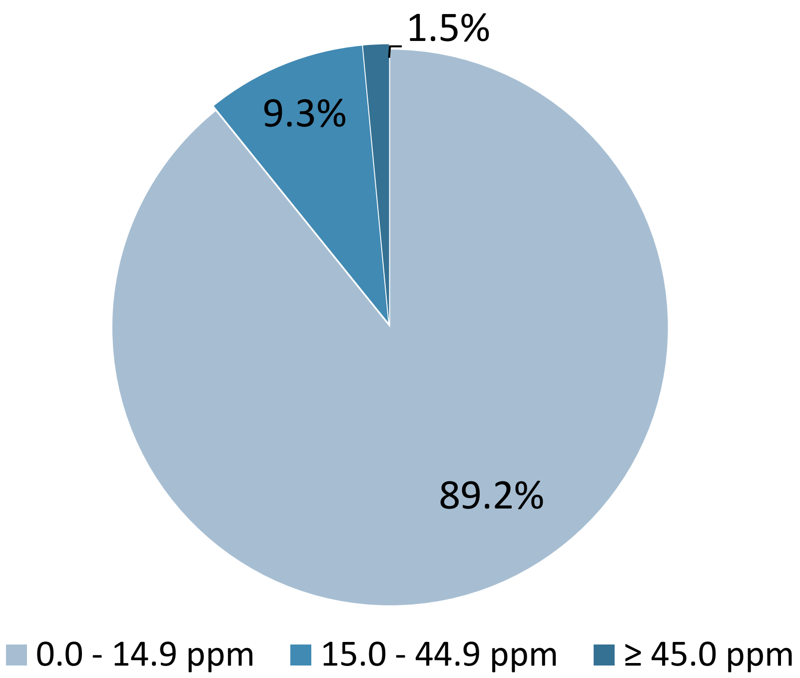


Figure 2. Proportion of iodine in salt at different levels, Gambia 2018

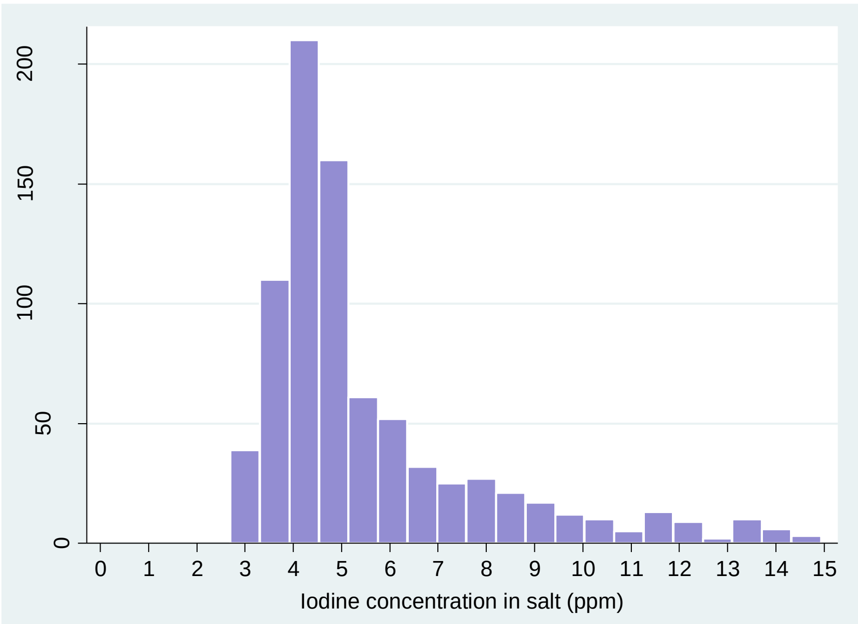


Figure 3. Distribution of salt iodine concentration in samples with less than 15ppm iodine, The Gambia 2018

3.2.2. Oil and rice consumption

When asked about the "the main edible oil" consumed by the household, more than 90% of households reported consuming vegetable oil (see Table 7). Approximately 55% and 40% purchased oil from an open market or retail shop, respectively. Only one-third of the households purchased oil in its original bottle, whereas the remainder of households purchased edible oil that was repackaged in another bottle or plastic bag. Nearly 85% households do not know the name of the oil brand that was routinely purchased.

Table 7. Oil consumption and purchase pattern for participating households, Gambia 2018

Characteristic	n	% [a]	(95% CI)[b]
Main type of oil purchased by household			
Vegetable oil	928	91.4	(87.4, 94.2)
Groundnut oil	10	1.0	(0.2, 5.0)
Refined palm oil	46	4.1	(2.7, 6.3)
Soybean oil	4	0.1	(0.0, 0.7)
Sunflower oil	6	0.4	(0.1, 2.7)
Olive oil	2	0.3	(0.1, 1.2)
Other	6	1.2	(0.4, 3.2)
I don't use it	10	1.4	(0.7, 2.9)
Unknown	2	0.0	(0.0, 0.2)
Location of oil purchase			
Supermarket	20	3.2	(1.6, 6.2)
Open market	482	55.4	(42.9, 67.2)
Retail shop	472	39.4	(27.0, 53.3)
Street stand	11	0.5	(0.1, 3.3)
Homemade	8	0.9	(0.1, 5.6)
Other	4	0.6	(0.2, 1.6)
Don't know	4	0.1	(0.0, 0.3)
Packaging of oil			
Original bottle	316	29.7	(23.7, 36.4)
Other bottle	536	43.9	(29.0, 59.9)
Plastic bag	149	26.5	(14.8, 42.8)
Brand of oil			
Yete Jallow	73	2.8	(1.5, 5)
Victoria	61	5.6	(3.3, 9.5)
Other brands	55	7.1	(4.4, 11.3)
Don't know brand	813	84.5	(80, 88.1)

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] CI=confidence interval, calculated taking into account the complex sampling design.

[c] Consumption of those consuming oil

Almost all surveyed households consume rice, the vast majority of whom (97.1%) purchased imported rice (see Table 8). Of those households purchasing rice, nine out of ten buy short grain rice (broken rice). The rice is mainly purchased on the market or from a retailer or shop; only very few households produce rice on their own. Almost 90% of rice consuming households purchase packaged rice, whereas the remainder of households purchased open rice.

Table 8. Rice consumption and purchase pattern for participating households, The Gambia 2018

Characteristic	n	% [a]	(95% CI)[b]
Origin of main type of rice purchased by household			
Don't consume rice	10	1.4	(0.7, 2.9)
Local paddy	13	1.5	(0.7, 3.5)
Imported rice	990	97.1	(95.2, 98.2)
Type of rice			
Short grain (broken rice)	945	93.3	(89.0, 96.0)
Long grain	58	6.7	(4.0, 11.0)
Main source of rice			
Home production	7	1.3	(0.5, 3.4)
Market/ retailer/ shop	985	98.0	(96.2, 99.0)
Local village producer / other	11	0.7	(0.3, 1,4)
Packaging of rice			
Open	111	10.7	(7.7, 14.8)
Packaged	891	89.2	(85.2, 92.3)
Other	1	0.1	(0.0, 0.4)

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] CI=confidence interval, calculated taking into account the complex sampling design.
[c] Consumption of those consuming rice

The proportion of households consuming bread in The Gambia is high with more than 95%. Most of the households consume 'Tapalapa' (>80%), followed by 'Senfur' (>35%). Less than 10% of households consume Sweet bread (see Table 9). The most frequently consumed food other than bread but made out of wheat flour is pancakes (>50) and salty pies (>30%). One-third of the surveyed households do not purchase any flour, one-third purchased wheat flour and almost 20% millet flour. The proportion of households consuming maize flour is low with about 6%.

Table 9. Bread and flour consumption and purchase pattern for participating households, The Gambia 2018

Characteristic	n	% [a]	(95% CI) [b]
Households that consumed bread	984	96.7	(96.3, 98.5)
Type of bread consumed by household [c]			
Senfur	345	35.3	(22.3, 50.8)
Sweet bread	71	8.3	(5.3, 12.8)
Tapalapa	734	82.4	(67.0, 91.5)
Other	35	2.2	(1.0, 4.5)
Consumption of other foods made with wheat flour [d]			
Pancakes	559	51.2	(36, 66.2)
Doughnuts	203	23.1	(11.5, 40.9)
Salty pies	330	33.6	(20.8, 49.3)
Other	90	11.9	(6.2, 21.7)
Main type of flour purchased by household			
Don't use flour in household	287	34.7	(25.2, 45.6)
Wheat flour	351	36.9	(26.6, 48.5)
Maize flour	109	6.4	(3.9, 10.3)
Millet flour	182	17.2	(11.3, 25.2)
Other	1	0.2	(0.0, 1.3)
Don't know	89	4.7	(3.2, 6.9)

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] CI=confidence interval, calculated taking into account the complex sampling design.
[c] Data related to consumption of each bread type collected separately, thus, proportions do not sum to 100%
[d] Results based on multiple-response questions, thus, proportions do not sum to 100%

3.3. Preschool children

3.3.1. Characteristics

Table 10 describes the demographic characteristics of children who participated in the GMNS. In total 1354 children 0-59 months were enrolled into the GMNS. The GMNS enrolled approximately equal proportions of male and female children and children residing in urban and rural areas. While more children from rural areas were enrolled, rural children represented about 37% of the total sampling following statistical weighting. Similar numbers of children were recruited from all LGAs, except Banjul and Kanifing where fewer children were enrolled.

Table 10. Description of sampled pre-school age children (0 - 59 months), The Gambia 2018

	Survey Sample		
Characteristic	n	% [a]	(95% CI) [b]
Age Group (in months)			
0-5	143	10.6	(9.1, 12.3)
6-11	124	9.2	(7.6, 11.1)
12-23	273	20.2	(18.0, 22.5)
24-35	266	19.6	(17.5, 22.5)
36-47	297	21.9	(19.3, 24.8)
48-59	251	18.5	(16.5, 20.8)
Sex			
Male	689	51.1	(48.5, 53.6)
Female	660	48.9	(46.4, 51.5)
Residence			
Urban	550	41.0	(35.7, 46.5)
Rural	804	59.0	(53.5, 64.3)
LGA			
Banjul	107	8.2	(6.3, 10.5)
Kanifing	111	8.3	(5.9, 11.5)
Brikama	180	13.5	(11.9, 15.2)
Mansakonko	181	13.3	(10.5, 16.8)
Kuntaur	180	13.1	(9.9, 17.2)
Kerewan	150	10.9	(8.2, 14.6)
Janjanbureh	226	16.6	(13.4, 20.3)
Basse	219	16.0	(10.7, 23.3)
ALL CHILDREN	1,354	100.0%	--

Note: The n's are un-weighted numbers of subjects in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages are un-weighted and do not account for unequal probability of selection.

[b] CI=confidence interval, calculated taking into account the complex sampling design

3.3.2. Stunting

In total, stunting in The Gambia was only found in 15.7% of children 0-59 months of age, with 4.9% and 10.7% of children found to be severely and moderately stunted, respectively. No statistically significant association was found between stunting and child age nor between stunting and child sex. Stunting in rural children was significantly higher than stunting among urban children, and significant differences were found among the LGAs, with the lowest stunting prevalence found in Banjul and the highest prevalence found in Kuntaur. Stunting was also markedly higher in children residing in households with the lower wealth quintiles. Further, a significantly higher proportion of children living in households with inadequate sanitation were stunted, compared to those living in households with adequate sanitation (Table 11).

Figure 4 shows the distribution of the height-for-age z-score in the surveyed population of children 0-59 months of age. It clearly shows that the distribution is normal and has shifted towards the left of the WHO standard growth curve.

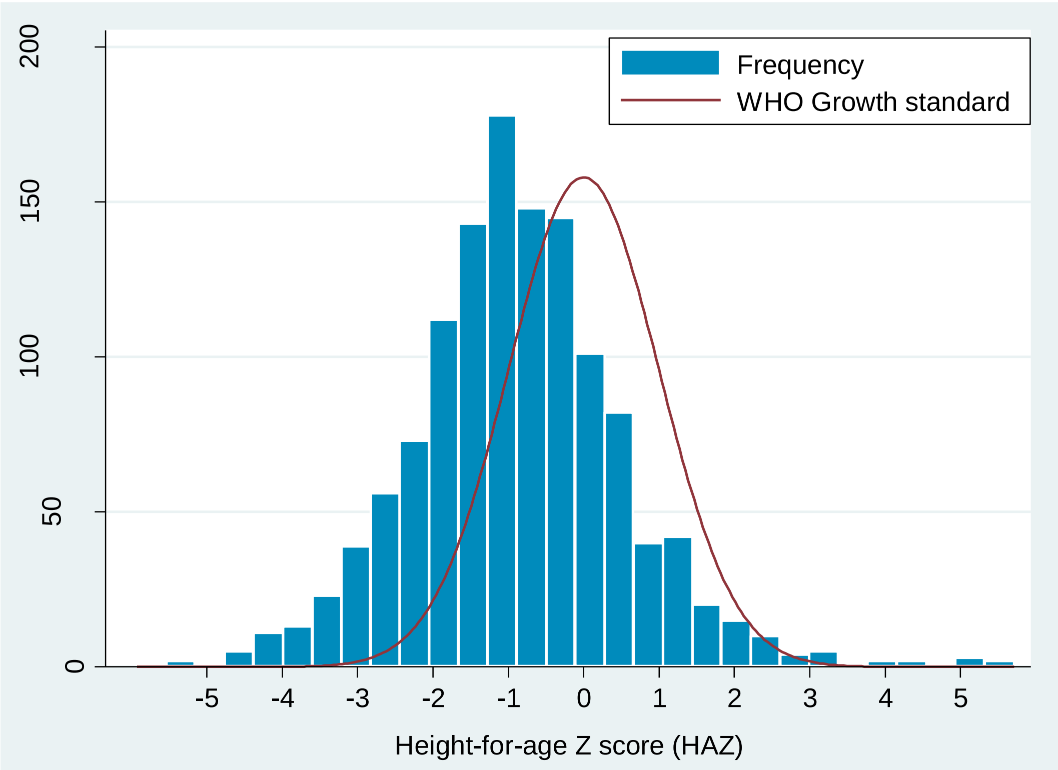


Figure 4. Histogram of height-for-age z-scores of the GMNS 2018 compared to the WHO growth curve, preschool-age children, The Gambia 2018

Table 11. Percentage of children (0-59 months) with stunting, The Gambia 2018

	Severe and Moderate Stunting						Any Stunting			
Characteristic	n	%[a] Severe[c]	(95% CI)[b]	n	%[a] Moderate[d]	(95% CI)[b]	n	%[a] Any[e]	(95% CI)[b]	P-value[f]
Age Group (in months)										
0-5	7	5.0	(1.7, 13.4)	10	3.7	(1.6, 8.2)	17	8.6	(3.7, 18.6)	0.47
6-11	6	2.0	(0.8, 4.7)	11	5.0	(8.6, 24.8)	17	17.0	(10.6, 26.0)	
12-23	13	5.6	(2.5, 12.2)	39	2.7	(8.5, 18.5)	52	18.3	(11.3, 28.2)	
24-35	21	6.1	(2.7, 13.4)	44	3.1	(7.7, 21.4)	65	19.2	(11.3, 30.7)	
36-47	16	4.6	(2.2, 9.1)	46	9.9	(7.3, 13.2)	62	14.4	(10.6, 19.3)	
48-59	11	4.7	(1.8, 11.6)	27	9.9	(4.8, 19.4)	38	14.6	(8.7, 23.5)	
Sex										
Male	47	5.7	(3.6, 8.7)	99	11.5	(9.6, 13.7)	146	17.1	(14.7, 19.9)	0.20
Female	27	4.2	(2.3, 7.3)	78	10.0	(7.9, 12.6)	105	14.1	(10.6, 18.6)	
Residence										
Urban	19	3.9	(1.9, 8.0)	46	7.8	(6.0, 10.1)	65	11.8	(8.9, 15.3)	
Rural	55	6.6	(5.1, 8.5)	131	15.9	(13.5, 18.6)	186	22.4	(19.1, 26.2)	
LGA										
Banjul	2	2.8	(0.7, 10.4)	6	6.9	(2.3, 19.0)	8	9.7	(3.1, 26.8)	
Kanifing	7	6.7	(2.9, 15.0)	13	11.0	(6.4, 18.5)	20	17.8	(12.8, 24.2)	<0.0001
Brikama	8	3.7	(1.3, 10.0)	11	6.8	(4.7, 9.7)	19	10.5	(7.0, 15.6)	
Mansakonko	6	4.2	(2.0, 8.4)	29	17.0	(11.3, 24.9)	35	21.2	(15.0, 29.1)	
Kuntaur	15	9.5	(4.9, 17.4)	38	21.3	(14.5, 30.0)	53	30.7	(22.1, 41.0)	
Kerewan	10	6.6	(3.9, 10.9)	23	16.0	(12.0, 21.0)	33	22.6	(16.2, 30.4)	
Janjanbureh	16	4.3	(1.6, 10.9)	37	14.9	(11.7, 18.7)	53	19.1	(13.4, 26.6)	
Basse	10	4.8	(2.9, 8.0)	20	9.9	(6.9, 14.2)	30	14.8	(10.6, 20.3)	
Wealth Quintile										
Lowest	41	7.6	(5.3, 10.7)	92	17.4	(13.1, 22.7)	133	25.0	(19.2, 31.8)	<0.01
Second	16	6.2	(2.7, 13.9)	41	13.2	(9.9, 17.5)	57	19.5	(12.9, 28.3)	
Middle	6	2.2	(0.6, 7.7)	21	6.1	(4.0, 9.3)	27	8.3	(5.2, 13.0)	
Fourth	4	2.7	(0.5, 13.0)	14	10.0	(3.9, 23.6)	18	12.7	(7.1, 21.8)	
Highest	7	5.1	(2.2, 11.4)	9	4.6	(2.0, 10.3)	16	9.8	(4.7, 19.2)	

Table continued on next page

	Severe and Moderate Stunting						Any Stunting			
Characteristic	n	%[a] Severe[c]	(95% CI)[b]	n	%[a] Moderate[d]	(95% CI)[b]	n	%[a] Any[e]	(95% CI)[b]	P-value[f]
Household sanitation [g]										
Inadequate	51	4.9	(3.2, 7.5)	132	13.8	(11.6, 16.3)	183	18.6	(15.6, 22.1)	
Adequate	23	5.0	(2.9, 8.5)	45	6.3	(4.3, 9.3)	68	11.3	(8.2, 15.5)	<0.005
ALL CHILDREN	74	4.9	(3.3, 97.3)	177	10.7	(9.4, 12.3)	251	15.7	(13.2, 18.4)	

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.
[a] Percentages weighted for unequal probability of selection
[b] CI=confidence interval, calculated taking into account the complex sampling design
[c] Severe stunting represents children who are below -3 standard deviations (SD; z-scores) from the WHO Child Growth Standards population median
[d] Moderate stunting includes children who are equal to or above -3 standard deviations (SD) and below-2 SD from the WHO Child Growth Standards population median
c Any stunting includes both severely and moderately stunted children
f P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups. Results are based on any stunting.
[g] Composite variable of toilet type and if toilet facilities are shared with non-household members; Adequate Sanitation = flush or pour flush toilet or pit latrine with slab not shared with another household. Inadequate sanitation= open pit, bucket latrine, hanging toilet/latrine, no facility, bush, field

As shown in Figure 5, most regions in The Gambia have a stunting prevalence below 20%, which is indicative of a low public health problem according to WHO classifications [12]. However, both Kuntaur and Kerewan have a stunting prevalence between 20% and 40%, which indicates a moderate public health problem.

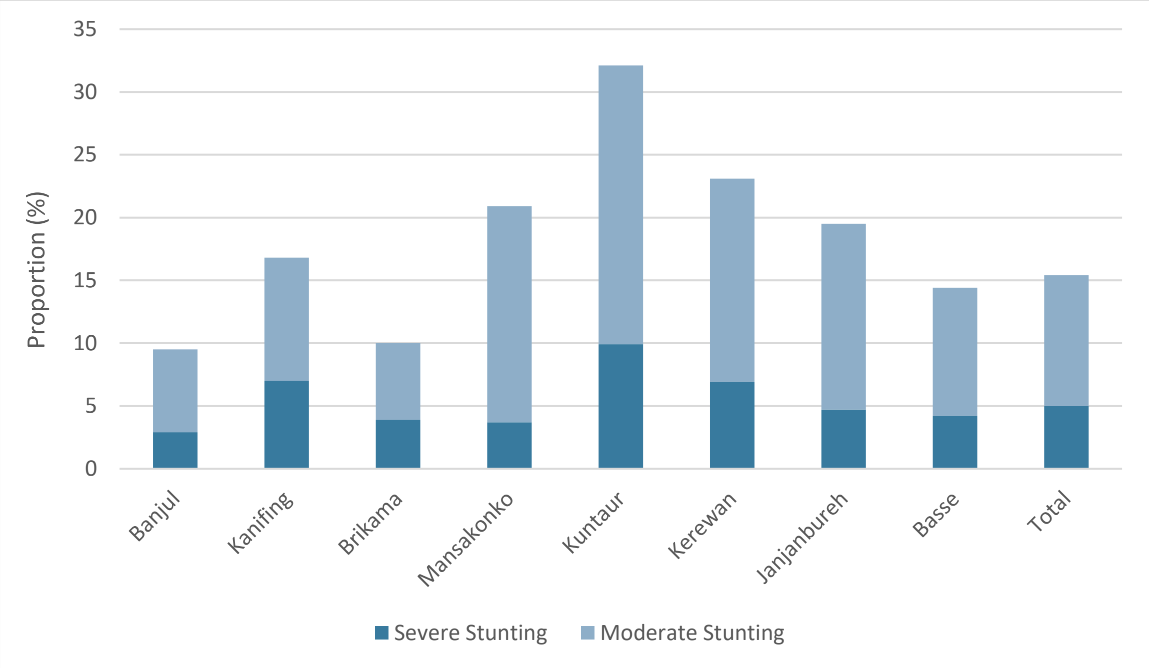


Figure 5. Prevalence of stunting by LGA, preschool-age children, The Gambia 2018

3.3.3. Wasting, Edema, and Overweight

The overall prevalence of wasting, also referred to as Global Acute Malnutrition (GAM), among children 0-59 months of age in The Gambia is 5.8%. No statistically significant differences were found between wasting prevalence and child age, child sex, LGA, or household wealth quintile. While the prevalence of wasting among rural children is higher than in urban children, the difference is not considered statistically significant (i.e. $p<0.05$). According to WHO classification [12] the public health situation of wasting in The Gambia is still “poor”, but close to “acceptable” (<5%).

Figure 6 shows the distribution of the weight for height z-score (WHZ) in the surveyed population of children 0-59 months of age. The distribution of WHZ is normal, and is only shifted slightly to the left of the WHO standard growth curve, with the majority of the distribution falling under the WHO standard curve. Only 24 children (2.7%; 95% CI: 1.8, 4.1) are classified as being overweight (i.e. WHZ >+2 SD).

Bilateral edema was found in only 3 children, or 0.1%. Due to the small number of children with edema, sub-group analysis by age group, sex, strata, and wealth quintile was not conducted.

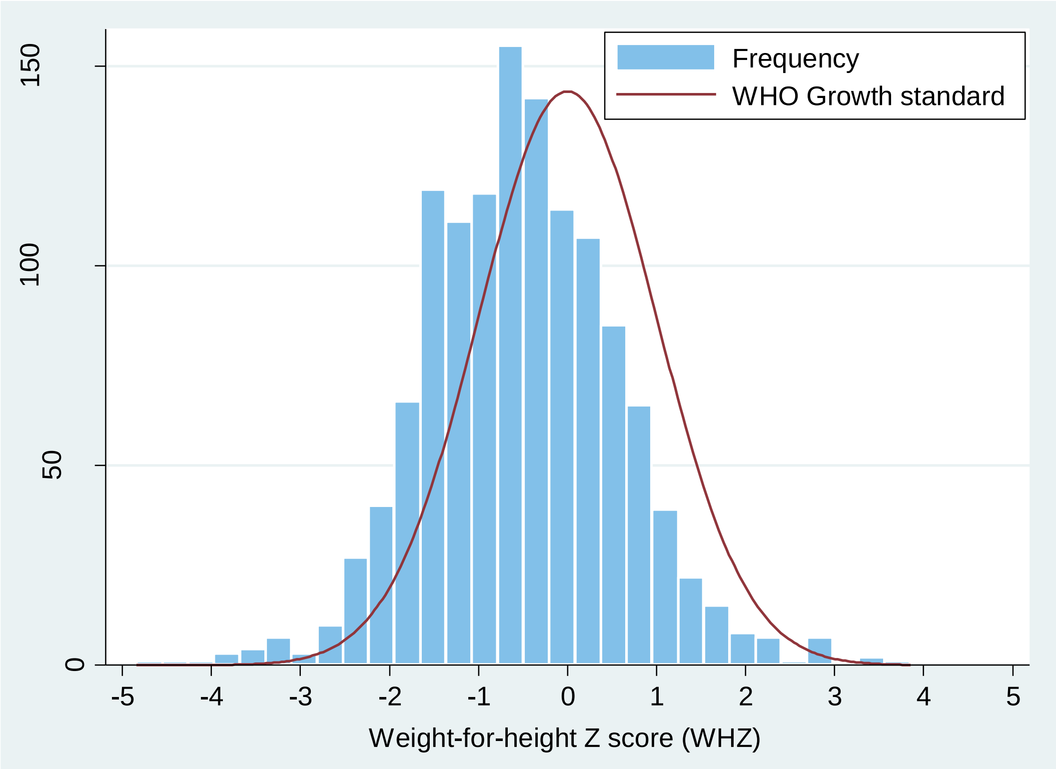


Figure 6. Histogram of weight-for-height z-scores of the GMNS 2018 compared to the WHO growth curve, preschool-age children, The Gambia 2018

Table 12. Percentage of children (0-59 months) with wasting, The Gambia 2018

	Severe and Moderate Wasting						Any Wasting			
Characteristic	n	%[a] Severe [c]	(95% CI)[b]	n	%[a] Moderate [d]	(95% CI)[b]	n	%[a] Any [e]	(95% CI)[b]	P-value [f]
Age Group (in months)										
0-5	3	2.1	(0.7, 6.8)	4	1.9	(0.6, 5.9)	7	4.1	(1.7, 9.5)	0.63
6-11	6	3.1	(1.0, 8.8)	8	4.0	(1.9, 8.6)	14	7.1	(3.9, 12.5)	
12-23	4	0.9	(0.3, 3.1)	17	7.0	(3.2, 14.7)	21	7.9	(3.9, 15.5)	
24-35	5	1.7	(0.5, 5.8)	12	3.4	(1.6, 7.3)	17	5.2	(2.6, 10.0)	
36-47	2	0.6	(0.1, 3.0)	17	5.7	(2.0, 15.5)	19	6.3	(2.3, 15.8)	
48-59	0	-	-	17	3.9	(2.0, 7.2)	17	3.9	(2.0, 7.2)	
Sex										
Male	11	1.4	(0.6, 3.0)	40	4.6	(2.8, 7.5)	51	6.0	(3.9, 9.1)	0.74
Female	9	1.0	(0.4, 2.6)	35	4.5	(2.8, 7.2)	44	5.5	(3.6, 8.4)	
Residence										
Urban	5	0.8	(0.3, 2.3)	23	3.6	(1.7, 7.5)	28	4.4	(2.4, 8.1)	0.07
Rural	15	1.9	(1.0, 3.6)	52	6.2	(4.7, 8.2)	67	8.2	(6.2, 10.7)	
LGA										
Banjul	0	-	-	0	-	-	0	-	-	0.18
Kanifing	4	3.1	(1.1, 8.5)	3	2.6	(0.9, 7.3)	7	5.7	(3.1, 10.0)	
Brikama	1	0.4	(0.0, 3.5)	7	3.2	(0.9, 10.2)	8	3.6	(1.2, 10.0)	
Mansakonko	2	0.7	(0.1, 3.6)	7	3.4	(1.4, 8.0)	9	4.1	(2.0, 8.1)	
Kuntaur	2	1.2	(0.2, 6.5)	15	8.5	(4.7, 14.9)	17	9.8	(7.0, 13.4)	
Kerewan	2	1.7	(0.4, 6.0)	11	7.4	(5.8, 9.5)	13	9.1	(6.3, 13.0)	
Janjanbureh	6	2.1	(0.8, 5.4)	16	6.3	(3.9, 9.9)	22	8.3	(5.2, 13.0)	
Basse	3	1.6	(0.4, 6.3)	16	5.9	(3.3, 10.4)	19	7.5	(4.1, 13.6)	
Wealth Quintile										
Lowest	8	1.6	(0.7, 3.5)	33	5.6	(3.4, 8.9)	41	7.2	(4.6, 11.0)	0.86
Second	5	1.2	(0.3, 4.9)	16	4.6	(1.9, 10.8)	21	5.9	(2.8, 12.0)	
Middle	2	0.2	(0.0, 1.3)	18	4.7	(2.0, 10.5)	20	5.1	(2.2, 11.2)	
Fourth	3	2.1	(0.6, 7.2)	5	3.8	(0.7, 16.9)	8	5.9	(2.0, 16.1)	
Highest	2	1.5	(0.4, 6.1)	3	2.8	(0.9, 8.3)	5	4.4	(2.0, 9.5)	

Table continued on next page

	Severe and Moderate Wasting						Any Wasting			
Characteristic	n	%[a] Severe [c]	(95% CI)[b]	n	%[a] Moderate [d]	(95% CI)[b]	n	%[a] Any [e]	(95% CI)[b]	P-value [f]
Household sanitation [g]										
Inadequate	17	1.5	(0.7, 2.8)	50	4.9	(3.3, 7.3)	67	6.4	(4.6, 8.9)	0.23
Adequate	3	0.9	(0.2, 3.2)	25	4.0	(2.2, 7.1)	28	4.8	(2.8, 8.2)	
ALL CHILDREN	20	1.2	(0.7, 2.2)	75	4.5	(3.0, 6.8)	95	5.8	(4.1, 8.1)	

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] Percentages weighted for unequal probability of selection

[b] CI=confidence, calculated taking into interval account the complex sampling design

[c] Severe wasting represents children who are below -3 standard deviations (SD; z-scores) from the WHO Child Growth Standards population median

[d] Moderate wasting includes children who are equal to or above -3 standard deviations (SD) and below-2 SD from the WHO Child Growth Standards population median

[e] Any wasting includes both severely and moderately wasted children

[f] P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups. Results are based on any wasting.

[g] Composite variable of toilet type and if toilet facilities are shared with non-household members; Adequate Sanitation = flush or pour flush toilet or pit latrine with slab not shared with another household. Inadequate sanitation= open pit, bucket latrine, hanging toilet/latrine, no facility, bush, field

3.3.4. Underweight

Underweight was observed in only 10.6% of children 0 to 59 months of age posing a moderate public health problem in The Gambia according to WHO classification [12]. No statistically significant differences in underweight prevalence were observed by child age or child sex. However, children residing in households in the lowest wealth quintile have significantly higher underweight prevalence. Geographic differences were also observed, with significantly higher underweight prevalence found among children in rural areas, and the highest prevalences were found in Kuntaur and Janjanbureh and lowest in Banjul (Table 13).

As shown in Figure 7, the distribution of WAZ is shifted to the left of the WHO standard growth curve, similar to the distributions of HAZ and WHZ.

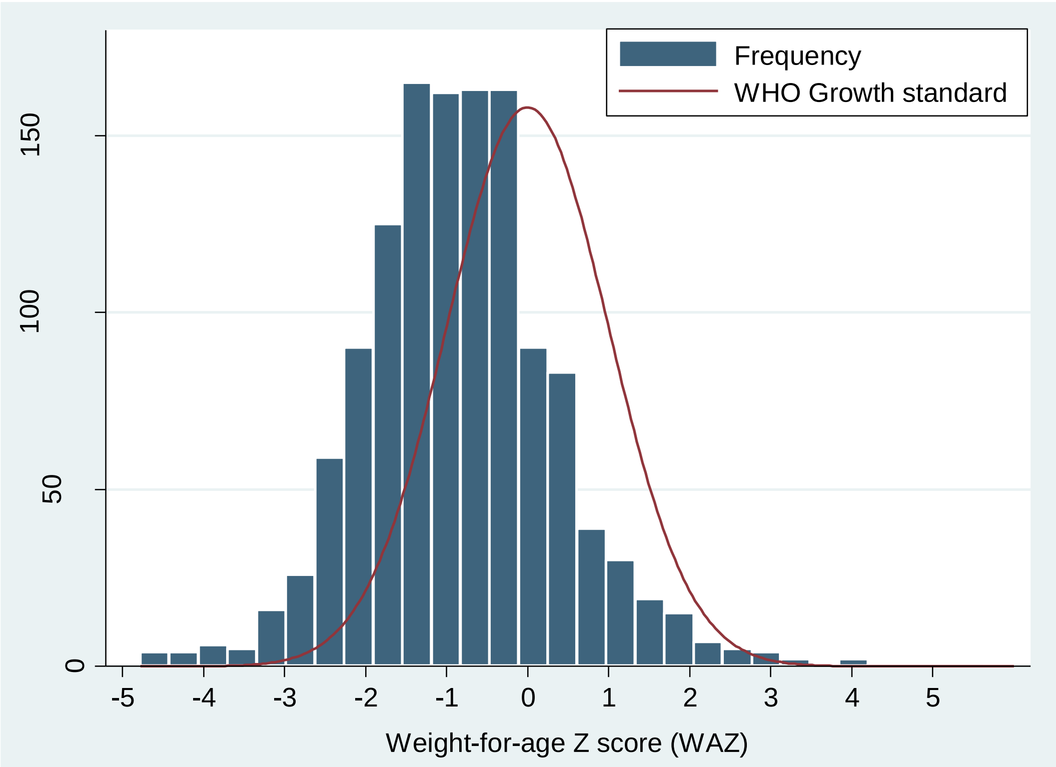


Figure 7. Histogram of weight-for-age z-scores of the GMNS 2018 compared to the WHO growth curve, preschool-age children, The Gambia 2018

Table 13. Percentage of children (0-59 months) underweight, The Gambia 2018

	Severe and Moderate Underweight						Any Underweight			
Characteristic	n	%[a] Severe [c]	(95% CI)[b]	n	%[a] Moderate [d]	(95% CI)[b]	n	%[a] Any [e]	(95% CI)[b]	P-value [f]
Age Group (in months)										
0-5	3	1.6	(0.5, 5.6)	4	1.0	(0.3, 2.8)	7	2.6	(1.0, 6.4)	0.11
6-11	6	3.0	(1.1, 8.2)	11	6.1	(2.7, 13.0)	17	9.1	(4.8, 16.8)	
12-23	6	1.5	(0.6, 3.7)	33	12.2	(7.7, 18.7)	39	13.7	(9.2, 20.0)	
24-35	9	2.3	(0.9, 6.1)	29	9.1	(4.8, 16.5)	38	11.4	(6.2, 20.2)	
36-47	8	2.3	(1.0, 5.1)	45	11.5	(6.3, 20.1)	53	13.8	(8.3, 22.0)	
48-59	4	1.0	(0.4, 2.9)	31	8.4	(4.8, 14.2)	35	9.5	(5.8, 15.2)	
Sex										
Male	20	2.1	(1.1, 3.7)	82	9.6	(6.4, 14.3)	102	11.7	(8.2, 16.5)	0.31
Female	16	1.7	(0.9, 3.3)	71	7.9	(6.4, 9.7)	87	9.6	(7.8, 11.8)	
Residence										
Urban	9	1.0	(0.4, 2.5)	37	6.7	(4.4, 10.2)	46	7.8	(5.1, 11.6)	<0.005
Rural	27	3.4	(2.2, 5.2)	116	12.3	(9.8, 15.4)	143	15.7	(3.2, 18.6)	
LGA										
Banjul	1	2.2	(0.4, 11.8)	1	0.9	(0.1, 6.9)	2	3.1	(0.8, 10.7)	<0.0001
Kanifing	2	1.5	(0.4, 5.5)	6	5.0	(2.7, 9.3)	8	6.5	(3.8, 10.9)	
Brikama	2	0.5	(0.1, 3.1)	13	6.2	(3.4, 11.1)	15	6.7	(3.8, 11.7)	
Mansakonko	3	2.2	(1.1, 4.2)	25	13.4	(0.0, 17.8)	28	15.6	(2.8, 18.9)	
Kuntaur	8	4.7	(1.9, 11.3)	34	19.5	(1.9, 30.1)	42	24.2	(7.0, 33.3)	
Kerewan	4	3.0	(1.1, 8.3)	17	10.7	(5.1, 21.1)	21	13.7	(8.0, 22.6)	
Janjanbureh	11	4.6	(3.3, 6.6)	38	15.5	(3.2, 18.1)	49	20.2	(7.7, 22.9)	
Basse	5	2.5	(1.0, 6.4)	19	8.0	(5.4, 11.7)	24	10.5	(7.1, 15.4)	
Wealth Quintile										
Lowest	19	3.3	(1.9, 5.4)	80	13.6	(9.9, 18.3)	99	16.8	(2.8, 21.9)	<0.05
Second	12	3.2	(1.6, 6.4)	36	10.0	(7.2, 13.6)	48	13.2	(9.8, 17.6)	
Middle	1	0.1	(0.0, 1.2)	21	4.6	(1.9, 10.8)	22	4.7	(2.0, 11.0)	
Fourth	3	1.9	(0.5, 6.5)	7	8.9	(3.0, 24.0)	10	10.8	(4.9, 22.1)	
Highest	1	0.7	(0.1, 5.0)	9	6.0	(2.3, 15.1)	10	6.7	(2.8, 15.3)	

Table continued on next page

	Severe and Moderate Underweight						Any Underweight			
Characteristic	**n**	**%[a] Severe [c]**	**(95% CI)[b]**	**n**	**%[a] Moderate [d]**	**(95% CI)[b]**	**n**	**%[a] Any [e]**	**(95% CI)[b]**	**P-value [f]**
Household sanitation [g]										
Inadequate	29	2.3	(1.4, 3.7)	112	10.3	(7.6, 13.7)	141	12.6	(9.5, 16.5)	0.08
Adequate	7	1.3	(0.5, 3.3)	41	6.5	(3.8, 10.9)	48	7.8	(4.8, 12.5)	
ALL CHILDREN	36	1.9	(1.2, 3.0)	153	8.8	(6.8, 11.2)	191	10.6	(8.4, 13.2)	

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] Percentages weighted for unequal probability of selection

[b] CI=confidence interval, calculated taking into account the complex sampling design

[c] Severe underweight represents children who are below -3 standard deviations (SD; z-scores) from the WHO Child Growth Standards population median

[d] Moderate underweight includes children who are equal to or above -3 standard deviations (SD) and below-2 SD from the WHO Child Growth Standards population median

[e] Any underweight includes both severely and moderately underweight children

[f] P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups. Results are based on any underweight.

[g] Composite variable of toilet type and if toilet facilities are shared with non-household members; Adequate Sanitation = flush or pour flush toilet or pit latrine with slab not shared with another household. Inadequate sanitation= open pit, bucket latrine, hanging toilet/latrine, no facility, bush, field

3.3.5. Microcephaly

The overall prevalence of microcephaly in children 0-59 months is 7.4%. No significant differences were found for child age, sex, household wealth or geographic location. Microcephaly is highly associated with other growth indicators, significantly more children who are stunted, wasted or underweight also suffer from microcephaly.

As shown in Figure 8, the distribution of HCAZ is shifted to the left of the WHO standard growth curve, similar to the distributions of HAZ, WHZ and WAZ.

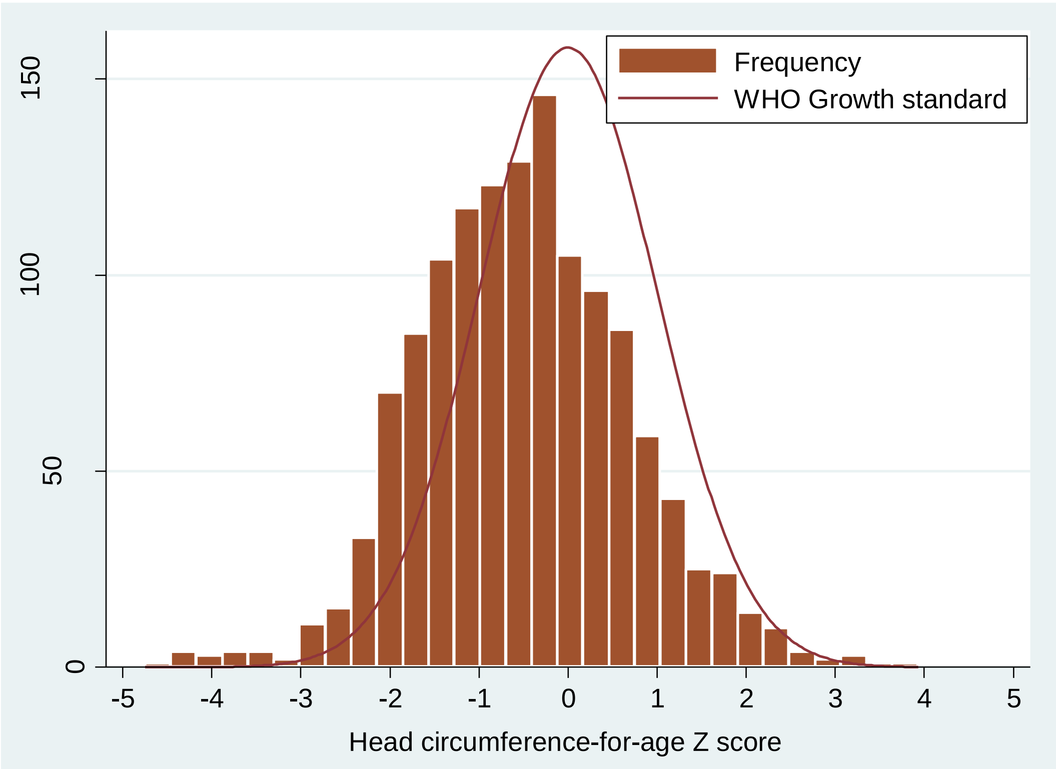


Figure 8. Histogram of head circumference-for-age z-scores of the GMNS 2018 compared to the WHO growth curve, preschool-age children, The Gambia 2018

Table 14. Percentage of children (0-59 months) microcephaly, The Gambia 2018

		Microcephaly [a]		
Characteristic	n	% [b]	(95% CI) [c]	P-value [d]
Age Group (in months)				
0-5	6	3.5	(1.3, 8.9)	0.68
6-11	7	6.4	(1.7, 20.9)	
12-23	24	7.3	(4.2, 12.5)	
24-35	23	9.3	(4.7, 17.7)	
36-47	27	8.3	(4.9, 13.6)	
48-59	22	6.7	(3.2, 13.5)	
Sex				
Male	65	8.8	(6.2, 12.5)	0.18
Female	44	5.9	(3.2, 10.7)	
Residence				
Urban	27	5.8	(2.9, 11.5)	0.13
Rural	82	10.0	(8.2, 12.1)	
LGA				
Banjul	2	4.4	(0.7, 22.3)	
Kanifing	4	3.9	(1.4, 10.5)	
Brikama	12	6.3	(2.3, 16.0)	
Mansakonko	20	10.4	(7.1, 15.0)	
Kuntaur	14	8.8	(7.1, 10.9)	
Kerewan	8	5.6	(3.0, 10.1)	
Janjanbureh	27	9.4	(5.8, 14.7)	
Basse				
Wealth Quintile				
Lowest	49	10.9	(7.3, 16.0)	0.27
Second	28	8.1	(5.4, 11.9)	
Middle	18	4.9	(1.9, 12.0)	
Fourth	9	7.9	(2.9, 19.6)	
Highest	5	4.5	(1.9, 10.4)	
Stunting				
No	60	5.4	(3.6, 8.1)	<0.0001
Yes	48	17.4	(2.1, 24.5)	
Wasting				
No	80	6.2	(4.1, 9.3)	<0.0001
Yes	29	26.1	(5.0, 41.3)	
Underweight				
No	54	4.9	(2.9, 8.2)	<0.0001
Yes	55	27.4	(8.9, 38.1)	
Household sanitation				
Inadequate	74	8.0	(5.1, 12.3)	0.46
Adequate	35	6.5	(4.0, 10.6)	
ALL CHILDREN	109	7.4	(5.1, 10.7)	--

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] Children with head circumference below -2 standard deviations (SD; z-scores) from the WHO Child Growth Standards population median were classified having microcephaly.

[b] Percentages weighted for unequal probability of selection

[c] CI=confidence interval, calculated taking into account the complex sampling design

[d] P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups.

3.3.6. Mid-upper arm circumference

Figure 9 below shows the distribution of MUAC measurements taken from children 6-59 months of age. In total, mean MUAC is 14.9 cm, and only nine children are classified as having SAM when using the cut-off of 11.5 cm. Despite the small number of children with SAM identified, a statistically significant difference is found between urban and rural areas, as all children with SAM are located in rural area. Child age and household sanitation are not significantly associated with SAM, however, younger children and children in households with inadequate sanitation tend to have higher levels of SAM. In addition, no statistical difference was observed by child sex, LGA, or wealth quintile (Table 15).

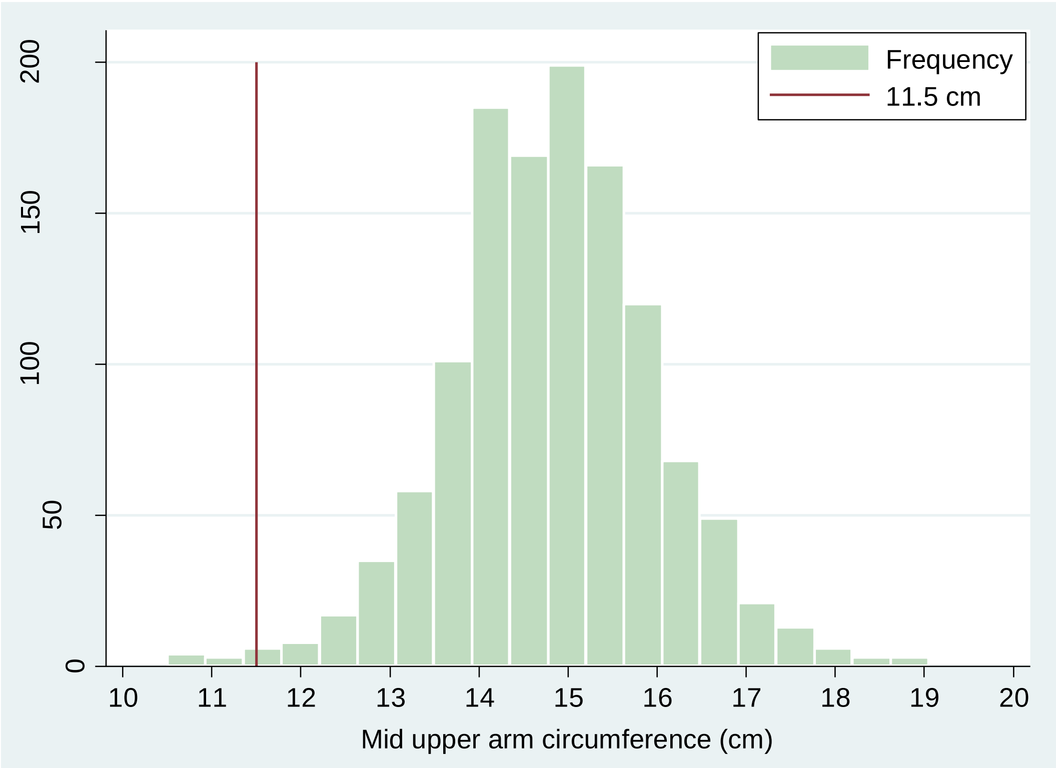


Figure 9. Histogram of mid upper arm circumference (MUAC) of the GMNS 2018, children 6 - 59 months of age, The Gambia 2018

Table 15. Severe Acute Malnutrition (SAM) by various characteristics in children, The Gambia 2018

Characteristic	n	% SAM [a, b]	(95% CI)[c]	P Value[d]
Age Group (in months)				
6-11	4	2.0	(0.6, 6.5)	0.08
12-23	4	0.9	(0.3, 3.0)	
24-35	1	0.1	(0.0, 0.7)	
36-47	0	-	-	
48-59	0	-	-	
Sex				
Male	3	0.4	(0.1, 1.3)	0.62
Female	6	0.5	(0.2, 1.3)	
Residence				
Urban	0	-	-	<0.001
Rural	9	1.2	(0.6, 2.4)	
LGA				
Banjul	0	-	-	0.41
Kanifing	0	-	-	
Brikama	0	-	-	
Mansakonko	1	0.5	(0.1, 4.4)	
Kuntaur	1	0.7	(0.1, 3.5)	
Kerewan	1	0.9	(0.1, 6.7)	
Janjanbureh	5	1.7	(0.6, 4.7)	
Basse	1	0.7	(0.1, 4.3)	
Wealth Quintile				
Lowest	6	1.1	(0.8, 0.2)	0.81
Second	3	0.8	(0.2, 3.0)	
Middle	0	-	-	
Fourth	0	-	-	
Highest	0	-	-	
Household sanitation [f]				
Inadequate	9	0.7	(0.3, 1.7)	0.07
Adequate	0	-	-	
ALL CHILDREN	9	0.4	(0.2, 1.0)	--

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] SAM includes children who are equal to or above -3 standard deviations (SD) from the WHO Child Growth Standards population median

[b] Percentages weighted for unequal probability of selection

[c] CI=confidence interval, calculated taking into account the complex sampling design

[d] P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups.

3.3.7. Malaria

When investigating the combined prevalence of all malaria species, the GMNS found that less than 1% of children are malaria positive using the RDT. Based on this measurement, malaria is observed in only 1 female child in Mansakonko. The child is between 36 and 47 months of age. A detailed sub-group was not conducted due to the small number of children identified with malaria using the RDT.

Analysis of blood pellets using PCR, however, identified a higher number of children with malaria than the RDTs. As shown in Table 16, approximately 8% of children are identified as malaria positive using this methodology. No statistically significant differences were found by child age, residence, LGA, or wealth quintile. However, female children have a significantly higher malaria prevalence than male children.

Table 16. Percentage of children (6-59 months) with malaria based on PCR analysis, The Gambia 2018

	Malaria [a]			
Characteristic	**n**	**% [b]**	**(95% CI) [c]**	**P-value [d]**
Age Group (in months)				
6-11	11	11.5	(4.2, 27.8)	0.09
12-23	14	4.5	(2.0, 9.6)	
24-35	23	13.7	(5.6, 29.6)	
36-47	20	5.8	(3.1, 10.5)	
48-59	18	6.4	(3.2, 12.2)	
Sex				
Male	37	6.3	(3.5, 11.3)	<0.05
Female	49	9.8	(5.3, 17.2)	
Residence				
Urban	30	7.9	(3.2, 18.3)	0.98
Rural	56	8.1	(4.8, 13.2)	
LGA				
Banjul	2	1.8	(0.4, 7.1)	0.38
Kanifing	2	3.1	(0.9, 10.2)	
Brikama	5	6.9	(1.5, 26.5)	
Mansakonko	12	8.8	(4.2, 17.4)	
Kuntaur	7	7.1	(1.4, 29.5)	
Kerewan	15	15.7	(6.2, 34.7)	
Janjanbureh	30	15.9	(8.9, 26.6)	
Basse	13	6.4	(3.1, 12.6)	
Wealth Quintile				
Lowest	41	8.4	(4.5, 15.2)	0.50
Second	17	9.4	(3.7, 21.7)	
Middle	15	11.1	(4.5, 25.0)	
Fourth	9	3.7	(0.6, 18.9)	
Highest	4	4.9	(1.8, 12.4)	
ALL CHILDREN	86	8.0	(4.4, 14.0)	--

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.
[a] Malaria detected via detection of the cytochrome oxidase III (COX-III) gene of *Plasmodium* parasites
[b] Percentages weighted for unequal probability of selection
[c] CI=confidence interval, calculated taking into account the complex sampling design
[d] P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups.

3.3.8. Anemia severity and hemoglobin distribution

More than 50% of Gambian children are anemic (hemoglobin <110 g/L). The WHO classifies the population public health severity of anemia as high for prevalence rates of ≥40% [11].

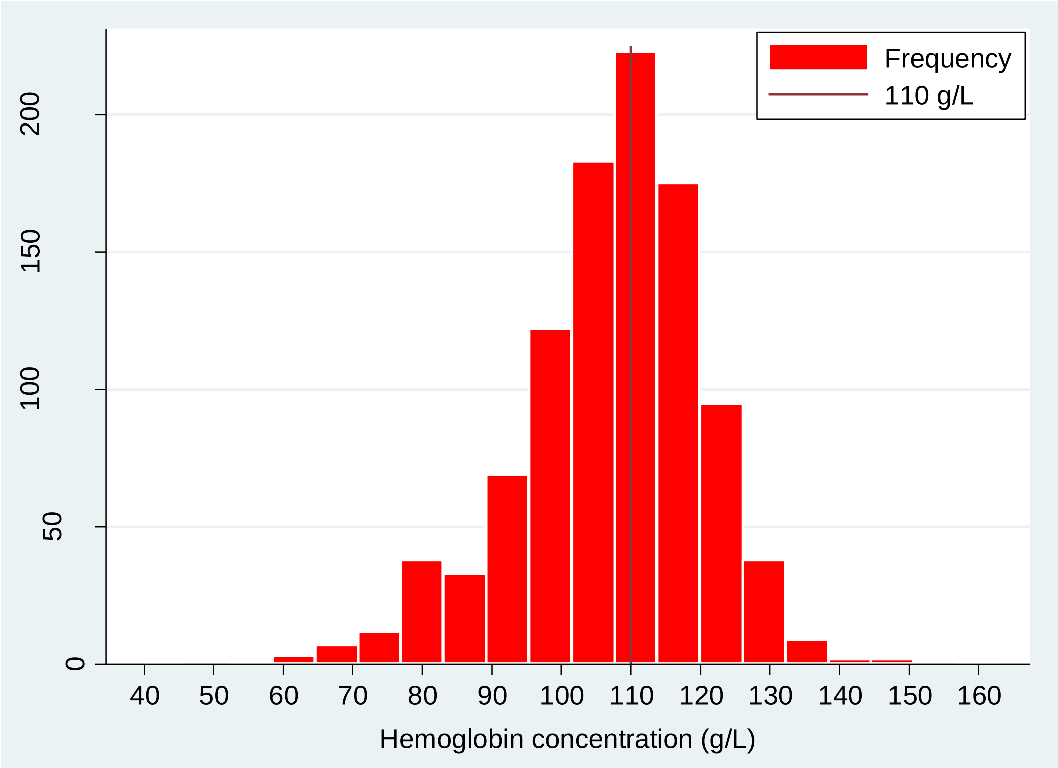


Figure 10. Histogram of hemoglobin concentration (g/L) in children 6-59 months of age, The Gambia 2018

Table 17. Proportion of mild, moderate and severe anemia in children 6-59 months of age, The Gambia 2018

	Mild anemia[c]			Moderate anemia[c]			Severe anemia[c]		
Characteristic	**n**	**% [a]**	**(95% CI)[b]**	**n**	**% [a]**	**(95% CI)[b]**	**n**	**% [a]**	**(95% CI)[b]**
Age Group (in months)									
6-11	35	37.2	(20.4, 57.8)	20	26.8	(19.7, 35.2)	1	0.3	(0.0, 2.0)
12-23	84	27.2	(21.5, 33.8)	74	30.3	(19.0, 44.5)	3	1.5	(0.3, 6.6)
24-35	74	34.1	(27.3, 41.6)	58	27.4	(20.6, 35.5)	4	1.6	(0.4, 6.5)
36-47	61	20.5	(13.1, 30.6)	49	15.2	(10.0, 22.3)	3	0.5	(0.1, 1.8)
48-59	46	26.9	(18.4, 37.5)	24	8.5	(5.4, 13.3)	1	0.2	(0.0, 1.3)
Sex									
Male	159	30.0	(24.4, 36.2)	125	23.9	(18.8, 30.0)	9	1.5	(0.6, 3.9)
Female	141	26.0	(21.8, 30.6)	100	18.8	(14.6, 23.8)	3	0.2	(0.0, 0.9)
Residence									
Urban	116	26.0	(20.4, 32.6)	74	19.0	(14.6, 24.4)	3	0.9	(0.2, 3.3)
Rural	184	31.3	(27.6, 35.3)	151	25.3	(20.0, 31.4)	9	0.9	(0.5, 1.7)
LGA									
Banjul	28	28.4	(19.3, 39.7)	11	13.5	(7.8, 22.4)	0	--	--
Kanifing	28	29.9	(21.3, 40.3)	16	17.0	(9.2, 29.2)	0	--	--
Brikama	34	25.1	(17.4, 34.7)	20	16.8	(12.2, 22.7)	1	0.6	(0.1, 5.0)
Mansakonko	40	30.0	(21.2, 40.5)	32	24.3	(16.0, 35.0)	2	0.5	(0.1, 2.6)
Kuntaur	42	34.6	(31.1, 38.2)	38	31.8	(24.5, 40.1)	2	1.7	(0.4, 7.3)
Kerewan	36	38.8	(31.5, 46.6)	18	18.7	(11.5, 28.9)	0	--	--
Janjanbureh	49	24.5	(18.3, 32.1)	45	31.9	(19.8, 47.0)	5	4.2	(1.6, 10.9)
Basse	43	26.8	(21.0, 33.4)	45	28.5	(17.9, 42.2)	2	0.8	(0.2, 2.8)

Table continued on next page

	Mild anemia[c]			Moderate anemia[c]			Severe anemia[c]		
Characteristic	**n**	**% [a]**	**(95% CI)[b]**	**n**	**% [a]**	**(95% CI)[b]**	**n**	**% [a]**	**(95% CI)[b]**
Wealth Quintile									
Lowest	113	31.0	(25.1, 37.6)	94	26.0	(20.2, 32.9)	8	1.4	(0.7, 2.9)
Second	64	25.0	(18.9, 32.3)	56	25.3	(16.9, 36.0)	1	1.0	(0.1, 7.0)
Middle	51	31.8	(19.8, 47.0)	40	20.3	(13.7, 29.0)	2	0.3	(0.1, 1.4)
Fourth	31	29.3	(17.7, 44.4)	14	14.2	(8.8, 22.0)	1	1.6	(0.3, 8.3)
Highest	41	21.2	(10.4, 38.3)	21	17.8	(10.1, 29.5)	0	--	--
Household sanitation [g]									
Inadequate	189	28.0	(23.3, 33.2)	146	22.9	(18.5, 28.1)	8	0.9	(0.4, 2.3)
Adequate	111	28.1	(22.9, 34.0)	79	19.3	(13.3, 27.3)	4	0.8	(0.2, 4.1)
Inflammation									
None	196	28.4	(23.5, 34.0)	142	18.4	(14.2, 23.5)	9	0.9	(0.3, 2.3)
Any inflammation	103	27.6	(17.7, 40.4)	82	29.1	(23.2, 35.9)	3	1.0	(0.2, 5.0)
ALL CHILDREN	300	28.1	(24.2, 32.3)	225	21.4	(18.0, 25.4)	12	0.9	(0.4, 2.1)

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] CI=confidence interval, calculated taking into account the complex sampling design.

[c] Mild, moderate, and severe anemia defined as hemoglobin 100-109 g/L, 70-99 g/L, and <70 g/L, respectively.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups.

[e] Any inflammation includes incubation=CRP only; early convalescence=CRP and AGP; late convalescence=AGP only.

3.3.9. Anemia, iron deficiency, and iron deficiency anemia

Approximately half of all children are anemic (hemoglobin< 110g/L ≤ 110g/L), close to 60% of children are iron deficient (serum ferritin < 12 ug/l) and more than one-third suffered from iron deficiency anemia (hemoglobin< 110g/L ≤ 110g/L and serum ferritin < 12 ug/l) (IDA, see Table 18). The overlap of anemia and iron deficiency (ID) is considerable, with approximately 75% of anemic children displaying concurrent ID (see Figure 10).

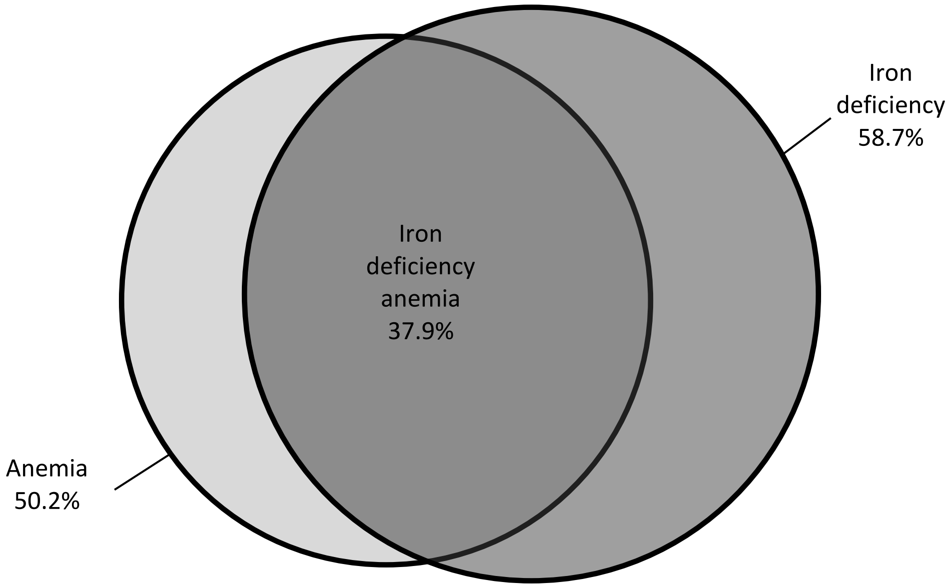


Figure 11. Venn diagram showing overlap between anemia and iron deficiency in children 6-59 months of age, The Gambia 2018

Anemia prevalence is higher in male children than in female children and in children living in rural areas compared to those living in urban areas. Anemia prevalence is lowest in Banjul, Kanifing and Brikama with below 50% and highest in Kuntaur with almost 70%. Further, the anemia prevalence is higher in children with inflammation than in children without. Neither household wealth nor household sanitation are predictors of anemia (see Table 18). Anemia prevalence is about half in children consuming vitamin/mineral supplements and/or infant formula at the time of the MICS compared to those not consuming. Neither breastfeeding nor fortified baby foods have an effect on anemia (see Table 19).

The overall prevalence of ID and IDA are 59.0% and 38.2%, respectively in Gambian children 6-59 months of age. Similar to anemia, the prevalence of ID and IDA is higher in rural areas and data shows that more children living in Kuntaur and Janjanbureh are iron deficient and iron deficient anemic, compared to Banjul, Kanifing and Brikama; although for ID the difference is not significant. Neither sex and inflammation nor household wealth and household sanitation are determinants of ID and IDA (see Table 18). Similar to anemia, less children consuming vitamin and/or mineral supplements and/or infant formula have anemia. Also, breastfeeding and fortified baby foods have no effect on anemia. For ID, none of the investigated feeding indicators affect its occurrence (see Table 19).

Table 18. Anemia, iron deficiency, and iron deficiency anemia in pre-school age children 6-59 months of age, The Gambia 2018

	Anemia[b]				Iron deficiency[e]				Iron deficiency anemia[f]			
Characteristic	n	% [a]	(95% CI)[c]	P value [d]	n	% [a]	(95% CI)[c]	P value [d]	n	% [a]	(95% CI)[c]	P value [d]
Age Group (in months)												
6-11	56	64.2	(43.0, 81.0)	<0.005	56	51.0	(40.4, 61.6)	<0.0001	37	40.6	(31.3, 50.5)	<0.001
12-23	61	59.0	(43.7, 72.7)		185	77.9	(67.3, 85.7)		139	50.9	(36.0, 65.6)	
24-35	36	63.2	(57.3, 68.7)		162	70.8	(58.4, 80.7)		112	49.7	(42.9, 56.6)	
36-47	13	36.2	(28.0, 45.3)		143	50.3	(41.2, 59.3)		88	25.9	(19.8, 33.1)	
48-59	71	35.6	(26.4, 46.1)		90	37.4	(28.5, 47.2)		51	24.2	(18.3, 31.2)	
Sex												
Male	293	55.4	(48.9, 61.8)	<0.05	341	59.9	(51.2, 68.0)	0.62	232	41.5	(34.1, 49.4)	0.19
Female	244	44.9	(38.8, 51.2)		295	58.1	(49.4, 66.3)		195	34.6	(28.7, 41.0)	
Residence												
Urban	193	46.0	(40.1, 51.9)	<0.05	218	54.0	(42.5, 65.1)	<0.05	146	33.5	(27.5, 40.1)	<0.01
Rural	344	57.5	(51.1, 63.6)		418	67.1	(61.2, 72.5)		281	45.8	(39.8, 51.9)	
LGA												
Banjul	39	41.9	(32.4, 52.1)	<0.005	39	44.1	(37.9, 50.5)	0.09	27	31.0	(20.8, 43.5)	<0.005
Kanifing	55	42.5	(35.8, 49.6)		51	56.4	(42.3, 69.6)		36	38.0	(25.5, 52.4)	
Brikama	74	54.7	(44.0, 65.0)		66	50.9	(35.4, 66.3)		37	28.4	(21.8, 35.9)	
Mansakonko	82	68.1	(62.2, 73.4)		97	69.4	(54.9, 80.8)		64	47.3	(33.7, 61.4)	
Kuntaur	54	57.5	(50.0, 64.7)		88	72.0	(61.5, 80.5)		70	57.4	(50.0, 64.5)	
Kerewan	99	60.7	(44.7, 74.6)		60	65.5	(54.1, 75.3)		38	41.4	(32.4, 51.0)	
Janjanbureh	90	56.1	(45.3, 66.3)		130	71.6	(65.8, 76.7)		82	49.9	(36.4, 63.4)	
Basse	37	50.4	(45.9, 54.8)		105	63.6	(52.4, 73.5)		73	45.5	(34.9, 56.6)	

Table continued on next page

	Anemia[b]				Iron deficiency[e]				Iron deficiency anemia[f]			
Characteristic	n	% [a]	(95% CI)[c]	P value [d]	n	% [a]	(95% CI)[c]	P value [d]	n	% [a]	(95% CI)[c]	P value [d]
Wealth Quintile												
Lowest	215	58.4	(49.6, 66.8)	0.26	268	68.9	(57.5, 78.4)	0.33	182	47.6	(38.8, 56.6)	0.31
Second	121	51.3	(40.0, 62.6)		139	60.2	(47.2, 71.9)		92	37.2	(26.8, 48.9)	
Middle	93	52.5	(37.9, 66.6)		104	48.8	(31.2, 66.6)		72	37.7	(25.0, 52.3)	
Fourth	46	45.1	(31.5, 59.5)		58	59.2	(43.5, 73.3)		37	35.8	(22.9, 51.1)	
Highest	62	39.0	(29.9, 48.8)		67	57.0	(39.9, 72.6)		44	28.8	(20.0, 39.5)	
Household sanitation [g]												
Inadequate	343	51.9	(45.2, 58.5)	0.41	405	58.2	(49.2, 66.7)	0.75	269	37.5	(31.6, 43.9)	0.73
Adequate	194	48.3	(42.9, 53.7)		231	60.2	(49.1, 70.3)		158	39.2	(32.1, 46.9)	
Inflammation												
None	347	47.7	(43.5, 51.9)	<0.05	462	60.2	(53.2, 66.8)		283	36.6	(31.0, 42.5)	0.08
Any inflammation	188	57.7	(48.3, 66.7)		174	56.1	(45.0, 66.6)		144	43.0	(36.5, 49.8)	
ALL CHILDREN	537	50.4	(45.9, 54.8)		636	59.0	(51.3, 66.4)		427	38.2	(33.6, 43.0)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] Anemia defined as hemoglobin < 110 g/L.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

[e] ID= Iron deficiency defined as serum ferritin < 12 µg/l, values are adjusted for inflammation according to BRINDA.

[f] IDA= Iron deficiency anemia, defined as low Hb (< 110 g/L) with low serum ferritin (< 12.0 µg/L).

[g] Composite variable of toilet type and if toilet facilities are shared with non-household members; Adequate Sanitation = flush or pour flush toilet or pit latrine with slab not shared with another household. Inadequate sanitation= open pit, bucket latrine, hanging toilet/latrine, no facility, bush, field.

[h] Any inflammation includes incubation=CRP only; early convalescence=CRP and AGP; late convalescence=AGP only.

Table 19. Anemia, iron deficiency and iron-deficiency anemia in children 6-23 months of age, by key feeding indicators, The Gambia 2018

Characteristic	Anemia[a]				Iron deficiency[e]				Iron deficiency anemia[f]			
	n	% [b]	(95% CI)[c]	P value [d]	n	% [a]	(95% CI)[c]	P value [d]	n	% [a]	(95% CI)[c]	P value [d]
Child is currently being breastfed												
No	139	64.3	(56.4, 71.5)	0.98	165	77.1	(61.8, 87.6)	0.34	117	53.4	(42.0, 64.5)	0.69
Yes	197	64.5	(51.4, 75.7)		212	68.2	(54.9, 79.1)		157	50.0	(37.9, 62.0)	
Consumed fortified baby foods at time of MICS												
No	190	63.0	(50.8, 73.7)	0.84	203	68.2	(55.7, 78.5)	0.28	151	49.1	(37.5, 60.9)	0.75
Yes	27	64.9	(46.8, 79.6)		34	78.8	(58.3, 90.8)		24	53.4	(30.7, 74.8)	
Consumed vitamin/mineral supplements at time of MICS												
No	197	66.9	(54.4, 77.4)	<0.005	208	68.6	(55.5, 79.3)	0.41	157	52.1	(40.7, 63.2)	<0.05
Yes	20	33.1	(18.0, 52.6)		29	77.6	(56.7, 90.1)		18	30.3	(16.0, 49.6)	
Consumed infant formula at time of MICS												
No	211	64.7	(52.6, 75.2)		227	70.0	(57.6, 80.1)	0.62	170	51.2	(40.0, 62.3)	
Yes	5	31.2	(13.5, 56.9)		9	59.5	(19.7, 89.8)		4	17.0	(4.7, 45.8)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Anemia defined as hemoglobin < 110 g/L.

[b] Percentages weighted for unequal probability of selection.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

[e] Iron deficiency defined as serum ferritin < 12 µg/l, values are adjusted for inflammation according to BRINDA

[f] Iron deficiency anemia, defined as low Hb (< 110 g/L) with low serum ferritin (< 12.0 µg/L).

Nutritional factors such as ID and vitamin A deficiency (VAD) are the main predictors of anemia in children. More than twice as many children with ID suffer from anemia compared to those without and the prevalence of anemia is almost 30% higher in children with VAD than in children without VAD. Surprisingly, infectious diseases and inflammation are not associated with anemia, except for fever, which increases the likelihood of anemia (see Table 20). Please note, malaria is not added to the analyses as only one child has been diagnosed with malaria.

Table 20. Anemia in pre-school age children 6-59 months of age, by infection-related characteristics and vitamin A and iron status, The Gambia 2018

		Anemia[b]		
Characteristic	n	% [a]	(95% CI) [c]	P value [d]
Child had any type of diarrhea in past 2 weeks during MICS				
No	383	50.7	(46.4, 55.0)	0.79
Yes	143	51.7	(42.6, 60.7)	
Child had a cough in past 2 weeks during MICS				
No	414	50.5	(44.5, 56.5)	0.91
Yes	112	51.1	(43.8, 58.3)	
Child had fever in past 2 weeks during MICS				
No	391	48.5	(43.2, 53.7)	<0.05
Yes	135	57.7	(50.8, 64.4)	
Inflammation [f]				
None	347	47.7	(43.5, 51.9)	0.11
Incubation (elevated CRP only)	25	74.7	(39.2, 93.1)	
Early convalescence (elevated CRP and AGP)	62	68.6	(45.3, 85.3)	
Late convalescence (elevated AGP only)	101	49.0	(38.0, 60.1)	
Vitamin A deficient [g]				
No (RBP≥0.7 µmol/L)	398	48.4	(43.9, 53.1)	<0.005
Yes (RBP<0.7 µmol/L)	137	60.6	(51.9, 68.5)	
Iron deficient [h]				
No (ferrtin≥12 µg/L)	108	29.4	(22.7, 37.1)	<0.0001
Yes (ferritin<12 µg/L)	427	65.7	(59.2, 71.6)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] Anemia defined as hemoglobin < 110 g/L adjusted for altitude.
[c] CI=confidence interval, calculated taking into account the complex sampling design.
[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly
[e] Positive malaria status identified using rapid diagnostic tests during GMS data collection
[f] Incubation=CRP only; early convalescence=CRP and AGP; late convalescence=AGP only.
[g] Vitamin A deficiency (VAD) defined as retinol binding protein (RBP) <0.70 µmol/L; RBP concentrations adjusted for inflammation using the BRINDA approach
[h] Iron deficiency defined as serum ferritin < 12 µg/l, values are adjusted for inflammation according to BRINDA

3.3.10. Vitamin A Deficiency

Nationally, 18.2% of children have VAD using RBP as the indicator of vitamin A status (see Table 21). This prevalence denotes a moderate public health problem according to WHO classifications [37]. The prevalence of VAD differs significantly by age, and is lower in younger children (6-23months) compared to older children (24-59 months). More male children have VAD compared to female and the prevalence of VAD is twice as high in rural areas than in urban centers. VAD is also associated with place of residence, highest prevalences are found in Janjanbureh, Kuntaur and Basse, lowest in Banjul and Kanifing. Moreover, a higher proportion of children living in households with inadequate sanitary conditions have VAD, compared to children residing in households with adequate sanitary conditions.

Regarding supplementation status, the prevalence of vitamin A deficiency was significantly lower in children that received vitamin A supplements in the past 3 months. When comparing children who were supplemented in the past 6 months with other children, no significant difference was observed.

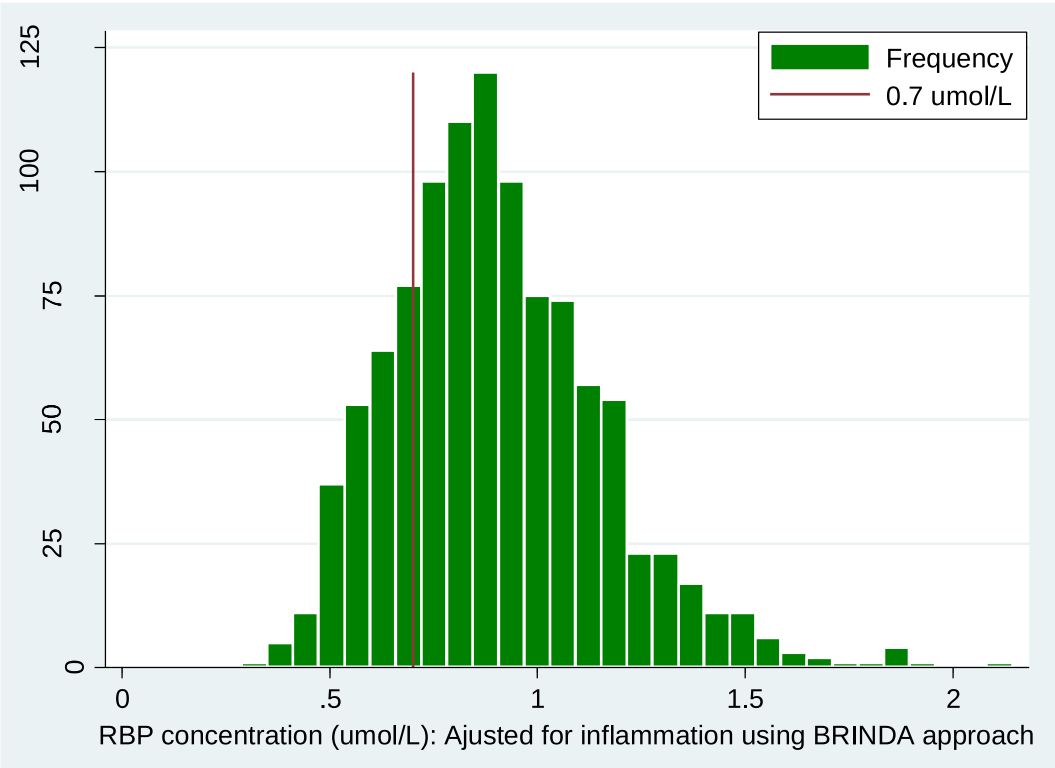


Figure 12. Distribution of retinol binding protein concentration, adjusted for inflammation using the BRINDA approach, children 6-59 months of age, The Gambia 2018

Table 21. Proportion of children 6-59 months of age with vitamin A deficiency, by various characteristics, The Gambia 2018

Characteristic	n	Vitamin A deficiency %[a, b]	(95% CI)[c]	P value [d]
Age Group (in months)				
6-11	4	3.9	(1.0, 13.7)	<0.001
12-23	25	8.6	(5.1, 14.0)	
24-35	68	24.0	(16.6, 33.4)	
36-47	70	23.8	(17.1, 32.2)	
48-59	51	24.7	(15.6, 36.7)	
Sex				
Male	124	22.2	(17.6, 27.6)	<0.005
Female	94	14.0	(10.3, 18.8)	
Residence				
Urban	52	13.8	(8.5, 21.6)	<0.05
Rural	166	25.5	(21.4, 30.0)	
LGA				
Banjul	10	9.1	(5.4, 14.9)	<0.05
Kanifing	11	12.4	(4.9, 28.3)	
Brikama	15	13.3	(6.5, 25.2)	
Mansakonko	27	20.1	(15.6, 25.5)	
Kuntaur	31	27.2	(19.5, 36.5)	
Kerewan	12	13.3	(7.7, 22.0)	
Janjanbureh	66	32.8	(24.8, 42.0)	
Basse	46	27.3	(19.2, 37.2)	
Wealth Quintile				
Lowest	107	27.7	(21.4, 35.0)	0.07
Second	54	24.0	(13.7, 38.7)	
Middle	30	12.4	(6.4, 22.6)	
Fourth	11	14.3	(4.5, 37.1)	
Highest	16	7.1	(3.3, 14.7)	
Received vitamin A supplementation in past 6 months				
No	148	20.3	(15.0, 27.0)	0.09
Yes	61	12.8	(8.3, 19.3)	
Received vitamin A supplementation in past 3 months				
No	188	20.1	(15.1, 26.3)	<0.05
Yes	21	7.1	(4.1, 12.2)	
Inflammation[e]				
None	42	16.6	(12.8, 21.2)	0.31
Incubation	8	30.8	(15.0, 53.0)	
Early convalescence	22	21.4	(11.9, 35.5)	
Late convalescence	46	21.4	(13.0, 33.1)	
Household sanitation				
Inadequate	151	22.2	(16.1, 29.6)	<0.05
Adequate	67	12.6	(8.3, 18.6)	
ALL CHILDREN	218	18.3	(14.5, 22.8)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] Vitamin A deficiency (VAD) defined as retinol binding protein (RBP) <0.70 µmol/L; RBP concentrations adjusted for inflammation using the BRINDA approach.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

[e] Incubation=CRP only; early convalescence=CRP and AGP; late convalescence=AGP only.

3.3.11. Vitamin A Supplementation

Nationally, approximately one-third (31.4%; 95%CI:28.5, 34.5) received vitamin A supplements in the past six months (data not shown). The date of supplementation was confirmed by interviewers who inspected the child's health card in 98.4% of children reported to receive a vitamin A supplement in the past six months. Coverage of vitamin A supplements in the past 3 months was 15.7% (95% CI: 13.4, 18.2). Notably, the prevalence of vitamin A supplementation in the past 6 months was statistically significantly higher ($p<0.001$) in children 6-23 months of age compared to older children (see Figure 13).

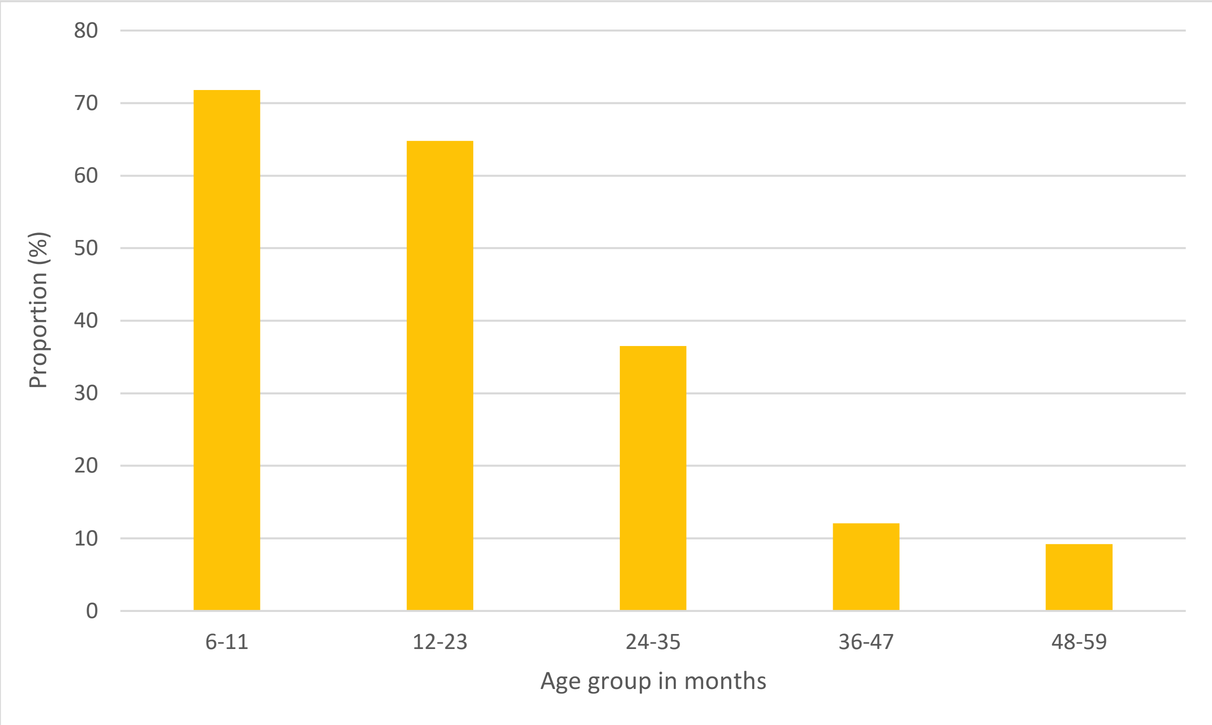


Figure 13. Prevalence of vitamin A supplementation in the past 6 months, children 6-59 months of age, The Gambia 2018

Table 22. Percentage of children (6-59 months) that received vitamin A supplements in the past 6 months, The Gambia 2018

Characteristic	n	% [a]	(95% CI) [b]	P-value [c]
Age Group (in months)				
6-11	89	68.9	(49.8, 83.1)	<0.001
12-23	177	60.2	(50.4, 69.3)	
24-35	97	33.3	(22.7, 45.9)	
36-47	36	6.8	(4.4, 10.4)	
48-59	23	4.3	(2.3, 8.1)	
Sex				
Male	214	33.7	(27.9, 40.0)	0.09
Female	208	28.7	(24.2, 33.6)	
Residence				
Urban	125	27.0	(22.1, 32.5)	<0.05
Rural	297	37.9	(30.9, 45.5)	
LGA				
Banjul	24	26.7	(17.6, 38.3)	0.12
Kanifing	26	28.9	(21.7, 37.5)	
Brikama	45	28.6	(22.5, 35.6)	
Mansakonko	66	37.9	(29.4, 47.3)	
Kuntaur	83	53.1	(41.1, 64.7)	
Kerewan	51	41.2	(28.3, 55.5)	
Janjanbureh	79	31.1	(20.1, 44.8)	
Basse	48	22.4	(11.2, 39.6)	
Wealth Quintile				
Lowest	185	35.2	(27.5, 43.8)	0.33
Second	97	30.3	(22.3, 39.7)	
Middle	66	36.8	(28.1, 46.4)	
Fourth	35	26.0	(19.9, 33.2)	
Highest	39	22.0	(13.7, 33.5)	
ALL CHILDREN	425	27.2	(24.2, 30.5)	

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] Percentages weighted for unequal probability of selection

[b] CI=confidence interval, calculated taking into account the complex sampling design

[c] P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups.

3.4. All Women

3.4.1. Characteristics

The GMNS has questionnaire data available for 1,873 women, of which 1,861 have complete data on their birth year and month. The age distribution is skewed, with more younger women enrolled in the GMNS than older women. To illustrate, women 15 to 19 years of age account for more than 20% of all women, whereas women 40 to 49 years old account for less than 15% of women. Slightly more women were recruited from urban areas, and more than 200 women were recruited from all LGAs except Kuntaur and Mansakonko. Approximately 8% of women self-reported as pregnant.

Table 23. Description of all sampled pregnant and non-pregnant women, The Gambia 2018

	Survey Sample		
Characteristic	n	% [a]	(95% CI)[b]
Age Group (in years)			
15-19	443	23.8	(22.0, 25.7)
20-24	350	18.8	(17.0, 20.8)
25-29	318	17.1	(15.1, 19.3)
30-34	266	14.3	(12.8, 15.9)
35-39	242	13	(11.3, 14.9)
40-44	133	7.1	(6, 8.5)
45-49	109	5.9	(4.9, 7.0)
Woman pregnant			
Yes	158	8.4	(7.1, 10.0)
No	1715	91.6	(90.0, 92.9)
Residence			
Urban	996	53.6	(48.8, 57.5)
Rural	877	46.8	(42.5, 51.2)
LGA			
Banjul	274	14.6	(12.6, 16.9)
Kanifing	219	11.7	(9.3, 14.6)
Brikama	262	14.0	(12.1, 16.2)
Mansakonko	198	10.6	(8.6, 12.9)
Kuntaur	186	9.9	(7.7, 12.7)
Kerewan	222	11.9	(10.0, 14.0)
Janjanbureh	261	13.9	(11.6, 16.7)
Basse	251	13.4	(8.8, 19.9)
ALL WOMEN	1,873	100.0	

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages are un-weighted and do not account for unequal probability of selection.

[b] CI=confidence interval, calculated taking into account the complex sampling design.

3.4.2. Knowledge and practices related to anemia and fortified food

About nine out of ten women have ever heard about anemia, most of them stating that the lack of iron and sickness/infection are the main causes of anemia. When asked how to prevent anemia, most of the women answered that the consumption of iron rich foods (57.6%) or the consumption of vitamin C rich foods (33.8%) prevent anemia. More than 80% of women drink tea or coffee during or shortly after the meal, the majority once a day.

Table 24. Extent of knowledge about anemia in all women (non-pregnant 15-49 years of age and pregnant), The Gambia 2018

Characteristic	n	% [a]	(95% CI)[b]
Have heard about anemia			
Yes	1733	90.6	(82.5, 95.2)
No	139	9.4	(4.8, 17.5)
Reported causes of anemia [c]			
Lack of iron in diet	1073	48.7	(43.5, 53.9)
Sickness/ infection (malaria, hookworms, HIV, etc.)	641	32.3	(25.5, 39.9)
Heavy bleeding during menstruation	102	5.6	(3.2, 9.7)
Other	120	5.7	(3.3, 9.6)
Don`t know	429	27.5	(21.2, 34.9)
Reported prevention of anemia [c]			
Eat iron rich foods/ diet rich in iron	1169	57.6	(50.9, 64.1)
Eat/ give vitamin- C rich foods during or right after meals	596	33.8	(25.8, 42.9)
Take/ give iron supplements	432	24.9	(20.5, 29.8)
Treat other causes of anemia (diseases or infections)	264	16.3	(11.0, 23.5)
Other	64	2.3	(1.3, 4.0)
Don´t know	279	17.1	(12.1, 23.6)
Drinks black/ green tea or coffee during or shortly after meals			
Yes	1525	82.1	(72.5, 88.9)
No	347	17.9	(11.1, 27.5)
Frequency of drinking black/ green tea or coffee during or shortly after meals			
Once a day	986	70.4	(59.8, 79.2)
2-3 times a day	512	28.5	(19.9, 39.0)
> 3 times a day	21	0.9	(0.5, 1.7)
Don´t know	6	0.2	(0.1, 0.5)

Note: The n’s are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] CI=confidence interval, calculated taking into account the complex sampling design.

[c] Anemia causes and anemia prevention questions permitted multiple responses, thus, percentages do not sum to 100%

Only a low proportion of the women have ever heard of fortified vegetable oil (6.6%). Of those more than 80% prefer to buy fortified oil over non-fortified vegetable oil. Awareness of the Enrichi logo is even lower with only about 2% of the surveyed women having ever seen it. Only about 1% of the women have ever heard of fortified wheat flour.

Table 25. Extent of knowledge about and use of fortified foods in all women (non-pregnant 15-49 years of age and pregnant), The Gambia 2018

Characteristic	n	% [a]	(95% CI) [b]
Have heard of fortified vegetable oil			
Yes	161	6.6	(4.6, 9.3)
No	1678	93.4	(90.6, 95.4)
Reported benefits of fortified vegetable oil [c]			
Prevents blindness	15	0.3	(0.1, 1.3)
Reduces mortality	14	0.3	(0.1, 0.9)
Prevents vitamin deficiency	24	0.6	(0.3, 1.1)
Improves health status	111	4.6	(2.8, 7.4)
Other	3	0.1	(0.0, 0.5)
Don´t know	24	1.2	(0.6, 2.4)
Preferred to buy vegetable oil which is labeled as fortified			
Yes	130	81.9	(66.0, 91.4)
No	23	13.7	(6.3, 27.3)
Don´t know	6	4.3	(0.9, 18.4)
Have seen Enrichi logo before			
Yes	79	2.2	(1.3, 3.8)
No	1781	97.5	(95.8, 98.5)
Have heard of fortified wheat flour			
Yes	45	1.2	(0.5, 2.7)
No	1804	98.2	(96.6, 99.0)
Reported benefits of fortified wheat flour [c]			
Gives blood	9	0.1	(0.0, 0.2)
Reduces anemia	7	0.3	(0.1, 1.4)
Reduces mortality	4	0.5	(0.1, 3.1)
Improves health status	34	1.0	(0.4, 2.6)
Better work output	2	0.0	(0.0, 0.2)
Other	2	0.0	(0.0, 0.1)
Don´t know	7	0.1	(0.0, 0.4)

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] CI=confidence interval, calculated taking into account the complex sampling design.
[c] Benefits of fortified vegetable oil and wheat flour only asked of women who had heard of fortified vegetable oil or wheat flour previously. Respondents could report more than one benefit.

3.4.3. Dietary diversity

As shown in Table 26, approximately 70% of women have minimum dietary diversity in the 24 hours prior to the GMNS. On average, women consume about 5 different foods, and there is little difference in the number of foods consumed by pregnant and non-pregnant women. Notably, very few women ate less than three food items, showing that the vast majority of woman ate three or more food groups in the past 24 hours.

Table 26. Minimum dietary diversity in all women 15-49 years, The Gambia 2018

Characteristic	n	Mean or % [a]	(95% CI)[b]
Mean number of food groups (all women)	1,839	5.3	(5.1, 5.6)
Mean number of food groups			
Pregnant	154	5.4	(4.9, 5.8)
Non-pregnant	1685	5.3	(5.1, 5.6)
Number of food groups			
0	1	0.0	(0.0, 0.0)
1-2	59	3.0	(2.0, 4.6)
3-4	476	26.2	(21.8, 21.2)
5-6	869	47.7	(44.1, 51.3)
7+	434	23.1	(18.2, 28.8)
Minimal dietary diversity (≥5 food groups)			
Yes	1303	70.8	(65.6, 75.5)
No	536	29.2	(24.5, 34.4)

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] CI=confidence interval, calculated taking into account the complex sampling design.

3.4.4. Physical activity

Only 13% of the surveyed women stated that their work involves vigorous to moderate activity. Amongst these women, vigorous to moderate work was conducted approximately 6 days a week about 3 hours per day.

Almost all women participating in the survey walk or bike to work, school or to do errands. On average the surveyed women spent about 4.6 hours sitting on weekdays and 6.8 hours on the weekends.

Table 27. Physical activity in non-pregnant women 15-49 years of age.

Characteristic	N	%[a], mean	(95% CI)[b]
Work involve vigorous/ moderate activity			
Yes	280	13.0	(6.5, 24.4)
No	1585	87.0	(75.6, 93.5)
Days in week work involves vigorous/ moderate activity (mean)	280	5.98	
Hours per day in vigorous/ moderate activity at work (mean)	280	2.62	
Usually walks or bikes to work, school, or errands			
Yes	1826	98.8	(97.6, 99.4)
No	41	1.2	(0.6, 2.4)
Hours on weekdays spent sitting (mean)	1870	4.63	(4.11, 5.14)
Hours on weekends spent sitting (mean)	1870	5.83	(5.06, 6.60)

[a] Weighted for unequal probability of selection among governorates
[b] CI=confidence interval calculated taking into account the complex sampling design.
[c] Only including those women who mentioned work involved vigorous or moderate activity

3.5. Non-pregnant women

3.5.1. Anthropometry

The prevalence of undernutrition and overnutrition/overweight, as measured by body mass index, is shown in Table 41 and Figure 14 below. Of the women weighed and measured, 15% are underweight (i.e. BMI < 18.5), and 55% are normal weight. Approximately 18% and 11% of women are overweight and obese, respectively.

Table 28 and Figure 14 illustrates that underweight (i.e. BMI<18.5) in woman is characterized primarily by women considered "at risk" of underweight, and that severe and moderate energy deficiencies in women are more rare.

As shown in Figure 15, the prevalence of overweight and obesity generally increase with age until 44 years of age, followed by a small decrease at 44 to 49 years of age. In addition, the prevalences of overweight and obesity are higher among women in urban areas, and the prevalence of obesity increases consistently by wealth quintile (see Table 29).

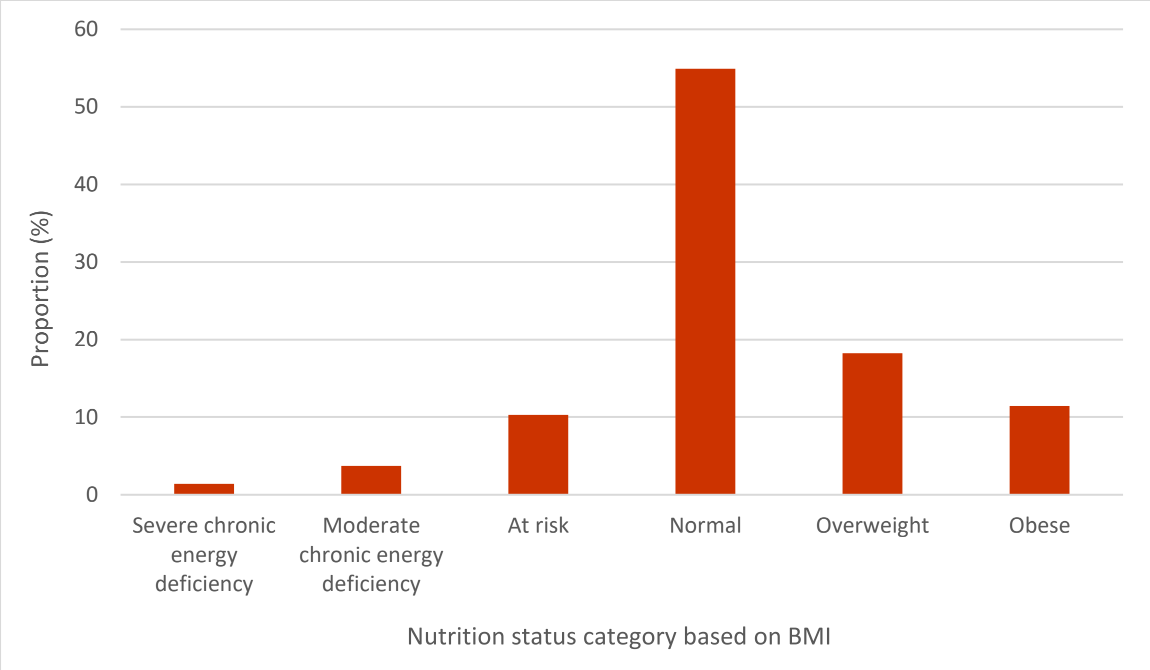


Figure 14. Prevalence of underweight, normal weight, and overweight and obesity in non-pregnant women, The Gambia 2018

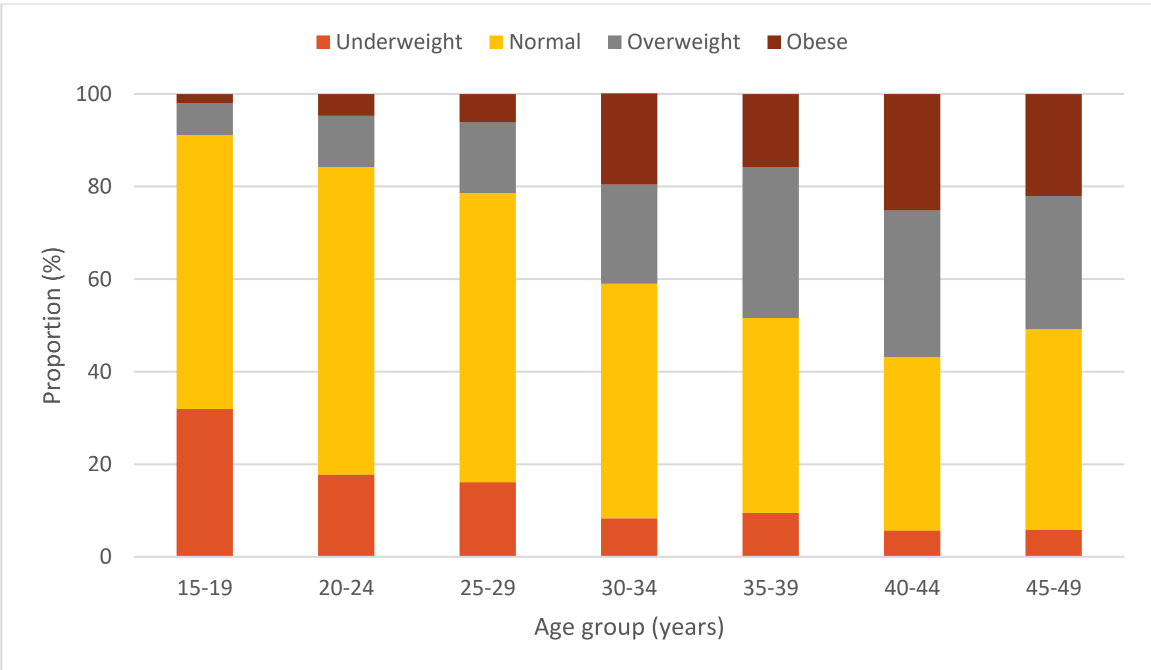


Figure 15. Prevalence of normal weight, overweight, and obesity in non-pregnant women by age group, The Gambia 2018

Table 28. Mean Body Mass Index (BMI) and percentage of specific BMI levels in non-pregnant women (15-49 years), The Gambia 2018

Characteristic	Mean BMI	Severe chronic energy deficiency (BMI: <16)			Moderate Chronic energy deficiency (BMI:16.0-16.9)			At risk (BMI: 17.0-18.4)		
		n	% [a]	95 CI[b]	n	% [a]	95 CI[b]	n	% [a]	95 CI[b]
Age in years										
15-19	20.4	18	2.5	(1.2, 5.0)	31	6.4	(3.4, 11.6)	80	20.9	(14.2, 29.6)
20-24	21.6	5	1.4	(0.4, 5.1)	15	6.0	(2.3, 14.7)	33	7.5	(4.6, 11.9)
25-29	22.3	1	0.4	(0.0, 2.7)	10	6.6	(3.2, 13.1)	31	12.7	(8.2, 19.3)
30-34	24.6	3	1.0	(0.3, 3.0)	3	0.4	(0.1, 1.9)	13	4.4	(1.9, 9.8)
35-39	25.4	2	0.4	(0.1, 1.7)	5	1.0	(0.3, 2.9)	14	8.7	(4.8, 15.3)
40-44	27.5	2	1.0	(0.2, 6.8)	1	0.4	(0.1, 3.2)	4	1.5	(0.5, 4.3)
45-49	27.3	2	3.9	(0.7, 19.0)	2	1.6	(0.4, 6.4)	2	1.5	(0.3, 6.3)
Residence										
Urban	23.5	18	1.3	(0.6, 3.0)	29	3.6	(1.9, 6.9)	92	10.0	(7.4, 13.5)
Rural	22.6	15	1.6	(0.8, 3.1)	38	4.4	(3.4, 5.8)	87	10.9	(7.8, 15.0)
LGA										
Banjul	24.6	4	1.8	(0.6, 5.8)	7	3.2	(1.3, 8.0)	27	11.9	(8.1, 17.3)
Kanifing	24.0	2	1.0	(0.3, 3.8)	4	2.7	(1.3, 5.4)	22	11.8	(7.6, 17.9)
Brikama	23.3	3	1.1	(0.3, 4.4)	11	4.0	(1.6, 9.7)	23	9.4	(5.9, 14.6)
Mansakonko	21.9	6	3.1	(1.4, 6.9)	9	5.0	(2.2, 10.7)	18	9.8	(5.5, 16.8)
Kuntaur	21.0	3	1.3	(0.4, 4.5)	11	7.8	(4.9, 12.4)	21	14.8	(9.2, 23.1)
Kerewan	23.0	4	1.6	(0.4, 6.5)	6	3.2	(1.8, 5.6)	30	15.8	(9.9, 24.1)
Janjanbureh	22.6	6	2.9	(1.3, 6.6)	10	4.5	(2.6, 7.5)	21	9.9	(8.0, 12.2)
Basse	23.4	5	1.6	(0.5, 4.9)	9	3.9	(2.4, 6.4)	17	6.6	(3.1, 13.5)

Table continued on next page

Characteristic	Mean BMI	Severe chronic energy deficiency (BMI: <16)			Moderate Chronic energy deficiency (BMI:16.0-16.9)			At risk (BMI: 17.0-18.4)		
		n	% [a]	95 CI[b]	n	% [a]	95 CI[b]	n	% [a]	95 CI[b]
Woman's Education										
Never at school	24.0	17	1.7	(0.8, 3.4)	25	2.6	(1.4, 4.9)	65	8.9	(6.5, 12.2)
Primary school	23.5	3	2.8	(0.8, 8.7)	8	2.5	(1.1, 5.6)	22	6.7	(3.8, 11.3)
Lower Secondary	23.0	8	0.8	(0.4, 1.7)	15	4.4	(2.0, 9.3)	43	13.8	(9.3, 19.9)
Upper Secondary	21.3	3	0.8	(0.2, 3.3)	15	5.7	(2.7, 11.9)	39	15.9	(9.7, 25.1)
Higher	23.6	1	0.1	(0.0, 0.6)	3	7.2	(1.3, 32.2)	7	4.5	(1.8, 10.8)
Wealth Quintile										
Lowest	22.0	7	1.6	(0.6, 3.7)	22	4.1	(2.6, 6.6)	62	14.0	(9.6, 19.8)
Second	23.0	7	0.7	(0.3, 2.1)	15	4.6	(2.6, 7.9)	27	10.1	(7.7, 13.3)
Middle	23.7	8	1.5	(0.5, 4.8)	12	3.6	(1.6, 8.1)	24	6.9	(4.5, 10.5)
Fourth	23.2	4	2.1	(0.6, 7.5)	8	4.8	(1.4, 14.9)	24	12.7	(7.0, 21.9)
Highest	23.9	7	1.3	(0.3, 4.8)	10	2.6	(1.1, 5.8)	42	10.0	(6.0, 16.1)
Minimal dietary diversity										
No (0-4)	22.5	10	1.3	(0.5, 3.1)	19	4.2	(1.4, 12.0)	49	12.5	(9.4, 16.5)
Yes (≥5)	23.6	21	1.2	(0.6, 2.3)	47	3.7	(2.6, 5.3)	125	9.2	(6.6, 12.8)
Household sanitation [d]										
Inadequate	23.0	17	1.5	(0.8, 2.9)	43	4.4	(2.9, 6.5)	110	11.4	(8.8, 14.7)
Adequate	23.6	16	1.3	(0.5, 3.6)	24	3.3	(1.4, 7.4)	69	9.0	(6.4, 12.3)
ALL WOMEN	23.2	33	1.4	(0.8, 2.6)	67	3.9	(2.5, 6.0)	179	10.3	(8.1, 12.9)

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] Percentages weighted for non-response and survey design.

[b] CI=confidence interval, adjusted for cluster sampling design.

[c] Chi-square p-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly

[d] Composite variable of toilet type and if toilet facilities are shared with non-household members; Adequate Sanitation = flush or pour flush toilet or pit latrine with slab not shared with another household. Inadequate sanitation= open pit, bucket latrine, hanging toilet/latrine, no facility, bush, field

Table 29. Mean Body Mass Index (BMI) and percentage of specific BMI levels in non-pregnant women (15-49 years), The Gambia 2018

Characteristic	Any underweight (BMI<18.5)			Normal (BMI: 18.5-24.9)			Overweight (BMI: 25.0-29.9)			Obese (BMI: ≥30.0)		
	n	% [a]	95 CI[b]	n	% [a]	95 CI[b]	n	% [a]	95 CI[b]	n	% [a]	95 CI[b]
Age in years												
15-19	129	29.7	(23.6, 36.6)	239	61.0	(53.3, 68.2)	28	7.2	(4.3, 11.8)	8	2.1	(0.7, 6.3)
20-24	53	14.8	(8.9, 23.7)	198	70.9	(63.8, 77.0)	33	10.2	(6.7, 15.1)	14	4.1	(1.8, 9.3)
25-29	42	19.7	(14.4, 26.4)	163	62.0	(53.9, 69.6)	40	13.6	(8.4, 21.2)	16	4.7	(2.3, 9.3)
30-34	19	5.8	(3.0, 11.0)	116	54.7	(44.4, 64.5)	49	23.5	(17.0, 31.5)	45	16.1	(9.0, 26.9)
35-39	21	10.1	(6.0, 16.5)	93	38.8	(31.6, 46.5)	72	36.0	(29.0, 43.6)	35	15.1	(10.4, 21.6)
40-44	7	3.0	(1.1, 7.6)	46	33.3	(24.7, 43.2)	39	31.6	(22.9, 41.7)	31	32.2	(24.7, 40.8)
45-49												
Residence												
Urban	139	14.6	(10.8, 19.4)	423	52.8	(48.6, 56.7)	189	20.0	(15.8, 25.1)	121	12.3	(9.6, 15.6)
Rural	140	17.3	(13.3, 22.2)	484	59.9	(54.9, 66.3)	103	14.1	(10.9, 17.9)	52	8.3	(5.2, 13.0)
LGA												
Banjul	38	17.3	(10.9, 26.4)	98	42.1	(35.9, 48.5)	55	23.9	(20.2, 28.0)	43	17.1	(12.3, 23.2)
Kanifing	28	15.6	(11.2, 21.1)	92	48.0	(37.9, 58.3)	38	20.6	(14.0, 29.4)	29	15.8	(11.6, 21.2)
Brikama	37	13.8	(8.7, 21.3)	133	55.6	(50.9, 60.2)	39	18.6	(12.7, 26.3)	26	11.4	(7.8, 16.4)
Mansakonko	33	18	(13.9, 23.0)	118	66.9	(61.5, 71.9)	20	10.1	(5.6, 17.5)	9	5.1	(3.8, 7.0)
Kuntaur	35	24.5	(16.8, 34.2)	97	64.0	(59.6, 68.1)	20	9.3	(4.5, 18.2)	8	2.7	(1.0, 7.3)
Kerewan	40	20.2	(13.2, 29.8)	105	55.5	(49.2, 61.6)	31	14.4	(11.3, 18.1)	20	9.6	(5.8, 15.4)
Janjanbureh	37	18.3	(14.8, 22.4)	136	57.1	(47.9, 65.8)	40	18.8	(13.9, 24.9)	15	6.8	(4.2, 10.9)
Basse	31	11.5	(7.4, 17.4)	128	56.3	(45.2, 66.8)	49	20.9	(16.6, 26.0)	23	10.6	(5.4, 19.9)

Table continued on next page

	Any underweight (BMI<18.5)			Normal (BMI: 18.5-24.9)			Overweight (BMI: 25.0-29.9)			Obese (BMI: ≥30.0)		
Characteristic	**n**	**% [a]**	**95 CI[b]**	**n**	**% [a]**	**95 CI[b]**	**n**	**% [a]**	**95 CI[b]**	**n**	**% [a]**	**95 CI[b]**
Woman's Education												
Never at school	107	13.7	(10.5, 17.7)	374	49.0	(42.3, 55.8)	147	25.5	(19.5, 32.7)	74	12.2	(9.0, 16.4)
Primary school	33	11.8	(7.7, 17.7)	156	56.9	(51.4, 62.3)	40	20.4	(14.0, 28.7)	30	10.7	(7.4, 15.4)
Lower Secondary	66	18.9	(13.9, 25.1)	169	56.8	(50.1, 63.3)	45	11.5	(8.7, 15.2)	31	12.7	(7.3, 21.4)
Upper Secondary	57	21.6	(13.1, 33.3)	141	64.4	(52.6, 74.7)	29	6.8	(3.1, 14.1)	19	6.3	(3.5, 11.1)
Higher	11	9.9	(2.2, 34.7)	53	56.5	(44.4, 68.0)	23	20.9	(14.7, 28.8)	13	10.8	(4.3, 24.4)
Wealth Quintile												
Lowest	91	21.1	(14.6, 29.6)	258	58.8	(52.8, 64.6)	49	17.7	(10.9, 27.3)	18	3.9	(2.5, 5.9)
Second	49	14.8	(12.5, 17.5)	203	60.2	(52.7, 67.3)	51	15.4	(9.5, 24.1)	27	8.9	(6.7, 11.7)
Middle	44	11.8	(8.8, 15.8)	160	54.8	(48.1, 61.3)	62	20.1	(16.3, 24.6)	34	13.0	(8.6, 19.2)
Fourth	36	18.9	(10.2, 32.5)	113	51.7	(43.1, 60.2)	43	16.6	(10.4, 25.6)	34	12.2	(7.9, 18.3)
Highest	59	13.2	(7.4, 22.5)	173	49.8	(39.1, 60.6)	87	20.6	(13.2, 30.7)	60	15.7	(10.3, 23.3)
Minimal dietary diversity												
No (0-4)	78	18.1	(13.6, 23.7)	83	16.7	(12.4, 22.1)	83	16.7	(12.4, 22.1)	38	6.8	(3.7, 12.1)
Yes (≥5)	193	13.8	(10.7, 17.5)	208	19.2	(15.8, 23.2)	208	19.2	(15.8, 23.2)	131	12.8	(10.2, 16.0)
Household sanitation [d]												
Inadequate	170	17.5	(13.3, 22.6)	531	56.1	(50.8, 61.3)	143	16.6	(12.4, 21.9)	89	10.0	(7.5, 13.2)
Adequate	109	12.9	(9.4, 17.3)	376	53.8	(47.8, 59.7)	149	20.2	(15.4, 26.0)	84	12.4	(9.1, 16.7)
ALL WOMEN	279	15.4	(12.4, 19.0)	907	55.0	(51.7, 58.3)	292	18.3	(15.1, 22.0)	173	11.1	(9.0, 13.7)

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] Percentages weighted for non-response and survey design.

[b] CI=confidence interval, adjusted for cluster sampling design.

[c] Chi-square p-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly

[d] Composite variable of toilet type and if toilet facilities are shared with non-household members; Adequate Sanitation = flush or pour flush toilet or pit latrine with slab not shared with another household. Inadequate sanitation= open pit, bucket latrine, hanging toilet/latrine, no facility, bush, field

3.5.2. Malaria

Malaria is observed in only 1 woman in a rural area of Janjanbureh. This woman is between 15 to 19 years of age. Due to the small number of women with malaria, a detailed sub-group analysis by age group, strata, and wealth quintile was not conducted.

Analysis of blood pellets using PCR, however, identified a higher number of women with malaria than the RDTs. As shown in Table 30, approximately 6% of women are identified as malaria positive using this methodology. No statistically significant differences were found by residence or wealth quintile, but significant, or nearly significant, differences are found by age and LGA. Regarding age, women 25-29 years of age have the highest rate of malaria infection. Infection is highest in Kerewan LGA.

3.5.3. Anemia, iron deficiency, and iron deficiency anemia

Approximately half of non-pregnant women are anemic (see Table 32). Less than 2% of non-pregnant women are severely anemic, whereas moderate and mild anemia is present in 23% and 27% of women, respectively (see Table 31). Anemia in women was not associated with age, urban rural residence, LGA, woman's education, household wealth and household sanitation (Table 32). The distribution of hemoglobin concentration is shown in Figure 16, showing that hemoglobin is almost normally distributed except for a slight tail of lower hemoglobin concentrations.

ID and IDA affect about 41% and 28% of non-pregnant women, respectively. There is an overlap of anemia and ID (see Figure 17), and among women with anemia, 55.5% had concurrent ID (data not shown). A significantly higher proportion of women living in rural areas have ID compared to those in urban areas. Moreover, highest prevalence can be found in Kuntaur and Mansakonko and lowest in Kanifing and Banjul. Almost of 50% of women who never attended school have ID, compared to only to 23% of those attending upper secondary school. The prevalence of ID continuously decreases with increasing household wealth, and the ID prevalence was significantly higher among women living in households with inadequate sanitation (Table 32).

Similar to ID, a higher proportion of women living in rural areas have IDA compared to those living in urban centers. The IDA prevalence in Kuntaur and Mansakonko is more than twice as high as in Banjul and Kanifing. Moreover, the highest prevalence of IDA can be found in women who never attended school; a prevalence two-times higher than found in women who completed upper secondary school. Similar to ID, the prevalence of IDA gradually increased with decreasing household wealth (Table 32).

Table 30. Percentage of non-pregnant women with malaria based on PCR analysis, The Gambia 2018

		Malaria [a]		
Characteristic	n	% [b]	(95% CI) [c]	P-value [d]
Age Group (in years)				
15-19	19	5.9	(2.8, 11.7)	<0.05
20-24	11	6.2	(2.8, 13.2)	
25-29	19	12.7	(5.7, 26.0)	
30-34	7	3.5	(1.3, 9.1)	
35-39	8	4.7	(1.6, 12.8)	
40-44	6	3.6	(1.2, 10.2)	
45-49	2	0.7	(0.1, 3.7)	
Residence				
Urban	37	5.2	(2.3, 11.4)	0.473
Rural	36	7.6	(3.8, 14.6)	
LGA				
Banjul	7	2.8	(1.0, 7.7)	0.054
Kanifing	6	2.7	(1.0, 6.9)	
Brikama	6	5.6	(1.3, 20.8)	
Mansakonko	2	1.3	(0.4, 4.3)	
Kuntaur	6	6.1	(1.2, 25.5)	
Kerewan	31	18.0	(6.6, 40.8)	
Janjanbureh	6	4.0	(2.0, 7.6)	
Basse	9	5.3	(2.6, 10.3)	
Wealth Quintile				
Lowest	22	8.3	(3.7, 17.4)	0.270
Second	11	5.4	(2.7, 10.6)	
Middle	11	5.4	(2.4, 11.5)	
Fourth	22	12.1	(3.9, 31.8)	
Highest	7	1.2	(0.3, 4.1)	
ALL WOMEN	73	6.0	(3.4, 10.4)	--

Note: The n's are un-weighted numerators for each subgroup; subgroups that do not sum to the total have missing data.

[a] Malaria detected via detection of the cytochrome oxidase III (COX-III) gene of *Plasmodium* parasites

[b] Percentages weighted for unequal probability of selection

[c] CI=confidence interval, calculated taking into account the complex sampling design

[d] P-value <0.05 indicates that the variation in the values of the subgroup are significantly different from all other subgroups

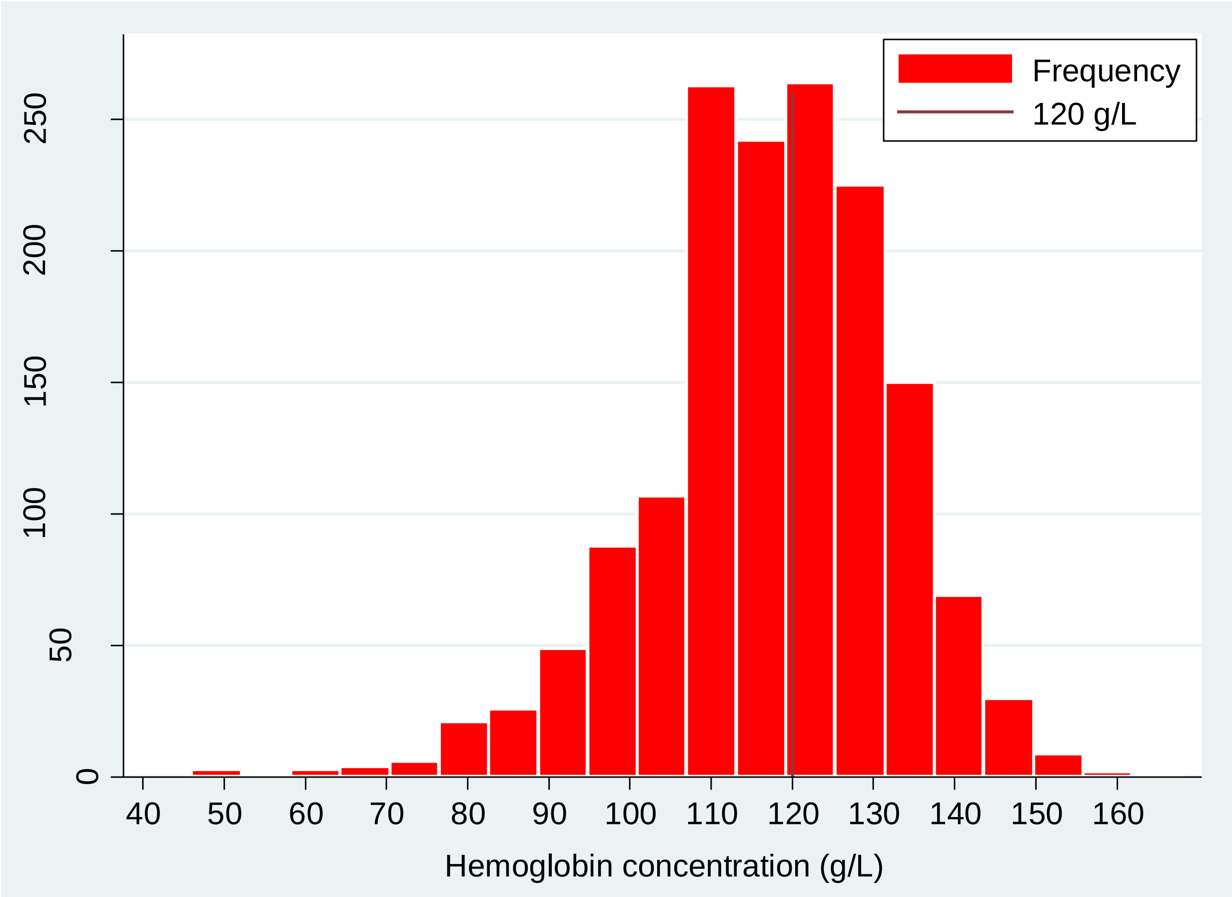


Figure 16. Histogram of hemoglobin concentration (g/L) in non-pregnant women of reproductive age, The Gambia 2018

Table 31. Proportion of mild, moderate and severe anemia in non-pregnant women, The Gambia 2018

	Mild anemia [b]			Moderate anemia [b]			Severe anemia [b]		
Characteristic	n	%[a]	(95% CI)[c]	n	%[a]	(95% CI)[c]	n	%[a]	(95% CI)[c]
Age Group (in years)									
15-19	88	23.3	(17.6, 30.1)	58	16.1	(11.5, 22.2)	7	1.6	(0.6, 3.9)
20-24	64	28.9	(23.1, 35.5)	59	22.1	(15.5, 30.4)	3	0.5	(0.2, 1.8)
25-29	56	28.6	(19.3, 40.3)	55	23.5	(15.6, 33.9)	3	3.1	(0.5, 16.0)
30-34	47	31.0	(19.4, 45.7)	60	26.4	(17.9, 37.2)	3	0.9	(0.3, 2.7)
35-39	46	18.2	(11.2, 28.3)	47	27.6	(18.1, 39.8)	2	1.0	(0.2, 4.0)
40-44	36	36.1	(22.1, 52.9)	26	25.7	(20.9, 31.1)	2	0.6	(0.1, 3.7)
45-49	23	22.4	(13.8, 34.3)	19	25.9	(15.6, 39.8)	1	0.8	(0.1, 6.0)
Residence									
Urban	185	27.2	(22.2, 33.0)	153	21.7	(17.4, 26.7)	7	1.0	(0.3, 3.2)
Rural	176	25.4	(21.6, 29.6)	173	26.1	(21.5, 31.2)	14	2.0	(1.0, 4.3)
LGA									
Banjul	43	20.0	(15.0, 26.1)	32	11.1	(6.0, 19.5)	2	1.3	(0.3, 4.6)
Kanifing	47	22.8	(16.7, 30.3)	43	20.7	(16.6, 25.5)	1	0.5	(0.1, 3.7)
Brikama	53	30.1	(22.5, 39.0)	40	22.4	(15.9, 30.6)	2	1.1	(0.2, 5.5)
Mansakonko	55	34.0	(27.0, 41.8)	42	24.7	(18.5, 32.1)	1	0.6	(0.1, 4.1)
Kuntaur	35	34.1	(22.3, 48.2)	35	30.2	(21.1, 41.1)	4	2.7	(0.8, 8.5)
Kerewan	31	19.4	(12.2, 29.4)	23	16.7	(10.2, 26.2)	3	1.8	(0.6, 5.3)
Janjanbureh	50	25.1	(17.5, 34.8)	50	24.6	(20.5, 29.2)	2	1.0	(0.3, 3.7)
Basse	47	23.7	(19.4, 28.5)	61	30.5	(21.0, 42.0)	6	2.9	(0.8, 9.4)
Woman's Education									
Never at school	161	30.0	(25.8, 34.5)	159	27.3	(22.9, 32.1)	11	1.0	(0.4, 2.5)
Primary school	47	19.1	(11.0, 31.2)	53	25.9	(18.6, 34.8)	3	1.1	(0.3, 4.4)
Lower Secondary	76	27.8	(17.5, 41.0)	47	16.1	(9.7, 25.5)	5	3.3	(0.9, 11.5)
Upper Secondary	52	24.9	(18.2, 33.0)	37	19.0	(12.1, 28.7)	2	0.6	(0.2, 1.6)
Higher	15	23.2	(12.9, 38.2)	24	25.0	(13.9, 40.8)	0	-	-
Wealth quintile									
Lowest	96	28.5	(21.3, 37.1)	103	29.1	(25.1, 33.6)	10	2.4	(1.1, 5.3)
Second	79	33.1	(25.6, 41.6)	56	18.6	(14.4, 23.6)	4	1.1	(0.4, 3.0)
Middle	61	23.1	(19.1, 27.6)	61	25.1	(21.2, 29.5)	3	2.1	(0.5, 8.1)
Fourth	43	21.4	(13.6, 32.1)	43	24.3	(13.2, 40.4)	2	1.0	(0.2, 4.5)
Highest	82	27.5	(22.2, 33.6)	63	18.9	(13.4, 26.1)	2	0.1	(0.0, 0.3)
Household sanitation [f]									
Inadequate	213	27.0	(22.4, 32.3)	184	23.4	(19.0, 28.4)	12	1.6	(0.6, 4.2)
Adequate	148	26.3	(22.4, 30.6)	142	22.4	(17.7, 27.9)	9	1.0	(0.4, 2.3)
ALL WOMAN	361	26.7	(22.9, 30.9)	326	22.9	(19.5, 26.7)	21	1.3	(0.7, 2.6)

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] Mild, moderate, and severe anemia defined as hemoglobin 110-119 g/L, 80-109 g/L, and <80 g/L, respectively; after adjusting hemoglobin for smoking.
[c] CI=confidence interval, calculated taking into account the complex sampling design.
[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

Table 32. Anemia, iron deficiency, and iron deficiency anemia in non-pregnant women (15-49 years), The Gambia 2018

	Anemia [b]				Iron deficiency [e]				Iron deficiency anemia [f]			
Characteristic	n	%[a]	(95% CI)[c]	P value[d]	n	%[a]	(95% CI)[c]	P value[d]	n	%[a]	(95% CI)[c]	P value[d]
Age Group (in years)												
15-19	153	41.0	(34.6, 47.7)	0.06	140	36.7	(29.8, 44.1)	0.20	96	22.5	(17.9, 27.9)	0.35
20-24	126	51.5	(42.9, 60.1)		133	47.4	(39.8, 55.0)		83	26.9	(15.0, 43.5)	
25-29	114	55.3	(44.4, 65.7)		101	51.9	(39.7, 63.8)		72	39.3	(25.7, 54.8)	
30-34	110	58.3	(45.6, 70.0)		96	41.5	(31.2, 52.5)		71	33.1	(24.4, 43.2)	
35-39	95	46.8	(39.6, 54.2)		82	33.9	(23.6, 46.0)		58	24.2	(14.3, 37.9)	
40-44	64	62.4	(45.5, 76.7)		47	38.4	(25.9, 52.7)		30	22.3	(12.0, 37.7)	
45-49	43	49.1	(36.6, 61.7)		30	36.9	(23.8, 52.3)		22	28.6	(13.9, 49.9)	
Residence												
Urban	345	49.9	(42.8, 57.1)	0.45	285	37.0	(32.3, 41.9)	<0.001	176	24.4	(20.3, 29.0)	<0.005
Rural	363	53.5	(47.2, 59.7)		347	52.6	(45.2, 60.0)		258	37.1	(30.6, 44.2)	
LGA												
Banjul	77	32.3	(24.2, 41.7)	0.05	75	34.2	(28.9, 39.9)	<0.005	40	17.0	(11.5, 24.6)	<0.005
Kanifing	91	44.1	(37.6, 50.7)		63	32.5	(25.7, 40.1)		37	19.4	(13.3, 27.3)	
Brikama	95	53.6	(42.1, 64.8)		64	37.1	(30.4, 44.2)		47	26.0	(20.0, 33.1)	
Mansakonko	98	59.3	(54.3, 64.1)		89	55.1	(43.9, 65.8)		69	42.5	(32.8, 52.8)	
Kuntaur	74	67.0	(60.0, 73.3)		68	61.4	(51.8, 70.1)		54	49.7	(41.8, 57.6)	
Kerewan	57	37.9	(25.2, 52.6)		64	44.1	(33.2, 55.6)		36	24.6	(14.5, 38.6)	
Janjanbureh	102	50.8	(40.0, 61.5)		102	51.6	(43.6, 59.6)		73	33.7	(25.6, 42.9)	
Basse	114	57.1	(47.0, 66.6)		107	54.1	(39.7, 67.8)		78	38.5	(27.0, 51.4)	
Woman's Education												
Never at school	331	58.2	(53.6, 62.8)	0.06	299	49.4	(43.2, 55.6)	<0.005	210	34.0	(29.2, 39.2)	<0.005
Primary school	103	46.1	(35.1, 57.4)		108	45.1	(37.6, 52.9)		71	31.4	(24.3, 39.6)	
Lower Secondary	128	47.2	(36.6, 58.1)		111	40.8	(32.5, 49.7)		78	27.2	(20.1, 35.7)	
Upper Secondary	91	44.5	(33.4, 56.2)		71	22.7	(17.9, 28.2)		45	13.5	(8.5, 20.7)	
Higher	39	48.2	(36.5, 60.2)		30	38.1	(20.1, 60.0)		20	26.1	(13.8, 43.8)	

Table continued on next page

	Anemia [b]				Iron deficiency [e]				Iron deficiency anemia [f]			
Characteristic	**n**	**%[a]**	**(95% CI)[c]**	**P value[d]**	**n**	**%[a]**	**(95% CI)[c]**	**P value[d]**	**n**	**%[a]**	**(95% CI)[c]**	**P value[d]**
Wealth Quintiles												
Lowest	209	60.1	(51.5, 68.1)	0.36	201	53.7	(40.4, 66.4)	<0.05	155	40.3	(29.6, 52.0)	<0.05
Second	139	52.7	(43.2, 62.1)		136	47.2	(36.4, 58.2)		91	30.1	(24.5, 36.3)	
Middle	125	50.3	(44.2, 56.3)		113	43.3	(36.4, 50.5)		78	29.2	(23.3, 35.8)	
Fourth	88	46.7	(28.9, 65.4)		77	38.2	(31.3, 45.6)		50	25.1	(16.5, 36.1)	
Highest	147	46.5	(39.6, 53.7)		105	27.0	(17.5, 39.1)		60	17.7	(10.3, 28.7)	
Household sanitation [e]												
Inadequate	409	52.0	(46.2, 57.8)	0.51	376	46.8	(41.8, 52.0)	<0.01	264	31.5	(26.0, 37.6)	0.09
Adequate	299	49.7	(42.6, 56.8)		256	35.2	(29.1, 41.8)		170	23.9	(18.5, 30.2)	
ALL WOMEN	708	50.9	(45.5, 56.4)		632	41.4	(37.6, 45.4)		434	28.0	(24.4, 31.9)	

[a] Percentages weighted for unequal probability of selection.

[b] Anemia defined as hemoglobin < 120 g/L adjusted for smoking.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups.

[e] Iron deficiency defined as serum ferritin < 12 µg/l, values are adjusted for inflammation according to BRINDA

[f] Iron deficiency anemia, defined as low Hb (< 110 g/L) with low serum ferritin (< 15.0 µg/L).

[g] Composite variable of toilet type and if toilet facilities are shared with non-household members; Adequate Sanitation = flush or pour flush toilet or pit latrine with slab not shared with another household. Inadequate sanitation= open pit, bucket latrine, hanging toilet/latrine, no facility, bush, field

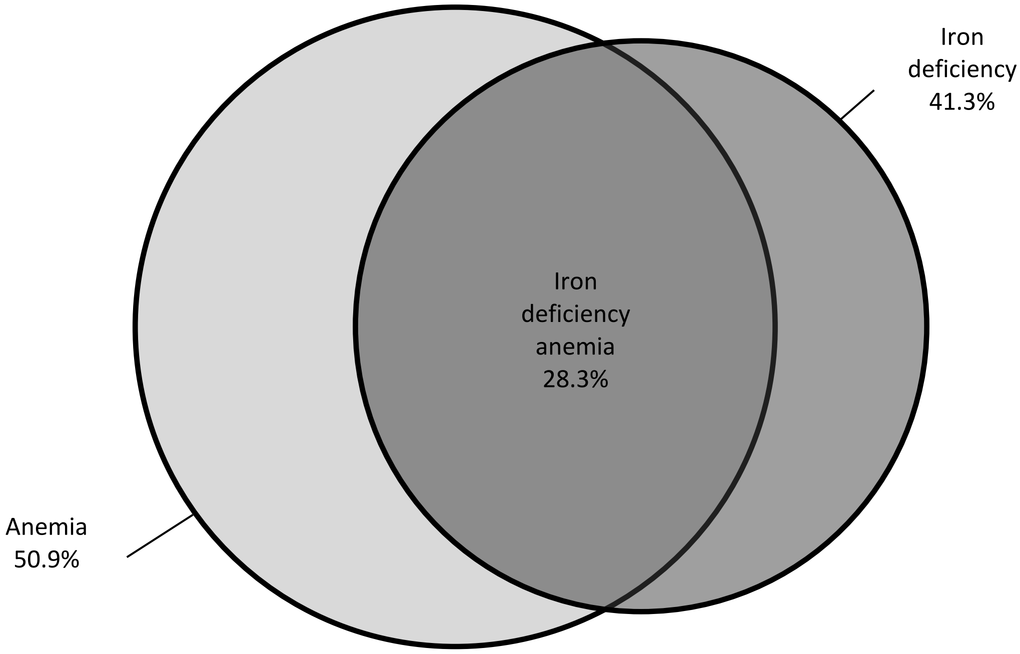


Figure 17. Venn diagram showing overlap between anemia and iron deficiency in non-pregnant women of reproductive age, The Gambia 2018

Similar to children, ID and VAD are the main predictors of anemia in women. Neither inflammation nor malaria are determinants of anemia, indicating that most of the anemia found in women is nutritionally induced. Both ID and IDA are strongly associated with VAD, but not with inflammation. Bivariate analyses showed that only IDA is associated with nutritional status, with a higher proportion of women having IDA compared to normal and overweight women (Table 33).

3.5.4. Vitamin A deficiency

Less than 2% of non-pregnant women have vitamin A deficiency when using RBP as the indicator. Age, education, minimum dietary diversity, and household sanitation are not significantly associated with vitamin A deficiency. Statistically, significantly higher prevalences are observed in women residing in households from the lowest wealth quintile, rural areas, and in the Kuntaur, Mansakonko, and Basse LGAs.

Table 33. Anemia, iron deficiency, and iron deficiency anemia in non-pregnant women (15-49 years) by micronutrient status, supplement consumption, malaria status, inflammation and nutritional status, The Gambia 2018

	Anemia [b]				Iron deficiency [e]				Iron deficiency anemia [f]			
Characteristic	n	%[a]	(95% CI)[c]	P value[d]	n	%[a]	(95% CI)[c]	P value[d]	n	%[a]	(95% CI)[c]	P value[d]
Iron status [e]												
Deficient (<15 µg/l)	434	67.8	(60.8, 74.1)	<0.0001								
Sufficient (≥15µg/L)	243	37.4	(30.1, 45.4)									
Vitamin A status [g]												
Deficient (<0.7nmol/L)	38	88.4	(72.5, 95.7)	<0.0005	32	74.7	(53.9, 88.2)	<0.005	31	74.0	(53.3, 87.7)	<0.0001
Sufficient (≥0.70nmol/L)	639	49.3	(43.7, 54.9)		600	40.8	(37.0, 44.8)		403	27.3	(23.7, 31.2)	
Inflammation [h]												
None	554	50.0	(43.4, 56.7)	0.44	533	42.2	(37.7, 46.9)	0.83	367	28.9	(24.8, 33.4)	0.60
Incubation	44	40.8	(29.4, 53.2)		44	38.6	(23.9, 55.6)		26	20.3	(11.7, 32.8)	
Early convalescence	42	55.2	(38.7, 70.6)		31	36.7	(25.9, 49.0)		21	25.0	(12.6, 43.6)	
Late convalescence	37	60.9	(38.8, 79.3)		24	37.6	(18.1, 62.1)		20	34.0	(15.0, 60.0)	
Nutritional status												
Underweight (BMI< 18.5)	127	53.1	(41.3, 64.6)	0.70	100	42.3	(33.1, 52.0)	0.08	77	34.4	(25.0, 45.1)	<0.05
Normal weight (BMI 18.5-24.9)	395	51.5	(45.5, 57.4)		375	45.0	(40.5, 49.5)		268	30.6	(26.7, 34.9)	
Overweight/Obesity (BMI >25)	182	49.2	(41.3, 57.2)		154	34.4	(26.3, 43.4)		87	20.3	(14.9, 27.0)	
Minimal dietary diversity												
No (0-4)	192	52.7	(42.8, 62.4)	0.53	178	43.1	(37.8, 48.5)	0.51	113	29.5	(23.8, 35.9)	0.43
No (≥5)	503	49.8	(44.9, 54.8)		445	40.7	(35.8, 45.8)		314	27.3	(23.7, 31.1)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] Anemia defined as hemoglobin < 120 g/L adjusted for smoking.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups.

[e] Iron deficiency defined as serum ferritin < 12 µg/l, values are adjusted for inflammation according to BRINDA

[f] Iron deficiency anemia, defined as low Hb (< 110 g/L) with low serum ferritin (< 15.0 µg/L).

[g] VAD= Vitamin A deficiency, defined as low retinol binding protein (<0.7 nmol/L), was not adjusted for inflammation according to BRINDA recommendations

[h] Incubation=CRP only; early convalescence=CRP and AGP; late convalescence=AGP only.

Table 34. Vitamin A deficiency in non-pregnant women (15-49 years), The Gambia 2018

Characteristic	n	%[a, b]	(95% CI)[c]	P value[d]
Age Group (in years)				
15-19	11	2.5	(1.2, 5.1)	0.15
20-24	8	1.7	(0.6, 4.5)	
25-29	9	2.4	(1.0, 5.4)	
30-34	10	3.3	(1.5, 7.2)	
35-39	5	0.9	(0.4, 2.2)	
40-44	1	0.2	(0.0, 1.3)	
45-49	0	-	-	
Residence				
Urban	7	0.6	(0.3, 1.4)	<0.0001
Rural	37	4.9	(2.9, 8.0)	
LGA				
Banjul	0	-	-	<0.0001
Kanifing	3	1.6	(0.6, 4.0)	
Brikama	1	0.1	(0.0, 0.6)	
Mansakonko	9	4.9	(1.7, 13.2)	
Kuntaur	10	7.5	(3.9, 13.7)	
Kerewan	3	1.8	(0.6, 5.6)	
Janjanbureh	7	3.5	(1.0, 11.0)	
Basse	11	5.3	(2.1, 13.0)	
Woman's Education				
Never at school	26	2.7	(1.6, 4.6)	0.14
Primary school	8	2.9	(1.2, 6.9)	
Lower Secondary	9	1.4	(0.6, 3.5)	
Upper Secondary	0	-	-	
Higher	1	1.1	(0.1, 8.3)	
Wealth quintile				
Lowest	25	5.7	(3.4, 9.4)	<0.0005
Second	13	2.6	(1.2, 5.6)	
Middle	3	0.4	(0.1, 1.5)	
Fourth	1	0.7	(0.1, 5.0)	
Highest	2	0.9	(0.2, 3.5)	
Minimal dietary diversity				
No (0-4)	7	1.3	(0.6, 2.8)	0.31
Yes (≥5)	37	2.1	(1.3, 3.5)	
Household sanitation [e]				
Inadequate	29	2.1	(1.2, 3.5)	0.57
Adequate	15	1.6	(0.7, 3.4)	
ALL WOMEN	44	1.8	(1.2, 2.8)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] Vitamin A deficiency defined as retinol binding protein (RBP) <0.70 µmol/L; RBP concentrations adjusted for inflammation.
[c] CI=confidence interval, calculated taking into account the complex sampling design.
[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

3.5.5. Median urinary iodine concentration

As shown in Table 35, median urinary iodine concentration is 143.1 ug/L, and significantly higher concentrations are found in women 15-19 years old, those with higher levels of education, and those in the highest wealth quintile. In addition, women residing in households with adequately iodized salt have median urinary iodine concentrations of nearly 220 ug/L. No statistically significant differences in iodine concentrations are found by residence or LGA.

Table 35. Median urinary iodine concentration in non-pregnant women (15-49 years), The Gambia 2018

Characteristic	n	Median UIC [a, b]	(IQR)[c]	P Value[d]
Age Group (in years)				
15-19	443	170.3	(101.4, 257.8)	<0.001
20-24	350	126.7	(64.9, 195.6)	
25-29	318	133.3	(77.9, 218.0)	
30-34	266	148.5	(84.6, 231.1)	
35-39	242	131.5	(77.4, 205.3)	
40-44	133	143.8	(81.4, 223.6)	
45-49	109	132.9	(87.7, 202.3)	
Residence				
Urban	996	166.3	(108.3, 246.3)	0.283
Rural	877	113.7	(61.5, 185.5)	
LGA				
Banjul	274	163.9	(110.8, 231.7)	0.936
Kanifing	219	177.5	(114.0, 257.8)	
Brikama	262	162.1	(99.1, 240.7)	
Mansakonko	198	99.0	(55.2, 170.7)	
Kuntaur	186	98.7	(48.2, 135.3)	
Kerewan	222	122.8	(76.6, 195.3)	
Janjanbureh	261	118.1	(60.3, 179.0)	
Basse	251	180.1	(110.6, 313.2)	
Woman's Education				
Never at school	803	131.9	(69.1, 212.3)	<0.01
Primary school	298	134.3	(83.2, 206.3)	
Lower Secondary	347	150.6	(82.2, 246.8)	
Upper Secondary	274	162.5	(104.0, 240.7)	
Higher	116	189.5	(117.2, 248.5)	

Table continued on next page

Characteristic	n	Median UIC [a, b]	(IQR)[c]	P Value[d]
Wealth quintile				
Lowest	475	86.9	(48.9, 148.8)	<0.001
Second	372	143.7	(84.4, 233.6)	
Middle	336	159.0	(101.4, 252.6)	
Fourth	256	156.7	(92.3, 236.0)	
Highest	434	173.8	(114.0, 242.7)	
Adequately iodized salt in household[d]				
No	1088	136.8	(79.5, 217.9)	<0.01
Yes	94	217.9	(136.2, 389.7)	
ALL WOMEN [a]	1287	143.1	(82.1, 229.6)	

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Median's unweighted; UIC = urinary iodine concentration

[b] IQR = Interquartile range, calculated using unweighted data.

[c] P-value <0.05 indicates that the geometric mean in at least one subgroup is statistically significantly different from the values in the other subgroups

[d] Adequately iodized salt > 15 ppm

3.5.6. Hyperglycemia

As shown in Table 36, approximately 8% of women were classified as being hyperglycemic, and 46% were classified with intermediate hyperglycemia. Hyperglycemia was significantly associated with age and nutritional status, with the highest prevalences found among women 45-49 years of age and obese women. Of note, the prevalence of hyperglycemia exceeded 10% in underweight woman. No significant difference in hyperglycemia was found by residence, LGA, wealth quintile, physical activity status, or hypertension status.

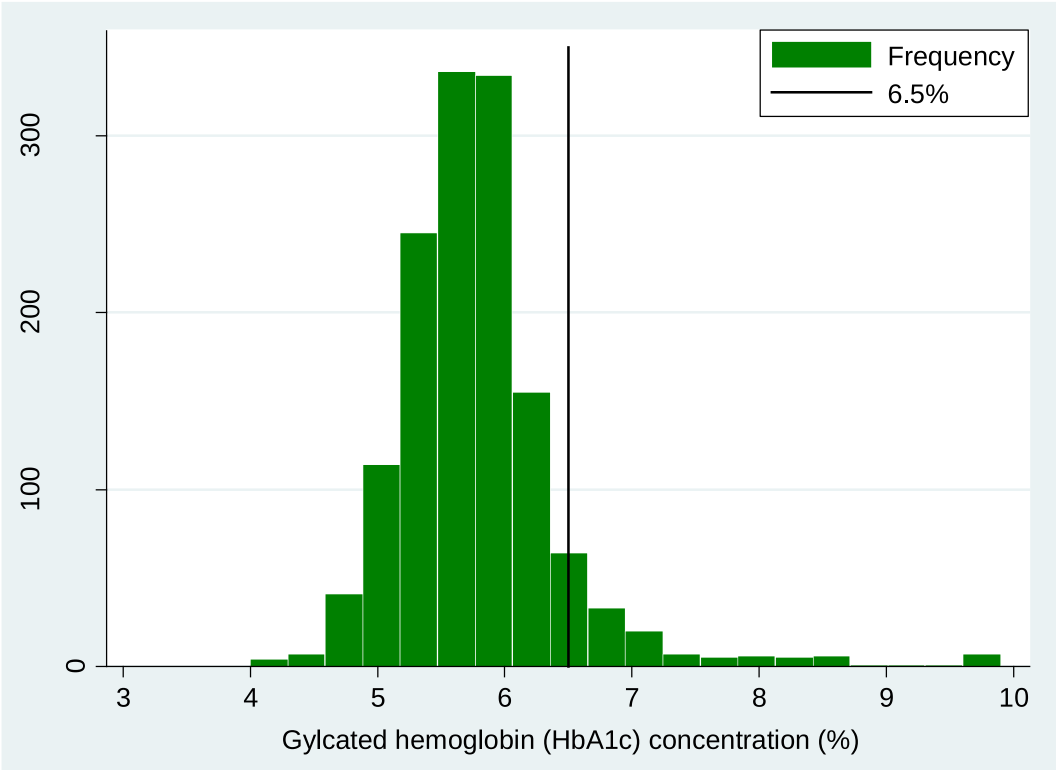


Figure 18. Distribution of glycated hemoglobin concentrations in non-pregnant women of reproductive age, The Gambia 2018

Table 36. Hyperglycemia in non-pregnant women (15-49 years), The Gambia 2018

	Hyperglycemia[a]				Intermediate Hyperglycemia[a]			
Characteristic	n	%[b]	(95% CI)[c]	P Value[d]	n	%[b]	(95% CI)[c]	P Value[d]
Age Group (in years)								
15-19	23	4.4	(2.3, 8.1)	<0.05	140	46.1	(38.0, 54.4)	0.49
20-24	26	7.9	(4.9, 12.5)		110	44.1	(36.7, 51.9)	
25-29	19	8.8	(4.5, 16.8)		93	42.2	(31.5, 53.6)	
30-34	21	5.7	(3.0, 10.5)		86	47.5	(34.6, 60.6)	
35-39	27	12.5	(6.8, 22.0)		90	44.1	(31.9, 57.1)	
40-44	8	5.2	(2.0, 12.8)		60	58.7	(47.2, 69.4)	
45-49	13	18.9	(8.8, 36.0)		41	46.3	(34.6, 58.4)	
Residence								
Urban	72	7.2	(4.2, 12.0)	0.22	344	46.4	(38.1, 54.9)	0.76
Rural	67	10.3	(7.7, 13.7)		282	46.3	(42.1, 50.5)	
LGA								
Banjul	29	13.6	(7.4, 23.8)	0.14	101	43.6	(34.6, 53.0)	0.21
Kanifing	15	7.8	(4.8, 12.4)		69	37.2	(28.2, 47.3)	
Brikama	16	7.1	(3.1, 15.8)		96	50.6	(37.9, 63.2)	
Mansakonko	5	2.8	(1.5, 5.3)		63	38.3	(29.1, 48.5)	
Kuntaur	22	19.8	(11.6, 31.8)		55	47.8	(37.2, 58.7)	
Kerewan	18	12.9	(7.7, 20.8)		74	50.4	(45.5, 55.3)	
Janjanbureh	23	8.9	(4.1, 18.5)		86	47.9	(38.5, 57.4)	
Basse	11	5.4	(2.9, 10.0)		82	44.5	(36.9, 52.5)	

Table continued on next page

Characteristic	Hyperglycemia[a] n	%[b]	(95% CI)[c]	P Value[d]	Intermediate Hyperglycemia[a] n	%[b]	(95% CI)[c]	P Value[d]
Wealth quintile								
Lowest	40	11.9	(7.6, 18.4)	0.57	162	50.2	(44.3, 56.1)	0.19
Second	26	7.8	(4.1, 14.5)		121	49.6	(35.6, 63.6)	
Middle	13	7.5	(2.8, 18.6)		110	52.6	(42.2, 62.7)	
Fourth	22	5.9	(3.3, 10.4)		93	49.0	(37.9, 60.2)	
Highest	38	7.6	(4.3, 12.9)		140	31.8	(16.0; 53.2)	
Physical activity[e]								
Physically active	22	8.3	(3.3, 19.0)	0.96	84	55.0	(46.5, 64.1)	0.05
Physically inactive	117	8.1	(5.6,11.4)		538	45.2	(38.7, 51.9)	
Nutritional status								
Underweight (BMI<18.5)	22	10.9	(5.4, 20.9)	<0.01	99	49.7	(39.9, 59.5)	<0.05
Normal weight (BMI 18.5-24.9)	56	5.1	(3.3, 7.9)		336	41.8	(33.3, 50.9)	
Overweight (BMI 25-29.9)	33	8.3	(5.0, 13.5)		116	59.7	(47.0, 71.3)	
Obesity (>29.9)	25	15.7	(8.7, 26.5)		73	44.5	(36.1, 53.2)	
Hypertension[f]								
Yes	12	4.2	(1.8, 9.4)	0.13	55	59.4	(46.3, 71.3)	<0.05
No	124	8.1	(5.6, 11.7)		570	45.3	(39.0, 51.8)	
ALL WOMEN	139	8.1	(5.7, 11.3)		626	46.4	(40.2, 52.7)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Hyperglycemia defined as HbA1c ≥6.5%; intermediate hyperglycemia 5.7%-6.4% [34]

[b] Percentages weighted for unequal probability of selection.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] Chi-square p-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups. P-values based on proportions for a) hyperglycemia compared to all other stages (i.e. intermediate and normal combined), and b) intermediate hyperglycemia compared to normal status

[e] physically active if > 150 min vigorous/ moderate intensity work/ per week (based on WHO recommendations on physical activity for health [38])

[f] Hypertension defined as systolic blood pressure ≥ 140mm Hg and diastolic blood pressure ≥ 90 mm Hg

3.5.7. Hypertension

Approximately 10% of non-pregnant women displayed hypertension (systolic blood pressure 140 mm Hg; diastolic blood pressure 90 mm Hg) at the time of the survey (see Table 37).

Table 37. Hypertension in non- pregnant women (15- 49 years), The Gambia 2018

		Hypertension [b]		
Characteristic	n	%[a]	(95% CI)[c]	P Value[d]
Age Group (in years)				
15-19	17	4.2	(1.5, 11.6)	<0.001
20-24	18	6.3	(2.3, 16.1)	
25-29	13	3.9	(1.7, 8.4)	
30-34	34	14.5	(8.9, 22.8)	
35-39	33	15.1	(8.4, 25.7)	
40-44	32	23.2	(9.8, 45.5)	
45-49	25	31.2	(20.7, 44.0)	
Residence				
Urban	87	10.0	(6.4, 15.4)	0.295
Rural	86	13.3	(8.7, 19.9)	
LGA				
Banjul	25	12.2	(8.0, 18.3)	<0.05
Kanifing	8	4.9	(2.5, 9.4)	
Brikama	25	11.3	(6.3, 19.3)	
Mansakonko	15	8.0	(5.4, 11.6)	
Kuntaur	9	3.9	(1.9, 8.2)	
Kerewan	38	20.0	(9.2, 38.3)	
Janjanbureh	16	7.1	(6.2, 8.1)	
Basse	37	14.7	(10.4, 20.5)	
Wealth quintile				
Lowest	42	11.8	(7.5, 18.0)	<0.005
Second	35	9.4	(4.7, 17.9)	
Middle	42	16.5	(12.1, 22.2)	
Fourth	25	10.0	(6.6, 15.0)	
Highest	29	5.3	(3.7, 7.4)	
ALL WOMEN	173	10.6	(7.6, 14.4)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] Hypertension defined as systolic blood pressure ≥ 140 and diastolic blood pressure ≥ 90.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

The prevalence of hypertension increases by age group; few women 15-29 years are hypertensive, with a marked increase in hypertension occurring at 30 years of age. Significant differences are also found by LGA, with Banjul, Brikama, Kerewan, and Basse displaying higher levels of hypertension. Significant differences in hypertension were also found by wealth quintile, with the highest wealth quintile displaying less hypertension.

As shown in Table 38, hypertension is not associated with physical activity, but is significantly higher in overweight and obese women compared to those that are underweight or normal weight. In addition, the prevalence of hypertension is higher among women with intermediate hyperglycemia. The prevalence of hypertension was also examined by women's smoking status and alcohol consumption. However, the current practice of smoking (n=2) and consumption of alcohol in the past month (n=3) were both very rare, and thus the sub-group analysis showing the proportion of women with hypertension that engaged in these practices was not calculated.

Table 38. Hypertension in non- pregnant women (15 - 49 years) by lifestyle and nutritional status, The Gambia 2018

		Hypertension		
Characteristic	n	%[a, b]	(95% CI)[c]	P Value[d]
Physical activity[e]				
Physically active	14	5.4	(1.9, 14.9)	0.13
Physically inactive	154	11.3	(8.4, 15.0)	
Nutritional status				
Underweight (BMI<18.5)	11	2.9	(1.2, 6.9)	<0.001
Normal weight (BMI 18.5-24.9)	50	5.7	(3.3, 9.5)	
Overweight (BMI 25-29.9)	42	15.9	(9.6, 25.2)	
Obesity (>29.9)	37	19.5	(11.5, 31.2)	
Hyperglycemia[f]				
Normal	47	6.5	(3.9, 10.8)	<0.05
Intermediate hyperglycemia	55	10.5	(7.1, 15.2)	
Hyperglycemia	12	4.4	(1.9, 9.9)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] Hypertension defined as systolic blood pressure ≥ 140mm Hg and diastolic blood pressure ≥ 90 mm Hg.
[c] CI=confidence interval, calculated taking into account the complex sampling design.
[d] Chi-square p-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups
[e] Physically active if > 150 min vigorous/moderate intensity work per week (based on WHO recommendations on physical activity for health[38])
[f] Hyperglycemia defined as HbA1C ≥ 6.5%

3.6. Pregnant women

3.6.1. Characteristics

In Table 39, the characteristics of the pregnant women participating in the survey are described. Approximately equal number of pregnant women are recruited from urban and rural areas, and notably more pregnant women reside in households in the lowest wealth quintile. No age restriction is used for the recruitment of pregnant women, and the age range of pregnant women surveyed is 15.1 to 44.9 years. Most pregnant women included in the survey sample were between 20 and 29 years of age, and only two women are older than 40 years of age. As such, the age groups 30-39 and 40-44 have been merged for the following tables in Section 3.6.

Table 39. Description of pregnant women, Gambia 2018

	Survey Sample		
Characteristic	**n**	**%** [a]	**(95% CI)** [b]
Age Group (in years)			
15-19	25	15.8	(11, 22.2)
20-29	87	55.1	(47.7, 62.2)
30-39	44	27.8	(21, 35.9)
40-44	2	1.3	(0.3, 5.1)
Residence			
Urban	78	49.4	(34.2, 64.6)
Rural	80	50.6	(35.4, 65.8)
LGA			
Banjul	24	15.2	(7.3, 28.9)
Kanifing	11	7	(2.7, 17.0)
Brikama	17	10.8	(4.5, 23.4)
Mansakonko	18	11.4	(5.0, 23.9)
Kuntaur	26	16.5	(7.3, 33.1)
Kerewan	13	8.2	(3.5, 18.1)
Janjanbureh	30	19	(8.4, 37.3)
Basse	19	12	(4.9, 26.7)
Wealth quintile			
Lowest	54	34.2	(22.1, 48.7)
Second	33	20.9	(12.4, 33.0)
Middle	24	15.2	(8.2, 26.3)
Fourth	21	13.3	(7.6, 22.2)
Highest	26	16.5	(9.0, 28.3)
ALL WOMEN	158	100.0	--

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages are un-weighted and do not account for unequal probability of selection.

[b] CI=confidence interval, calculated taking into account the complex sampling design

3.6.2. Mid-upper arm circumference

As shown in Table 40, nearly 8% of pregnant woman had a MUAC <23 cm and are therefore considered undernourished. The proportion was markedly higher in pregnant women 15-19 years of age.

Mean MUAC among all pregnant women was nearly 27cm. Women 15-19 years of age had the lowest mean of any sub-group (25.1 cm), and rural pregnant women had lower mean MUAC (26.0 cm) than that found in urban pregnant women (27.6 cm) (data not shown).

Table 40. Percentage undernourished by various characteristics in pregnant women, The Gambia 2018

Characteristic	n	% under-nourished [a, b]	(95% CI) [b]	P value [c]
Age Group (in years)				
15-19	4	22.2	(8.1, 48.0)	<0.05
20-29	4	6.1	(2.4, 14.6)	
30-44	1	2.8	(0.4, 15.3)	
Residence				
Urban	3	6.0	(2.0, 16.8)	0.58
Rural	6	8.6	(4.1, 16.9)	
ALL WOMEN	9	7.5	(4.1, 13.4)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] %Undernourished= % of women with MUAC < 23 cm
[c] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

3.6.3. Malaria

Malaria in pregnant women is rare. Only three pregnant women were found with malaria, resulting in a malaria prevalence of 2.3% (95% CI: 0.7, 6.8). The three women reside in Kanifing, Kuntaur, and Janjanbureh, respectively. Two of the women are 15-19 years of age, whereas the third woman is 20-29 years of age. Due to the very small number of women with malaria, no subgroup analysis of malaria was conducted. As blood samples were only collected on-site from pregnant women (i.e. no blood tube was collected), no blood samples were available in pregnant women for detection of *Plasmodium* genes using PCR analysis.

3.6.4. Anemia, urinary iodine and hypertension

As shown in Table 41, nearly 57% of pregnant women are diagnosed anemic, posing a severe public health problem according to WHO classification [11]. No significant differences were observed for age nor residence. As shown in Figure 19, hemoglobin concentration is normally distributed among pregnant women, however, a large number of women have concentration around 110 g/L, the threshold for anemia in pregnant women.

Table 41. Anemia in pregnant women, The Gambia 2018

Characteristic	n	Anemia %[a, b]	(95% CI)[c]	P Value[d]
Age Group (in years)				
15-19	17	73.9	(50.4, 88.7)	0.11
20-29	41	57.7	(44.1, 70.3)	
30-44	17	44.7	(29.4, 61.1)	
Residence				
Urban	31	50.0	(35.5, 64.5)	0.18
Rural	44	62.9	(50.9, 73.4)	
ALL WOMEN	75	56.8	(47.2, 66.0)	--

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] Anemia defined as hemoglobin < 110 g/L adjusted for smoking.
[c] CI=confidence interval, calculated taking into account the complex sampling design.
[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

3.6.5. Median urinary iodine concentration

As shown in Table 42, median urinary iodine concentrations among all pregnant women, and among all sub-groups, is below 150 ug/L. This suggests an insufficient iodine status according to WHO recommendations [39]. Of note, pregnant women residing in rural areas have significantly lower urinary iodine concentrations than their urban counterparts. Pregnant women in households with adequately iodized salt have a median urinary iodine concentration of more than 270 ug/L, a level significantly higher than women residing in households with inadequately iodized salt.

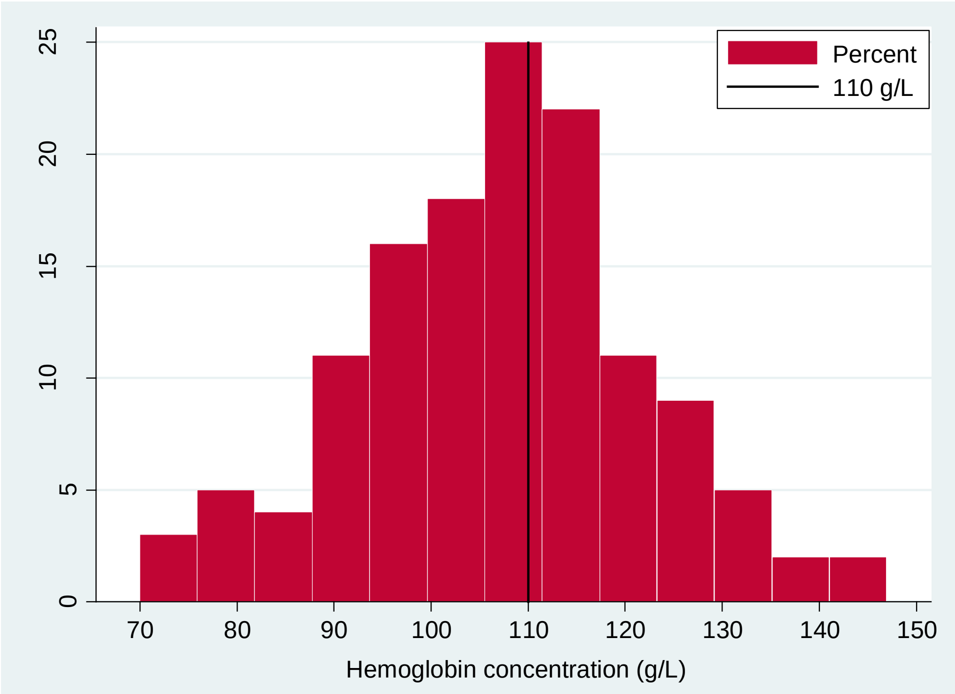


Figure 19. Histogram of hemoglobin concentration (g/L) in pregnant women, The Gambia 2018

Table 42. Median urinary iodine concentration in pregnant women, The Gambia 2018

	Urinary iodine concentration (ug/L)			
Characteristic	n	Median [a]	IQR [b]	P Value [c]
Age Group (in years)				
15-19	19	96.7	(49.9, 213.9)	0.62
20-29	63	123.3	(65.0, 215.4)	
30-44	36	106.7	(42.4, 202.9)	
Residence				
Urban	54	136.5	(92.6, 236.7)	<0.01
Rural	64	88.8	(41.3, 162.8)	
Adequately iodized salt in household [d]				
No	102	105.3	(49.9, 199.9)	<0.05
Yes	8	273.5	(117.5, 433.9)	
TOTAL RESPONDING [a]	118	113.5	(50.1, 205.9)	

Note: The n's are un-weighted denominators in each subgroup; the sum of subgroups may not equal the total because of missing data.
[a] Percentages weighted for unequal probability of selection.
[b] IQR=interquartile range
[c] P-value <0.05 indicates that the geometric mean in at least one subgroup is statistically significantly different from the values in the other subgroups
[d] Adequately iodized salt > 15 ppm

3.6.6. Hypertension

Nearly 10% of pregnant women display hypertension (systolic blood pressure 140 mm Hg; diastolic blood pressure 90 mm Hg) at the time of the GMNS. There is no significant difference found in hypertension by physical activity and nutritional status. Of note, hypertension is significantly higher in older and urban women.

Table 43. Hypertension in pregnant women by demographic, lifestyle and nutritional status characteristics, The Gambia 2018

Characteristic	n	Hypertension %[a, b]	(95% CI)[c]	P Value[d]
Age Group (in years)				
15-19	1	4.0	(0.5, 25.0)	<0.05
20-29	5	5.9	(2.4, 13.5)	
30-44	9	19.6	(9.7, 35.5)	
Residence				
Urban	12	15.8	(8.7, 27.0)	<0.05
Rural	3	3.8	(0.9, 14.4)	
Physical activity [e]				
Physically active	0	--	--	0.14
Physically inactive	15	11.6	(6.6, 19.8)	
Nutritional status				
Underweight (MUAC< 23cm)	0	--	--	0.39
Normal weight (MUAC ≥23cm)	9	8.1	(3.9, 16.3)	
TOTAL RESPONDING	15	9.6	(5.3, 16.7)	

Note: The n's are un-weighted numerators in each subgroup; the sum of subgroups may not equal the total because of missing data.

[a] Percentages weighted for unequal probability of selection.

[b] Hypertension defined as systolic blood pressure ≥ 140 and diastolic blood pressure ≥ 90.

[c] CI=confidence interval, calculated taking into account the complex sampling design.

[d] P-value <0.05 indicates that the proportion in at least one subgroup is statistically significantly different from the values in the other subgroups

[e] Physically active if > 150 min vigorous/ moderate intensity work/ per week (based on WHO recommendations on physical activity for health[38])

4. DISCUSSION

4.1. Consumption of vegetable oil, wheat flour and rice

The GMNS assessed consumption of vegetable oil, rice and bread, in tandem with individual micronutrient status, to determine if large-scale food fortification can be a viable approach in The Gambia. More than 90% of households consumed vegetable oil. While few women were vitamin A deficient and at present, may not require vitamin A-fortified oil, oil fortification can potentially address vitamin A deficiency in children in these households. Moreover, regular consumption of vitamin A-fortified oil may augment the vitamin A intake of children between biannual supplementation.

Iron deficiency is a severe problem in women and children in The Gambia. Survey findings showed that rice and wheat flour constitute suitable iron fortification vehicles to reduce iron deficiency in women and children as both are consumed by the vast majority of surveyed households. About 99% of households in The Gambia consume rice and about 97% consume bread. Of those consuming bread, 4 out of 5 most frequently buy Tapalapa, which is normally produced from wheat flour. In addition, more than one-third of surveyed households use wheat flour. Moreover, the home production of wheat flour and rice is low, which should allow a wide coverage of commercial fortified products.

4.2. Nutritional status

Children in The Gambia have relatively low levels of wasting, underweight, and stunting, denoting mild to moderate public health problems. This survey's findings demonstrate a decline for all three indicators of malnutrition compared to prior assessments. The GMNS found prevalences of stunting, wasting and underweight of 15.7%, 5.8% and 10.6%, respectively, compared to 25%, 12% and 16.2% in 2013 [4]. Despite this low national prevalence, stunting, wasting and underweight are more prevalent in certain sub-groups. Malnutrition is elevated in the rural population and LGAs with a high proportion of rural population. Highest prevalence of malnutrition is found in Kuntaur, denoting a severe public health problem according to WHO in that LGA. Furthermore, stunting and underweight, but not wasting, are more prevalent in children living in poorer households. Stunting is highly associated with inadequate household sanitation in The Gambia; poor sanitary conditions and the sustained exposure to enteric pathogens have been identified as one of the risk factors for stunting in other studies. Evidence suggests that interventions reducing the exposure to enteric pathogens by a combination of biological as well as social and economic mechanisms potentially reduce the prevalence of stunting [40]. Stunting and wasting prevalence in The Gambia are similar to those recently found in Ghana, where 21.4% and 7.0% of children were stunted and wasted, respectively [41].

Microcephaly, which affects brain size and thus cognitive development [42,43], is present in 7.4% of the surveyed children. The survey did not allow distinguishing between primary (congenital) microcephaly (the brain fails to grow during pregnancy) and secondary (perinatal) microcephaly (head circumference normal at birth, but fails to grow appropriately thereafter). Our data suggests that undernutrition is highly associated with microcephaly in The Gambia, as a significantly higher proportion of stunted, wasted and underweight children also suffered from microcephaly. This finding is in agreement with previous findings from other countries [42]. The authors are not aware of other national surveys in West Africa that have measure the prevalence of microcephaly on a nationally-representative sample of children.

Undernutrition in women, both pregnant and non-pregnant, does not appear to be of major concern in The Gambia. Nationally about 16% of women are undernourished, the majority (10%) are at risk for chronic energy deficiency and only a small proportion (2.1%) have severe chronic energy deficiency. By far, the larger problem in this group is overweight and obesity, affecting about one-third of the surveyed non-pregnant women. High BMIs are especially prevalent in older women, in women living in urban areas and women living in wealthier households. Overweight and obesity are key risk factors of type-2 diabetes mellitus, which has been confirmed by the GMNS findings. Further, obesity is a risk factor of cardiovascular diseases and various cancers [44,45]. Obesity in Gambian non-pregnant women is associated with hyperglycemia and overweight with intermediate hyperglycemia. Moreover, significantly more overweight (about 15%) and obese (about 20%) women suffer from hypertension, compared to normal weight (about 6%) and underweight women (about 3%).

4.3. Anemia and micronutrient status

4.3.1. Anemia

The GMNS found that approximately 50% of children were anemic. Anemia in children was markedly higher in Mansakonko (68.1%), Kerewan (60.7%) and Kuntaur (57.5%) compared to Banjul (41.9%) and Kanifing (42.5%). A similar disparity was observed with ID and IDA as the prevalence in the predominantly rural LGAs was substantially higher than in the other LGAs. To illustrate, IDA was more than 70% in Kuntaur and less than 45% in Banjul. About 75% of anemic children had concurrent iron deficiency, which is substantially higher than the 28% estimate in a recent meta-analysis for countries in sub-Saharan Africa [46] and indicates that iron deficiency continues to play an important role in the etiology of anemia in Gambian children. The current analysis clearly suggests that although there are multiple factors contributing to anemia, anemia in Gambian children is mainly nutritionally-induced. Subgroup analyses show that ID as well as VAD are highly associated with anemia. Furthermore, for children consuming vitamin and mineral supplements, only 33% are anemic, compared to 66% of children not consuming supplements. As anemia in children is also associated with inflammation and fever, it is likely that anemia due to infection and anemia due to chronic disease contribute to the overall anemia prevalence, but to a lower extent. Other potential contributors to anemia such as the helminth infection have not been examined in the GMNS.

Further research will be required to assess if deworming campaigns can effectively increase hemoglobin levels and reduce anemia prevalence.

It has to be noted that the overall anemia prevalence of about 50% is lower than the 73% reported in 2013 [4]. The observed difference is surprisingly large considering the relatively short time interval between the two surveys. The prevalence of malaria and LRI found by the GMNS is lower than reported by the 2013 DHS. The lower levels of LRI and – in particular – malaria, could account for the lower anemia prevalence found by the GMNS. However, this speculation is confounded by the fact that the prevalence of fever (25.7%) and diarrhea (25.4%) were higher in the GMNS than DHS. Further, there are some technical differences that may partly explain the differences, although it is not known to what extent. For example, the GMNS used the Hemocue Hb 301 device, which has been described to yield slightly higher hemoglobin readings [47] than the Hb 201+, although such a bias is not described consistently [48].

Contrarily, the prevalence of anemia found in non-pregnant women (50.9%) and pregnant women (56.8%) are similar to the prevalence found in 2013 (58%). Similar to children, although not as distinct, high prevalences of anemia can be found in Kuntaur and Mansakonko and low prevalence in Banjul. The survey showed that nutritional anemia is also the main cause of anemia in women. More than 55% of anemic women had concurrent iron deficiency, and bivariate analyses show that ID and VAD are highly associated with anemia. Similar to the findings in children, inflammation was not associated with anemia in non-pregnant women.

4.3.2. Iron deficiency

The national prevalence of ID in children and women is high with 59.0 % and 41.4%. Compared to the survey in 1999 the prevalence in children almost tripled and doubled in women. Of note, a direct comparison of the results is not possible since the survey in 1999 a) did not adjust ID for increased concentrations of AGP and CRP; b) used a different cut-off for ID in children and women; c) only included lactating and pregnant women. ID and IDA in women and children are elevated in rural areas as well as in certain LGAs, such as Kuntaur and Mansakonko. Further, increased rates can be found in women and children living in poor households as well as in women with low educational level, indicating that both conditions, although universally present across the whole population, mainly affect the poor and uneducated. The main reasons for the high prevalence of ID are low intake of iron and/or low bioavailability of iron in the diet. The majority of the population, especially the poor subsist, on a plant-based diet, which is low in bioavailable iron and high in iron absorption inhibitors. More than 80% of the surveyed women stated that they consumed tea or coffee during or shortly after the meal, both beverages known to inhibit iron absorption. Surprisingly, very few women had ever heard about fortified wheat flour, fortified vegetable oil, or had seen the Enrichi logo. This low awareness of fortified products could possibly be a factor contributing to the high prevalence of iron deficiency.

4.3.3. Vitamin A deficiency

VAD affects more than 18% of children in The Gambia, representing a moderate public health problem according to WHO criteria [49]. Although this prevalence is lower than the prevalence reported in 1999 (64%), there are still subgroups where VAD poses a severe public health problem (prevalence ≥20%). The prevalence of vitamin A deficiency is significantly higher in children between 24-59 months (>20%), compared to younger children (<10%) and higher in males than in females. In addition, the prevalence in children residing in rural areas is almost twice as high as in children living in urban centers. Further, significant differences can be seen between the LGAs. Banjul has by far the lowest prevalence (<10%), whereas in Kuntaur and Basse more than 25% of children suffer from VAD, increasing to above 30% in Janjanbureh. The GMNS only found a significantly lower prevalence of vitamin A deficiency among children that received vitamin A supplements in the past 3 months, but no significant difference among children receiving supplements in the past 6 months. This finding corresponds to literature which suggest retinal levels only remain elevated for 2-3 months following vitamin A supplementation [50]. Furthermore, as supplementation coverage was markedly higher among children 6-23 months of age, increasing awareness of the importance of vitamin A supplementation for all children 6-59 months of age should be communicated via public health channels and BCC campaigns. [51].

4.3.4. Iodine deficiency

Only a small proportion of the salt in The Gambia was adequately iodized as per international standards. Nevertheless, the high median urinary iodine concentration in non-pregnant women demonstrates an absence of iodine deficiency in this group. Nonetheless, median urinary iodine concentrations are below levels of adequacy in certain population groups. Women residing in Kuntaur and Mansakonko have a median urinary iodine concentration below the cut-off for sufficiency. Also, women living in households of the lowest wealth quintile have median urinary iodine concentrations below levels of adequacy, which might be related to home production of salt. The situation is more severe in pregnant women, as in this population group the median urinary iodine concentration indicates iodine deficiency; especially for pregnant women living in rural areas and in households with inadequately iodized salt. Extending the coverage of iodized salt to all regions and specific target groups and strengthening the salt iodization surveillance system including quality assurance at the point of production and entry can improve iodine intake.

4.4. Non-communicable diseases

HbA1c was used to determine hyperglycemia in women, reflecting plasma glucose levels of the last 8-12 weeks. As such, HbA1c is a robust measure of hyperglycemia as it is not prone to day-to-day variations encountered when assessing hyperglycemia with other indicators (e.g. fasting blood glucose) [34,52]. Nationally, the prevalence of hyperglycemia in non-pregnant women was 7.4%. Although, this is somewhat lower than in European women (9.6%) [53], the prevalence is higher than the estimated prevalence in African countries in

2013 (about 5%) [54] and warrants attention. Especially alarming is the increase compared to a survey in 1993, which found that only 0.1% of Gambian women were diabetic [55].

The survey findings showed a high prevalence of intermediate hyperglycemia, which is characterized by an impaired glucose tolerance or impaired fasting glucose and can already increase the risk of morbidity and mortality [56,57]. Of note, these findings have to be used with caution as there is no consensus on the cut-off for intermediate hyperglycemia. Both, hyperglycemia and intermediate hyperglycemia are associated with nutritional status, and significantly more obese women suffer from hyperglycemia and significantly more women with intermediate hyperglycemia are overweight. This is not surprising as overweight and obesity, together with physical inactivity and age, belong to the main risk factors of type 2 diabetes or pre diabetes [58]. The survey findings indicate that physical inactivity is not associated with hyperglycemia, but a significantly higher proportion of women in the age group 44-49 years have intermediate hyperglycemia.

Similar to hyperglycemia, hypertension in Gambian women is related to nutritional status and age, which is in accordance with literature. However, the survey found 10.6% of non-pregnant women with hypertension, which is much lower compared to an assessment from 1993, estimating a prevalence of more than 27% [55].

4.5. Limitations

As the GMNS is a cross-sectional survey, it cannot establish causality between any outcome of interest and a potential risk factor. In addition, as the GMNS data collection occurred between March and May, data was collected in The Gambia's dry season, and could have influenced the prevalence of distribution of certain variables, such as malaria prevalence.

5. Recommendations

5.1. Reduce undernutrition in children

While national undernutrition denotes only a mild to moderate public health problem, the situation is severe in some parts of the country and warrants immediate attention. The GMNS did not collect data on IYCF indicators, and thus their impact on nutritional status in Gambian children requires further research. IYCF indicators were collected by the 2018 MICS; and further analysis should be explored with this data set. However, the survey findings show that inadequate sanitary conditions lead to increased rates of stunting and underweight. Thus, WASH programs targeted at increasing access to safe drinking water and adequate sanitation facilities, including a behavioral change component such as the "Gambia Climate Smart Rural WASH Development Project" should be further strengthened.

5.2. Reduce anemia and iron deficiency in children and women

The GMNS contained many of the major anemia risk factors, and the results suggest that mainly iron deficiency contributes to anemia in children and woman; and vitamin A deficiency to anemia in children. Other risk factors such as malaria or inflammation seem not or only marginally affect anemia. Although anemia, ID and IDA affect the whole population, programs to reduce their prevalence in Gambia should prioritize activities in rural areas of Kuntaur, Janjanbureh, Mansakonko and Basse LGAs, targeting poor households. Interventions should include the promotion of age-appropriate infant and young child feeding practices, including the promotion of foods (fortified or unfortified) rich in iron and vitamin A. In women, programs to promote the consumption of iron-rich foods and iron supplements can be considered. Moreover, to decrease the prevalence of ID, mandatory iron fortification programs should be implemented, such as iron fortification of rice and/or wheat flour. These programs should be implemented in combination with behavior change communication strategies in order to increase awareness of and knowledge about fortified foods. In addition, a rigorous surveillance programs that ensure a high coverage of adequately fortified food vehicles is needed.

5.3. Reduce vitamin A deficiency in children

While vitamin A deficiency affects almost 20% of children in Gambia, the severity of this deficiency is significantly higher in some LGAs. To address this severe public health problem, multiple approaches should be used. First, Gambia's vitamin A supplementation program should be improved to strengthen childrens' immune systems and reduce the risk of mortality due to measles, diarrhea, and other illnesses. Secondly, to increase the vitamin A stores, mandatory vitamin A fortification programs should be implemented, such as vitamin A fortification of vegetable oil. These programs should be implemented in combination with behavior change communication strategies in order to increase awareness of and knowledge about fortified foods. In addition, a rigorous surveillance programs that ensures a high coverage of adequately fortified food vehicles would increase quantity of retinol consumed by children on a daily basis. Thirdly, programs to improve consumption of vitamin A-rich foods, other than fortified products, should be pursued. This is particularly relevant in rural areas where vitamin A deficiency is high. This type of intervention could include promoting local food products rich in vitamin A, or introducing vitamin A-biofortified staple foods that could be readily cultivated in The Gambia.

5.4. Reduce the prevalence of overweight and obesity in women

Nearly one-third of non-pregnant women were classified as either overweight or obese, with the highest proportions in urban area and among women residing in wealthier households. As globally the prevalence of overweight and obesity has risen in the past decade and obesity and overweight are a key risk factors of type-2 diabetes mellitus as well as cardiovascular diseases and are associated with hypertension there is an imperative need to educate urban women about approaches to maintaining healthy weight to prevent the prevalence of overweight and obesity from rising further. As increased parity has been associated with increased overweight and obesity prevalence in other studies [59], it is recommended that antenatal and postnatal care provided by doctors and nurses be expanded to include behavior change messages and counseling for mothers. Although the survey did not find any association between overweight/obesity and physical activity, it is recommended that physical activity be promoted nonetheless as it has the potential to counteract obesity and overweight [60].

5.5. Increase median urinary iodine concentration

Only a small proportion of the salt consumed in Gambia is adequately fortified, leading to low median urinary iodine concentrations in some population sub groups. To increase the coverage of adequately iodized salt, Gambia's salt-iodization regulatory monitoring system should be expanded to the level of production and importation. As nearly all salt samples did not contain the required levels of iodine, surveillance and regulatory enforcement is needed during the domestic production/process of salt and the importation of salt from other countries. An improved surveillance system will have to be accompanied by BCC campaigns to increase the awareness and with it demand for iodized salt, targeting mainly the poor population in rural areas. Thus, salt intake, iodine intake and median urinary iodine concentrations should be regularly monitored to ensure adequate iodine status in all population groups.

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7. APPENDICES

7.1. - LIST OF SELECTED ENUMERATION AREAS

LGA Code	LGA	District Code	EA Code	Urban/ Rural	Settlement
1	Banjul	10	10103	1	Banjul
1	Banjul	10	10202	1	Banjul
1	Banjul	10	10207	1	Banjul
1	Banjul	11	11102	1	Banjul
1	Banjul	11	11107	1	Banjul
1	Banjul	11	11201	1	Banjul
1	Banjul	11	11204	1	Banjul
1	Banjul	11	11209	1	Banjul
1	Banjul	12	12101	1	Banjul
1	Banjul	12	12105	1	Banjul
1	Banjul	12	12207	1	Banjul
1	Banjul	12	12212	1	Banjul
1	Banjul	12	12304	1	Banjul
2	Kanifing	20	20104	1	Bakau New Town
2	Kanifing	21	21118	1	Latrikunda /Kanifing
2	Kanifing	21	21281	1	Ebo Town
2	Kanifing	22	22143	1	Bundung
2	Kanifing	22	22305	1	Serekunda
2	Kanifing	23	23211	1	Faji Kunda
2	Kanifing	23	23320	1	Latrikunda Sabiji
2	Kanifing	23	23473	1	Tallinding
2	Kanifing	24	24229	1	Dippa Kunda
2	Kanifing	24	24501	1	Kotu
3	Brikama	30	30305	1	Lamin
3	Brikama	30	30507	1	Wellingara
3	Brikama	30	30851	1	Busumbala
3	Brikama	31	31237	1	Gunjur Kunkujang
3	Brikama	32	32218	1	Medina Salanding
3	Brikama	33	33308	1	Kuloro
3	Brikama	34	34102	2	Bulock
3	Brikama	35	35220	2	Buniadou (Bakukul)
4	Mansakonko	43	43115	1	Soma
4	Mansakonko	43	43133	1	Soma
4	Mansakonko	40	40202	2	Bureng
4	Mansakonko	41	41108	2	Nemakuta
4	Mansakonko	42	42206	2	Sare Musa
4	Mansakonko	43	43208	2	Karantaba
4	Mansakonko	45	45102	2	Dongoroba
4	Mansakonko	45	45205	2	Sukuta
5	Kerewan	50	50114	1	Essau
5	Kerewan	53	53110	1	Kerewan
5	Kerewan	55	55337	1	Farafenni
5	Kerewan	50	50175	2	Berending
5	Kerewan	51	51127	2	Lamen
5	Kerewan	52	52211	2	Memmeh
5	Kerewan	54	54203	2	Kerr Pateh Koreh
5	Kerewan	56	56201	2	Daru Rahman
6	Kuntaur	60	60209	1	Kaur Touray Kunda

6	Kuntaur	63	63135	1	Waasu
6	Kuntaur	60	60108	2	Balanghar Njoben
6	Kuntaur	61	61120	2	Kerr Auldeh
6	Kuntaur	62	62110	2	Nioro Buba
6	Kuntaur	63	63125	2	Kataba Omar Ndow
6	Kuntaur	63	63228	2	Pallol Fula
6	Kuntaur	64	64209	2	Tabanani
7	Janjanbureh	73	73116	1	Darsilameh
7	Janjanbureh	74	74118	1	Bansang
7	Janjanbureh	70	70106	2	Bantang Koto Mandinka
7	Janjanbureh	72	72114	2	Batty Njoll
7	Janjanbureh	72	72234	2	Banni Daka
7	Janjanbureh	73	73235	2	Sare Babou (Taifa Amadou)
7	Janjanbureh	74	74202	2	Wellingara Demba Kandeh
7	Janjanbureh	74	74402	2	Misira
8	Basse	81	81112	1	Bassending
8	Basse	81	81139	1	Basse Santo-Su
8	Basse	82	82221	1	Kulari
8	Basse	80	80233	2	Basse/Mori Bugu
8	Basse	82	82132	2	Dingiri
8	Basse	83	83144	2	Sare Hendeng /Sendebu Kelepha
8	Basse	84	84223	2	Fadia Kunda
8	Basse	86	86218	2	Darsilame Mandinka

*1=urban, 2=rural

7.2. - A PRIORI SAMPLE SIZE CALCULATIONS

Sample size for children (0-59 months of age), non-pregnant women (15-49 years of age) and pregnant women per stratum and in the two strata (urban/ rural) taking into account desired precision and assumed design effect and individual response rate

Target group and indicator	Estimated prevalence	Desired precision for urban/ rural estimates (percentage points)	Assumed design effect	Assumed individual response	Number of persons to select in one stratum	Number of person to select in two strata
Households						
Adequately iodized salt	50%	±10	4.0	95%	550	1100
Children						
Anemia	70%	±5	1.7	85%	666	1332
Iron deficiency	50%	±10	1.5	85%	666	1332
Vitamin A deficiency	20%	±5	1.5	85%	666	1332
Wasting	12%	±3	1.8	95%	745	1490
Stunting	25%	±5	1.5	95%	745	1490
Non-pregnant women						
Anemia	60%	±10	1.8	90%	932	1864
Iron deficiency	50%	±10	1.5	90%	932	1864
Vitamin A deficiency	5%	±2.5	1.5	90%	932	1864
BMI <18.5	20%	±5	1.5	95%	983	1966
BMI >18.5	25%	±5	1.5	95%	983	1966
Pregnant women						
Anemia	70%	±10	1.8	90%	82	164
Low MUAC	50%	±10	1.5	90%	82	164

7.3. - ETHICAL APPROVAL

C/o MRC Unit: The Gambia at LSHTM,
Fajara
P.O. Box 273, Banjul
The Gambia, West Africa
Fax: +220 – 4495919 or 4496513
Tel: +220 – 4495442-6 Ext. 2308
~~Email: ethics@mrc.gm~~

The Gambia Government/MRC Joint
ETHICS COMMITTEE

21 February 2018

Mr Modou Cheyassin Phall
Executive Director
National Nutrition Agency
Bertil Harding High Way

Dear Mr Phall

R018 014, The Gambia Micronutrient 2018 Survey (GMNS)

Thank you for submitting your proposal for consideration by The Gambia Government/MRC Joint Ethics Committee.

I am happy to approve your proposed study.

With best wishes

Yours sincerely

Mr Malamin Sonko
Chairman, Gambia Government/MRC Joint Ethics Committee

Documents submitted for review:
- RePubliC approval letter – 13 February 2018
- Project proposal (including informed consent document)
- Cover letter – 5 February 2018

UNIVERSITY OF THE GAMBIA

SCHOOL OF MEDICINE & ALLIED HEALTH SCIENCES
RESEARCH AND PUBLICATION COMMITTEE (RePubliC)

13th February 2018

Modou Cheyassin Phall
Executive Director/Principal Investigator
National Nutrition Agency (NaNA)
Birtil Harind High Way, Mile 7

Dear Mr Phall,

Re: R018 014: "The Gambia Micronutrient 2018 Survey (GMNS)"

Thank you for submitting the above project for expedited review.

There are no substantive scientific queries. It is noted that you will provide treatment to participants with abnormal findings and/or refer them for appropriate management as indicated in the attached referral form.

I forward it to the Gambia Government/MRC Joint-Ethics Committee Chair for consideration. I note the limited time before last week of February in which to catch up with the ongoing MICS data collection.

Thank you.

SCHOOL OF MEDICINE & ALLIED HEALTH SCIENCE
SIGN............
DATE............
UNIVERSITY OF THE GAMBIA

Best wishes,

Prof Ousman Nyan
Chair

Cc: File

Enclosed:

1) Request letter for Scientific Review dated 5/2/18
2) GMNS 2018 Survey Protocol including
   Appendix 3: Consent form
   Appendix 4: Information Sheet
   Appendix 5: Referral Form

P.O. Box 1646 Banjul; Tel: +220 4201407; Mob: +220 3560003 Email: onyan@utg.edu.gm

7.4. - INFORMATION SHEET AND CONSENT FORM

INFORMATION SHEET (FOR THE PARTICIPANT TO KEEP)	
Title of Study:	**Gambia National Micronutrient Survey (Component of MICS)**
Principal Investigators:	Mr. Modou Cheyassin Phall
Certified Protocol Number	R018014

What is informed consent?

You are invited to take part in a survey. Participating in a survey is not the same as getting regular medical care. The purpose of normal medical care is to improve one's health. The purpose of a survey is to gather information that may be useful in the future for the whole population. It is your decision to take part and you can stop at any time without giving any reason.

Before you decide you need to understand why the suvey is being done and what will happen in it. Please take time to read the following information or get the information explained to you in your language. Listen carefully. You can ask questions if there is anything that you do not understand. Ask for it to be explained until you are satisfied. You may also wish to speak your spouse, family members or others before deciding to take part in the study.

If you decide to join the study, you will need to sign or thumbprint a consent form saying you agree to be in the study. You will receive a copy of the consent form.

Why is this study being done?

The Gambia MICS Micronutrient Component 2018 is conducted to understand the severity of various nutritional deficiencies, such as anemia, iron deficiency, vitamin A deficiency, and malnutrition in women and children. The survey is done by UNICEF, FAO, UNFPA, WHO, and GroundWork in Switzerland.

We will tell the results of this study to your community.

What does this study involve?

We will ask some questions about your household, and if there are selected women or children living in the household, we will ask individual questions to better understand their person and their food habits. The questionnaire for you usually takes about 10 minutes to complete.

Following the completion of the questionnaire, we will measure height and weight from all women 15 to 49 years of age and children 0 to 59 months of age. After this we would also request to draw a small amount of blood by pricking the finger- or heel from all women 15 to 49 years of age and children 6 to 59 months of age in a dedicated place near your household.

This small blood sample will be used to test if you have anemia or malaria, and these results will be provided to you directly. In addition, a small portion of blood will be collected to test for micronutrient deficiencies, such as iron, and vitamin A status.

Benefits/Risk of the study

For all non- pregnant women and children 6-59 months of age a small sample of blood (0.4 mL) will be collected from the finger or heel (children 6-12 months of age) via a small lancet by trained technicians. This might cause some pain, but is otherwise harmless. The blood draw should take less than 5 minutes, and the anemia and malaria results will be provided in less than 15 minutes following the taking of blood, and should you be diagnosed with severe anemia or malaria, we will provide you with a referral to a nearby health facility for further testing and treatment. This survey poses no serious risks to you or other participating family members.

Other than the information about your hemoglobin levels or malaria parasitemia and referral in case of diagnosis of severe anemia, we cannot promise that the survey will help you directly. But the information we get will help the Government to evaluate its nutrition and health services and if needed, adapt them.

Will you be compensated for participating in the study?

You will not get paid by the study.

Confidentiality

All information which is collected about you and your household during the course of the interview will be kept strictly confidential, and any information about you and the household address will not be included in the final report so that you cannot be recognized.

Only the personnel doing the interview and the principal researchers will have access to identifiable information and by providing your signature/thumbprint, you allow the research team in doing so.

Withdrawal from Study

Participation in this survey is voluntary, and if we should come to any question you do not want to answer, just let me know and I will go on to the next question; or you can stop the interview at any time, without any consequences to you or your household. However, we hope that you will participate in this survey since your views are important. There will not be any negative effects on you, if you decide that you no longer want to continue with the interview.

If you are younger than 18 years, your legal parent will have to give signed consent for your participation. This information sheet will be for you/your caretaker to keep. If you have any question, do not hesitate to contact the principal researchers.

Contact for additional Information

If you have any questions about the study, you are welcome to call Mr. Modou Cheyassin Phall from the National Nutrition Agency, who is in charge of this study, on *telephone number** and he will be happy to answer your questions. You can also call Dr. Samba Ceesay from the Ministry of Health and Social Welfare, on *telephone number**, if you have further concerns.

*Telephone numbers removed for report version

WRITTEN INFORMED CONSENT FORM

VOLUNTEER AGREEMENT

a. Child's participation

"I have read or have had someone read all information on the information sheet, have asked questions, received answers regarding participation in this study, and am willing to give consent for my child/ward to participate in this study. I have not waived any of my rights by signing this consent form and I confirm that my choice to let my child participate is entirely voluntary. Further, I agree for my child's blood sample to be shipped outside the Gambia. Prior to signing this consent form, I was given a copy of the information sheet for my personal records."

____

Name of mother or legal caregiver

____ ______________________

Signature or mark of mother or legal caregiver Date

b. Adult woman's own participation

" I have read or have had someone read all information on the information sheet, have asked questions, received answers regarding participation in this study, and am willing to give consent to participate in this study. I have not waived any of my rights by signing this consent form and I confirm that my choice to participate is entirely voluntary. Further, I agree for my blood sample to be shipped outside the Gambia. Prior to signing this consent form, I was given a copy of the information sheet for my personal records."

____

Name of participant

____ ______________________

Signature or mark participant Date

CLUSTER ID: ☐☐	**RESPONDENT Label:** **C** ☐☐☐☐
HH Label: **H** ☐☐☐☐	**RESPONDENT Label:** **W** ☐☐☐☐

If volunteers cannot read the form themselves, a witness must sign here:

I was present while the benefits, risks and procedures were read to the volunteer. All questions were answered and the volunteer has agreed to take part in the research.

____

Name of witness

____ ______________________

Signature of witness Date

I certify that the nature and purpose, the potential benefits, and possible risks associated with participating in this research have been explained to the above individual.

____ ☐ ☐

Name of Interviewer who obtained Consent Interviewer ID

____ _____________________

Signature of Interviewer who obtained Consent Date

7.5. – REFERAL FORM

The National Nutrition Agency (NaNA) and UNICEF are conducting a micronutrient component of the nation Multiple Indicator Cluster Survey (MICS) to better understand the various nutritional conditions. As part of this, the hemoglobin concentration and malaria parasitemia of women 15-49 years old, pregnant women, and children 6-59 months. In children, malnutrition is assessed.

Individuals with the following outcomes will be referred to the nearest health facility:

Malaria parasitemia

Severely anemic (<70 g/L for pregnant women and children; <80 g/L for non-pregnant women)

Child suffering from severe acute malnutrition (<-3 weight-for-height Z-scores)

This referral is to inform the health post about the results and possible treatment from on-site diagnostics:

Person is: Child ☐ Non-pregnant woman ☐ Pregnant woman ☐	Person's Name: ____________________ Person's Age: ____________________
	Hemoglobin = _______________g/dL
	Malaria testing result: ☐ Negative ☐ P.falc. ☐ Pan
	CHILDREN ONLY: Does child have SAM? (use SAM table to diagnose): ☐ NO ☐ YES Edema: ☐ NO ☐ YES
	PREGNANT WOMEN ONLY: **MUAC (cm)** = _______________ Less than 18.5 cm? ☐ NO ☐ YES
Name of referring phlebotomist: ____________________ Phone of referring phlebotomist: ____________________ Signature of referring phlebotomist: ____________________ Date: ____________	

7.6. – MICS HOUSEHOLD LIST

Household #15

Address: 19 HAGAN STREET
HH head: BABOUCARR TOURAY
Not yet assigned
Interviewed by: 012
Result: 01) Completed

Eligible individual questionnaires:

Line #	Name	Age	Questionnaire type	Interviewer	Result	H/W/Vacc
001	BABOUCARR TOURAY	43	M 15-49	016	01) Completed	
002	KADIJATOU BOJANG	35	W 15-49	012	01) Completed	
003	YASSIN TOURAY	14	C 5-17 (F)	012	01) Completed	
006	JARIATOU TOURAY	03	U5	012	01) Completed	Entered
007	FATOU TOURAY	00	U5	012	01) Completed	Entered

Household #16

Address: 24 HAGAN STREET
HH head: GIBRIL JOOF
Not yet assigned
Interviewed by: 012
Result: 01) Completed

No eligible individuals present in household listing

Household #17

Address: 69 HAGAN STREET
HH head: MUSTAPHA JOOF
Not yet assigned
Interviewed by: 014
Result: 01) Completed

Eligible individual questionnaires:

Line #	Name	Age	Questionnaire type	Interviewer	Result	H/W/Vacc
001	MUTAKA JOOF	41	M 15-49	016	01) Completed	

Household #18

Address: 69 HAGAN STREET
HH head: MODOBO MEGE
Not yet assigned
Interviewed by: 014
Result: 01) Completed
Water quality questionnaire: **Entered**

Eligible individual questionnaires:

Line #	Name	Age	Questionnaire type	Interviewer	Result	H/W/Vacc
002	MARIAMA TOURAY	32	W 15-49	014	01) Completed	
003	AMADOU MEGA	17	C 5-17 (M)	014	01) Completed	
007	SARATOU MEGA	04	U5	014	01) Completed	Entered
008	OUSMAN MEGA	02	U5	014	01) Completed	Entered
009	FATOUMATA MEGA	35	W 15-49	014	01) Completed	
011	MUHAMMADOU MEGA	04	U5	014	01) Completed	Entered
012	LAILA MEGA	02	U5	014	01) Completed	Entered
013	FATOUMATA MEGA	00	U5	014	01) Completed	Entered

7.7. - TEAMS, TEAM MEMBERS, AND SUPERVISORS

Team #	LGA	Team Supervisor (Leader)	Interviewer	Phlebotomist	Anthropometrists	Laboratory technician
1	Banjul	Alieu Kujabi	Ibra Ceesay	Hassan Sarr	Mariama Bojang Ansunama Dampha	Mamudou Dahaba
2	Kanifing	Lamin Y Darbo	Sally Fatajo	Amie Njie	Jawara Saidykhan Oumie Sarr	
3	Brikama	Musa B. Dahaba	Mustafa Sanyang	Awa Badjie	Ibraima Joof Jawara Touray	
4	Mansakonko	Fabakary Bass	Karamo Ceesay	Marie Nyassi	Salieu Darboe Buba Jatta	Ibraima Danso
5	Janjanbureh	Famara Nyabally	Malick Choi	Ismaila Gibba	Mafugi Jawara Seedy Saidykhan	Sulayman Kanteh
6	Basse	Musa Camara	Modou Chatty	Lamin Fatajo	Saikou Drammeh Sana Sowe	
7	Kuntaur	Saity Jatta	Aja Nyangado	Karamba Jabbie	Amadou Jallow Hadji Kumba Joof	Yahya Sallah
8	Kerewan	Michel Jammeh	Mustapha Saho	Chaku Touray	Alpha Mballow Fatoumata Sanyang	

7.8. - SURVEY QUESTIONNAIRES AND TOOLS

HOUSEHOLD QUESTIONNAIRE
GAMBIA MICRONUTRIENT SURVEY 2018

1. MICS Cluster number .... ☐☐☐	2. HH number on MICS household list ☐☐
3. Household label number.........H ☐☐☐☐	4. Interviewer Number_________
5. Date of the interview:	6. Name of household head:____________________
7. Consent given for interview:	Yes.......................................1 No.......................................2 → END

SALT PURCHASE / SAMPLE		
8. How often is salt purchased for consumption in this household on average? *Fill in number of times for only 1 time period.*	Number of times per: Week .... ☐☐ Month .... ☐☐ Year .... ☐☐ Never, we don't use it .... 00 Don't know / not sure 99	 ->Q10 ->Q10
9. What quantity is usually obtained whenever some salt is bought? *Fill in amount for either grams or kilograms, but NOT BOTH.*	Grams.............................. ☐☐☐☐ Kilograms .... ☐☐ Don't know / not sure...................9999 or 99	
10. Do you have salt in your house now?	Yes .... 1 No .... 2 Don't know 9	 ->Q16 ->Q16
11. Is this salt iodized?	Yes .... 1 No .... 2 Don't know 9	
12. May I see the salt container?	Yes, original package says iodized .... 1 Original package not mention iodization ... 2 Undetermined, not in original package ...... 3 Undetermined for other reasons 9	
13. May I have a small sample of the salt?	Yes .... 1 No .... 2 Don't know 9	-> Collect Salt ->Q16 ->Q16
14. Salt sample collected?	Yes .... 1 No 2	 ->Q16
15. If salt sample collected, ID number applied to sample.	HH Label number... H ☐☐☐☐	

VEGETABLE OIL PURCHASE AND CONSUMPTION		
16. What is the **main** edible **oil** consumed by your household (*clarify with: the oil that you use on most days in most meals in the home*)? *Select only **one** answer.*	Vegetable oil................1 Groundnut oil.....................2 Refined palm oil.......................................3 Soybean oil..4 Soy bean oil Sunflower oil...5 Olive oil...6 Don't use...0 Don't know...9 Other: ________________________......8	 → Q22 → Q22
17. Can you tell me where you usually get this [main oil type]? *Select only **one** answer.*	Purchased from supermarket.................1 Purchased from local open market........2 Retail Shop 3 Purchased from stand in the street.......4 Made it at home.......................................5 Received from food aid..........................6 Don't know...9 Other: ________________________......8	
18. This [main oil type] that you consume, when you get it, is it usually packaged or open?	In the original bottle................................1 In another bottle.......................................2 In a plastic bag..3 Don't know...9 Other: ________________________......8	
19. Can you tell me the brand of this oil? *Select only **one** answer.*	Yete Jalliw.................................1 Victoria.......................................2 Don't know...9 Other: ________________________......8	
20. For the most recent **major** purchase of [main oil type], how much did your household buy?	Quantity ☐☐☐☐ Liters..1 milliliters..2 Don't know...9	I cup is estimated to be equal to 250ml 4 cups is estimated to be equal to 1 litre
21. How long does this amount usually last in your household?	Duration ☐☐ Day(s)..1 Month(s)..2 Don't know...9	
RICE PURCHASE AND CONSUMPTION		
22. What is the main type of rice do you usually get? *Select only **one** answer.*	Local paddy...1 Imported rice...2 Don't use ....00 Don't know...88 Other: ________________________......99	 → Q24

23. Where does your household usually get rice from? *Select only <u>one</u> answer.*	Home production................................ 1 Market/ retailer/ shop.......................... 2 Local/ village producer......................... 3 Ration store.. 4 Gift...5 Don't know..88 Other: ____________________ 99	
24. Do you usually get open or packaged rice? *Select only <u>one</u> answer.*	Open...1 Packaged.. 2 Don't know..88 Other: ____________________ 99	
25. What type of rice do you usually get? *Select only <u>one</u> answer.*	Short grain (broken rice).......................1 Long grain..2 Don't know...8 Other (specify)________________ 9	
26. The last time your household got rice, how much did you get?	Quantity ☐☐☐☐ Kg...1 g..2 Don't know..9	
27. How long does this amount usually last in your household?	Duration ☐☐ Day(s)...1 Week(s)...2 Month(s)..3 Don't know..9	
BREAD AND FLOUR PURCHASE AND CONSUMPTION		
28. Do you usually eat bread in this household?	Yes .... 1 No... 2 Don't know 9	 Q40 Q40
29. Do you usually eat Senfurr (baguette type bread)	Yes .... 1 No... 2 Don't know 9	 Q32 Q32
30. How many times per day or week do you <u>usually</u> purchase Senfur? *Fill in number of times for only 1 time period.* *If respondent does not know, select*	Number of times a: A. Times per Day .... ☐ B. Times per Week .... ☐ C. Don't know / not sure..............................9	
31. What quantity is <u>usually</u> obtained whenever Senfur is bought? *Fill in number of loaves for either type, if MULTIPLE types are <u>USUALLY</u> bought, then fill in all relevant* *If respondent does not know, select "99"*	A. Number of Senfur .... ☐ B. Don't know / not sure 99	

32. Do you usually eat sweet bread in your household?	Yes .... 1 No .... 2 Don't know 9	 Q35 Q35
33. How many times per day or week do you usually purchase Sweet bread? *Fill in number of times for only 1 time period.* *If respondent does not know, select "99".*	Number of times a: A. Times per Day .... ☐ B. Times per Week .... ☐ C. Don't know / not sure........9	
34. What quantity is usually obtained whenever Sweet bread is bought? *Fill in number of loaves for either type, if MULTIPLE types are USUALLY bought, then fill in all relevant* *If respondent does not know, select "99".*	A. Number of Sweet........ ☐ B. Don't know / not sure 99	
35. Do you usually eat Tapalapa in your household?	Yes .... 1 No .... 2 Don't know 9	 Q38
36. How many times per day or week do you usually purchase Tapalapa? *Fill in number of times for only 1 time period.* *If respondent does not know, select "99".*	Number of times a: A. Times per Day .... ☐ B. Times per Week .... ☐ C. Don't know / not sure........9	
37. What quantity is usually obtained whenever Tapalapa is bought? *Fill in number of loaves for either type, if MULTIPLE types are USUALLY bought, then fill in all relevant* *If respondent does not know, select "99".*	A. Number of Tapalapa........ ☐ B. Don't know / not sure 99	
38. Do you eat any other types of bread	Yes .... 1 No .... 2 Don't know 9	
39. Specify "other" type of bread	____________	
40. What type of other food products made with wheat flour do you usually eat in this household? *Select all that apply.*	Pancakes........ A Doughnuts .... B Salty pies .... C Don't eat ....Y Don't know........Z Other: ____________ X	
41. What is the main edible flour consumed by your household (clarify with: the flour that you use on most days in most meals in the home)? *Select only one answer.*	Wheat flour........1 Maize flour........2 Millet flour........3 Sorghum flour........4 Don't use........00 Don't know / Don't remember........88 Other: ____________ 99	 → End → End → End

42. Can you tell me the brand of this flour?	Other: ____________ 9	
43. The last time your household [MAIN FLOUR TYPE], how much did you get?	Quantity ☐☐☐☐ Kg........1 g........2 Don't know........8888	
44. How long does this amount usually last in your household?	Duration ☐☐ Day(s)........1 Week(s) Month(s)........2 Don't know........88	

FINAL RESULT CODES: Completed interview ....1 No household member or no competent respondent at home at time of visit ....2 Entire household absent for long period or moved away 3	Refused .... 4 Dwelling vacant / Address not a dwelling .... 5 Dwelling destroyed .... 6 Dwelling not found .... 7 OTHER (SPECIFY) ____________ 8

S

Interviewer put household "H" label here	**BIOLOGICAL FORM - CHILD** **GAMBIA MICRONUTRIENT SURVEY 2018**	Phlebotomist put child "C" label here

1. MICS Cluster number .... ☐☐☐	2. HH number on MICS household list ☐☐
3. Child line number from MICS ...... ☐☐☐ *(look at household list from MICS)*	4. Name of child: ___________________________
5. Birth date of child: ☐☐/☐☐/☐☐ DAY MONTH YEAR	6. Age of child in completed months: ☐☐
9. Has child received a vitamin A capsule during the last **six** months *Show vitamin A capsule*	Yes.......................................1 No..2 → QUESTION 11 Not sure if it was vitamin A.... 3 → QUESTION 11 Don't know ....9 → QUESTION 11
10. When did (NAME) receive a vitamin A capsule? *Enter date only if within 6 months of the interview. If longer ago, go back to previous question*	A. From child welfare card ☐☐/☐☐/☐☐ DAY MONTH YEAR B. From recall ☐☐/☐☐/☐☐ DAY MONTH YEAR
11. Written informed consent given for anthropometry/phlebotomy?	Yes..........1 No............2 *If "No", stop here!*

ANTHROPOMETRY		
12. Anthorpometrist's code number?	Number............................ ☐☐	
13. Child's weight?	A.Kilograms (kg) ... ☐☐.☐ B.Kilograms (kg) ... ☐☐.☐ Not measured...................... ...99.9	
14. Was child mostly undressed (i.e. wearing only light under clothes)?	Yes.. 1 No... 2	
15. Child's length or height? ☐ Child <24 months old. ⇨ *Measure length (lying down).* ☐ Child ≥ 24 months old. ⇨ *Measure height (standing up).*	A. Centimeters (cm)..... ☐☐☐.☐ B. Centimeters (cm)..... ☐☐☐.☐ *Height not measured................................ 999.9*	→ **Check SAM chart. If child is <-3 SD, refer and complete Q27.**
16. Edema present in both feet?	Yes.. 1 No... 2	

17. Child's MUAC ? ⇨ *Measure in children 6-59 months of age only*	A.MUAC (cm) .... ☐☐.☐ B.MUAC (cm)............ ☐☐.☐ Not measured............................99.9	
18. Head circumference?	A. Head circ. (cm) .... ☐☐.☐ B. Head circ. (cm)............ ☐☐.☐ Not measured............................99.9	
19. Reason why weight, height, MUAC or head circumference measurement missing?	Disabled, cannot stand on scale ....1 Disabled, cannot measure height ....2 Uncooperative or uncontrollable..........3 Other (specify) ________________ 8 Refused.. 9	**Only ask this question in case of 99.9 or 99.9 in Q13 or Q15, otherwise skip**
PHLEBOTOMY / BLOOD SAMPLE COLLECTION		
20. Phlebotomist's code number:	Number............................. ☐☐	
21. Hemoglobin concentration (g/dL)	Hb (g/dL) ☐☐.☐	→ REFER IF <7.0 g/dL
22. Tube filling	None... 0 About ¼ .... 1 About ½ .... 2 About ¾ or more..................................3	
23. Malaria RDT result	Negative (after 15 minutes) .... 0 P. f positive .... 1 Pan positive. .... 2 P. f and Pan positive............................ 3 No valid test after 2 tries.......................4 Not done.. 9	REFER IF POSITIVE
24. Time of blood taking	☐☐ : ☐☐ HOUR MINUTE	
25. Date of blood taking	☐☐/☐☐/☐☐ DAY MONTH YEAR	
REFERRAL		
26. Coartem given to child if malaria infection identifies, and caretaker given instructions about how to administer coartem to the child?		Yes ....1 No 2
27. Child referred to health center for malaria, severe anemia and/or sever acute malnutrition:		Yes ....1 No 2

Measurer's Observations

WOMAN QUESTIONNAIRE
GAMBIA MICRONUTRIENT SURVEY 2018

1. MICS Cluster number	☐☐☐	2. HH number on MICS household list	☐☐
3. Woman line number from MICS survey......	☐☐☐	4. Household label number..........H	☐☐☐☐
4. Name of woman: ___________________________			
5. Is woman pregnant?	Yes................1 No..................2	6. Written informed consent given for questionnaire?	Yes...................1 No......................2

NUTRITION KNOWLEDGE AND ATTITUDES QUESTIONS		
7. Have you heard about anemia?	Yes .... 1 No 2	→ Next Q → Q10
8. What are the causes of anemia? *Do not prompt.* *Select all responses given by respondent*	Lack of iron in the diet/eat too little .... A Sickness/infection (malaria, hookworms, other infections such as HIV/AIDS) .... B Heavy bleeding during menstruation .... C Other (specify _____________)...................... X Don't know .... Z	
9. How can anemia be prevented? *Do not prompt.* *Select all responses given by respondent*	Eat iron-rich foods/diet rich in iron...................... A Eat/give vitamin-C-rich foods during or right after meals .... B Take/give iron supplements .... C Treat other causes of anemia (diseases and infections) .... D Other (specify _____________)...................... X Don't know .... Z	
10. Do you usually drink black tea, green tea or coffee during or shortly after your meals?	Yes .... 1 No 2	→ Next Q → Q12
11. How often do you usually drink black tea, green tea or coffee daily during or shortly after your meals?	Once a day ....1 2-3 times per day ....2 >3 times per day ....3 Don't know 8	
INDIVIDUAL BREAD OR RICE CONSUMPTION		
12. How often do you eat Senful (baguette type bread)?	Never or less than once a week..................1 Once a week...2 2-4 times per week.....................................3 5-6 times per week.....................................4 Once a day...5 2-3 times per day.......................................6	--> Q14
13. Per serving, how much Senful do you usually eat?	1/4 baguette..2 1/2 baguette..3 3/4 baguette..4 1 whole baguette ....5 1 1/2 baguetttes...6 2 or more baguettes....................................7 Don't know...8	

14. How often do you eat Sweet (loaf type bread)?	Never or less than once a week……………1 Once a week……………2 2-4 times per week……………3 5-6 times per week……………4 Once a day……………5 2-3 times per day……………6	→ Q16
15. Per serving, how much Sweet (loaf type bread) do you usually eat?	1 slice……………1 2 slices……………2 3 slices……………3 1/2 loaf ……………4 1 whole loaf or more……………5 Don't know……………8	
16. How often do you eat Tapalapa (heavy local baguette)?	Never or less than once a week……………1 Once a week……………2 2-4 times per week……………3 5-6 times per week……………4 Once a day……………5 2-3 times per day……………6	→ Q18
17. Per serving, how much Tapalapa (heavy local baguette) do you usually eat?	1/4 baguette……………1 1/2 baguette……………2 3/4 baguette……………3 1 whole baguette ……………4 1 1/2 baguetttes……………5 2 or more baguettes……………6 Don't know……………8	
FORTIFIED FOODS- KNOWLEDGE AND PRACTICES		
18. Have you ever heard about fortified vegetable oil (i.e. with nutrients added)?	Yes …………… 1 No …………… 2 Don't know …………… 8	→ NEXT Q → Q21 → Q21
19. Why do you think fortified oil (i.e. with added nutrients) is important? *Do not prompt*	Prevents blindness …………… A Reduces mortality …………… B Prevents vitamin deficiency …………… C Improve health status …………… D Other …………… Y (specify: ____________) Don't know …………… Z	
20. Do you prefer purchasing vegetable oil that is labeled as fortified (using the Enrichi logo or labeled otherwise)??	Yes …………… 1 No …………… 2 Don't know …………… 8	
21. Have you ever seen the Enrichi logo before ? *Show image of logo*	Yes …………… 1 No …………… 2 Don't know …………… 8	
22. Have you heard about fortified wheat flour (i.e. flour with nutrients added)?	Yes …………… 1 No …………… 2 Don't know …………… 8	→ NEXT Q → Q26 → Q26
23. Why do you think fortified flour (i.e. with added nutrients) is important?	Gives blood …………… A Reduces anemia …………… B Reduce mortality …………… C Improve health status …………… D Better school performance ……………E Better work output……………F Other (specify): ____________……………X Don't know …………… Z	
WOMAN DIETARY DIVERSITY		

24. **Please describe everything that you ate yesterday during the day or night, whether at home or outside the home.**

a) **Think about when you first woke up yesterday. Did you eat anything at that time? if yes: Please tell me everything that you ate at that time.**
Probe: **"Anything else?"** *until respondent says nothing else. If no, continue to question b).*

b) **What did you do after that? Did you eat anything at that time? If yes: Please tell me everything you ate at that time.**
Probe: **"Anything else?"** *until respondent says nothing else.*
Repeat question b) above until respondent says she went to sleep until the next day.
If respondent mentions mixed dishes like a porridge, sauce or stew, probe:

c) **What ingredients were in that (mixed dish)?**
Probe: **"Anything else?"** *until respondent says nothing else.*

A. Grains, white roots and tubers, and plantains	Yes .... 1 No .... 2 Don't know 8	
B. Pulses (beans, peas and lentils)	Yes .... 1 No .... 2 Don't know 8	
C. Nuts and seeds	Yes .... 1 No .... 2 Don't know 8	
D. Dairy	Yes .... 1 No .... 2 Don't know 8	
E. Meat, poultry and fish	Yes .... 1 No .... 2 Don't know 8	
F. Eggs	Yes .... 1 No .... 2 Don't know 8	
G. Dark green leafy vegetables (e.g. cassava leaves, spinach…	Yes .... 1 No .... 2 Don't know 8	
H. Other vitamin A-rich fruits and vegetables (e.g. pumpkin, carrots, squash, or sweet potatoes that are yellow or orange inside, ripe mango, ripe pawpaw, guava, or water melon)	Yes .... 1 No .... 2 Don't know 8	
I. Other vegetables	Yes .... 1 No .... 2 Don't know 8	
J. Other fruits	Yes .... 1 No .... 2 Don't know 8	
K. Red palm oil	Yes .... 1 No .... 2 Don't know 8	

PHYSICAL ACTIVITY		
25. Does your work involve vigorous/moderate-intensity activity that causes large increases in breathing or heart rate like carrying or lifting heavy loads, digging or construction work for at least 10 minutes continuously?	Yes ....1 No .... 2 Don't know 8	 → Q29 → Q29
26. In a typical week, on how many days do you do vigorous/moderate-intensity activities as part of your work?	Number of days.......................... ☐ Don't know 8	
27. How much time do you spend doing vigorous/moderate-intensity activities at work on a typical day? *Fill in either hours OR minutes, but not both.*	1. Hours.................................. ☐ OR 2. Minutes.......................... ☐☐ Don't know 8	
28. Do you usually walk or bike to work, school, or to do errands?	Yes...1 No...2 Unable to walk or bike.....................3 Don't know...8	
29. How many hours do you spend per day during the WEEKDAYS sitting?	1. Hours............................ ☐☐ Don't know 98	
30. How many hours do you spend per day during the WEEKENDS sitting?	1. Hours............................ ☐☐ Don't know 98	
31. Have you filled in the top part of the woman paper biological form?	Yes...1 No 2	Only allowed to continue if "yes"
32. Have you referred the woman to the anthropometry/phlebotomy site?	Yes...1 No 2	
Save questionnaire and continue when paper bio form is returned from anthropometry/ phlebotomy site		
BIOLOGICAL INFORMATION		
33. Record household label number here *enter label number from top right corner of biological form*	H☐☐☐☐	
34. Record label number here: *enter label number from top right corner of biological form*	W☐☐☐☐	
35. Record the line number form MICS household list *This number much match number entered in Q3*	☐☐☐	
URINE SAMPLE AND ANTHROPOMETRY		
36. Anthorpometrist's code number:	Number.............................. ☐☐	
37. Urine sample collected	Yes...1 No 2	→ Next Q → Q34

38. Approximate volume of urine collected (ml)	ml ……………………………… ☐☐ *Container more than half full enter 50* No urine……………………………………99	
39. Record urine ID from biological form *This number must match label number in top right corner of biological form*	Urine ID number W☐☐☐☐	
BLOOD PRESSURE (ALL WOMEN)		
40. Blood pressure (mmHg):	Systolic ☐☐☐ Diastolic ☐☐☐	→ NEXT Q (for non- pregnant) → Q 46 (for pregnant)
NON-PREGNANT WOMEN ONLY		
41. Woman's weight	A.Kilograms (kg)… ☐☐☐.☐ B.Kilograms (kg) … ☐☐☐.☐ Not measured……………………999.9	
42. Woman removed shoes and heavy clothing?	Yes………………………………………… 1 No………………………………………… 2	
43. Woman's height	A.Centimeters (cm) ☐☐☐.☐ B.Centimeters (cm) ☐☐☐.☐ Not measured…………………… 999.9	
44. Reason why weight or height measurement missing	Disabled, cannot stand on scale ……..1 Disabled, cannot measure height …….2 Uncooperative or uncontrollable……….3 Other (specify) ________________ 6 Refused…………………………………… 8	**Only ask this question in case of 999.9 in Q42 and/or Q44 otherwise skip**
MUAC (PREGNANT WOMEN ONLY)		
45. Woman's MUAC	A.MUAC (cm) ………… ☐☐.☐ B.MUAC (cm)………… ☐☐.☐ Not measured………………………99.9	
BLOOD SAMPLE COLLECTION		
46. Phlebotomist's code number:	Number………… ☐☐☐☐	
47. Hemoglobin concentration (g/L)	Hb ☐☐☐	→ REFER IF <70 G/L FOR PREGNANT WOMEN; <80 G/L FOR NON-PREGNANT WOMEN

48. Glycated hemoglobin (%)	HbA1c ☐☐.☐ %	
49. Malaria RDT result	Negative (after 15 minutes) ….. 0 P. f positive……………………… 1 Pan positive………………………. 2 P. f and Pan positive …………… 3 No valid test after 2 tries……….4 Not done……………………………8	→ Next Q (non-pregnant) → Q 52 (for pregnant)
50. Tube filling *Microtube collected from non-pregnant women ONLY!*	None……………………………… 0 About ¼ ……………………………… 1 About ½ ……………………………… 2 About ¾ or more…………………3	
REFERRAL- ALL WOMEN		
51. Woman referred to health center for malaria, anemia and/or acute malnutrition (i.e. low MUAC in pregnant women)		Yes …………..1 No ……………..2
52. Observations		

53. FINAL RESULT CODES: Completed interview and accepted participation in blood collection………………..1 Completed interview and refused participation in blood collection……………………2 No household member or no competent respondent at home at time of visit ……..3 Entire household absent for long period or moved away ………………………………4	Refused …………………………………………… 5 Dwelling vacant / Address not a dwelling …… 6 Dwelling destroyed ………………………………… 7 Dwelling not found ………………………………… 8 OTHER (SPECIFY) ___________________ 6

Interviewer put household "H" label here	**BIOLOGICAL FORM - WOMAN** **GAMBIA MICRONUTRIENT SURVEY 2018**	Phlebotomist put woman "W" label here

1. Cluster number .... ☐☐☐		**2.** HH number on MICS household list ☐☐	
3. Woman ID from MICS survey...... ☐☐☐		**4.** Name of woman:	
5. Written informed consent given for anthropometry/phlebotomy?	Yes...............1 No..................2	**6.** Is woman pregnant?	Yes...................1 No....................2

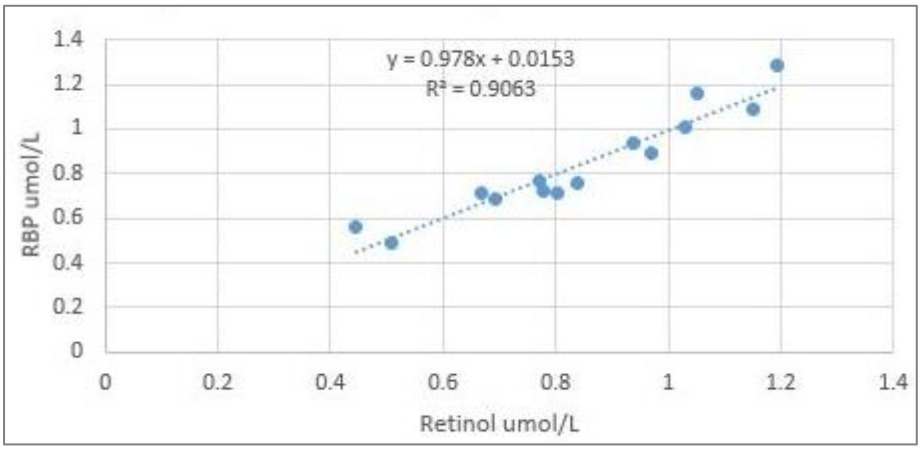
URINE SAMPLE RECEPTION AND ANTHROPOMETRY		
7. Anthorpometrist's code number:	Number.............................. ☐☐	
8. Urine sample collected	Yes..1 No 2	→ Next Q → Q 10 (for ALL women)
9. Approximate volume of urine collected (ml)	ml .... ☐☐ *Container more than half full enter 50* No urine..99	
BLOOD PRESSURE (ALL WOMEN)		
10. Blood pressure (mmHg):	Systolic ☐☐☐ Diastolic ☐☐☐	→Q11 (FOR NON- PREGNANT) →Q15 (FOR PREGNANT)
NON- PREGNANT WOMEN ONLY		
11. Woman's weight	A.Kilograms (kg)... ☐☐☐.☐ B.Kilograms (kg) ... ☐☐☐.☐ Not measured...........................999.9	
12. Woman removed shoes and heavy clothing?	Yes.. 1 No.. 2	
13. Woman's height	A.Centimeters (cm) ☐☐☐.☐ B.Centimeters (cm) ☐☐☐.☐ Not measured........................ 999.9	
14. Reason why weight or height measurement missing	Disabled, cannot stand on scale ....1 Disabled, cannot measure height ....2 Uncooperative or uncontrollable..........3 Other (specify) ________________ 8 Refused.. 9	ONLY ASK THIS QUESTION IN CASE OF 999.9 IN Q12 AND/OR Q14

MUAC (PREGNANT WOMEN ONLY)		
15. Woman's MUAC	A.MUAC (cm) .... ☐☐.☐ B.MUAC (cm)............ ☐☐.☐ Not measured...........................99.9	→ REFER IN Q22 IF < 18.5 CM
PHLEBOTOMY / BLOOD SAMPLE COLLECTION		
16. Phlebotomist's code number:	Number............................ ☐☐	
17. Record urine ID from container *This number must match label number in top right corner of biological form*	Urine ID number **W** ☐☐☐☐	
18. Hemoglobin concentration (g/dL)	Hb (g/dL) ☐☐.☐	→ REFER IF: <7.0 G/DL FOR PREGNANT WOMEN <8.0 G/DL FOR NON-PREGNANT WOMEN
19. Glycated hemoglobin:	HbA1c ☐☐.☐ %	
20. Malaria RDT result	Negative (after 15 minutes) ..... 0 P. f positive .... 1 Pan positive. .... 2 P. f and Pan positive................. 3 No valid test after 2 tries..........4 Not done...............................9	REFER IF POSITIVE
21. Tube filling	None .... 0 About ¼ .... 1 About ½ .... 2 About ¾ or more.....................3	
REFERRAL (ALL WOMEN)		
22. Coartem given to woman if malaria infection identified, and instructions about how to take coartem also given to woman?		Yes ....1 No 2
23. Woman referred to health center for anemia and/or malaria infection and/or underweight (pregnant only)?		Yes ....1 No 2

Measurer's Observations

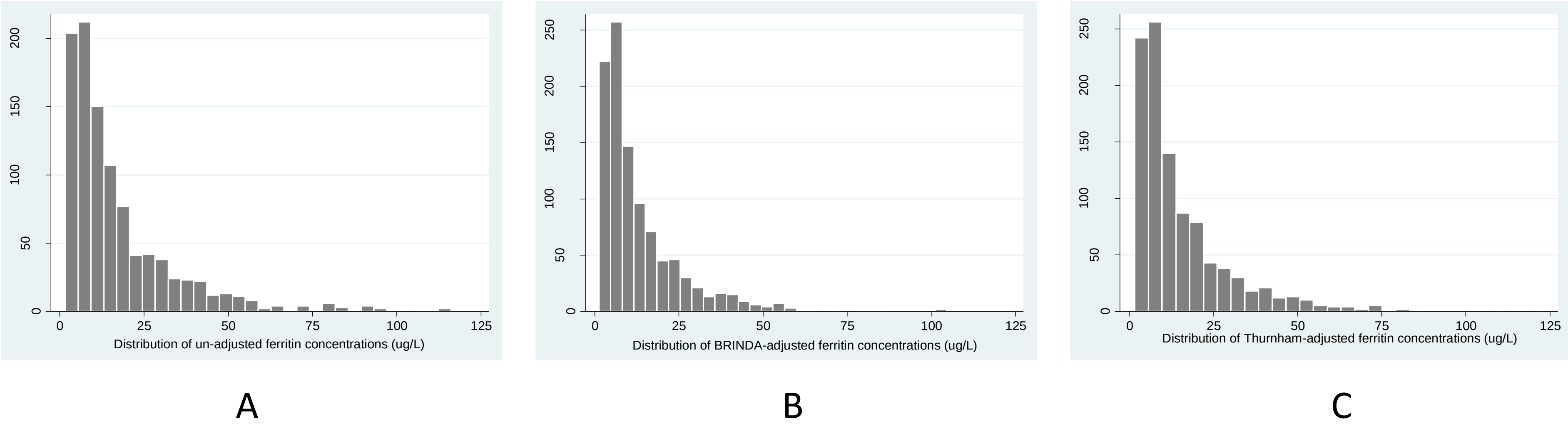
7.9. - COMPARISON OF SERUM RETINOL AND RETINOL BINDING PROTEIN

Figure A9 - 1. Correlation of retinol and retinol-binding protein concentrations in 14 samples, The Gambia 2018



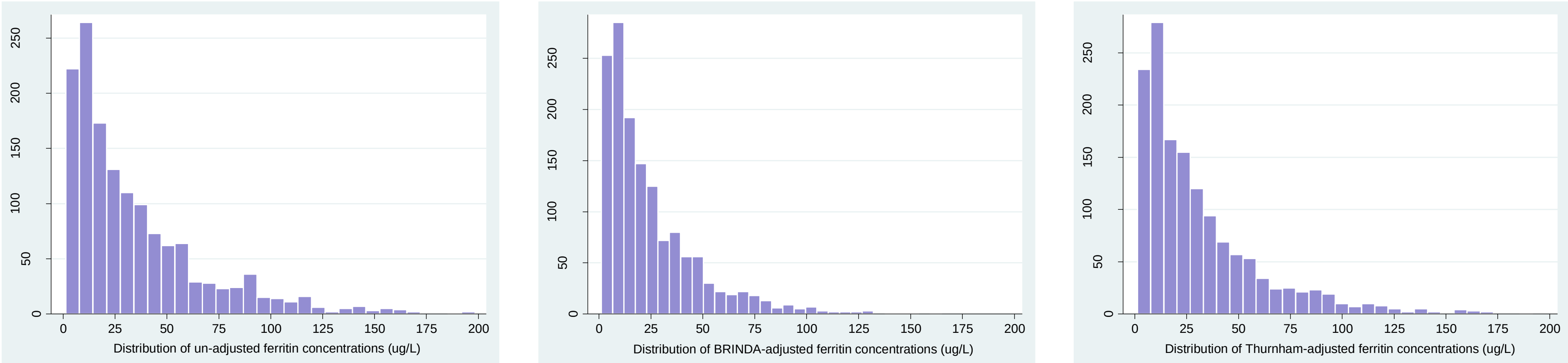
7.10. ADDITIONAL CHILD TABLES AND FIGURES

Figure A12 - 1. Distribution of ferritin concentrations with no inflammation adjustment (A), BRINDA inflammation adjustment (B), and Thurnham inflammation adjustment (C) in children 6-59 months of age, The Gambia 2018.



7.11. ADDITIONAL WOMAN TABLES AND FIGURES

Figure A13-1. Distribution of ferritin with no inflammation adjustment (A), BRINDA inflammation adjustment (B), and Thurnham inflammation adjustment (C) in non-pregnant women 15-49 years of age, The Gambia 2018.



A B C